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A CONFORMANCE HARNESS FOR VERIFIABLE MODULAR OPEN SYSTEMS OUTCOMES IN DEFENSE ACQUISITION: DESIGN, IMPLEMENTATION, AND CONTROLLED EVALUATION

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A CONFORMANCE HARNESS FOR VERIFIABLE MODULAR OPEN SYSTEMS OUTCOMES IN DEFENSE ACQUISITION: DESIGN, IMPLEMENTATION, AND CONTROLLED EVALUATION

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Abstract

The Department of Defense (DoD) requires systems that can accept new technology, suppliers, and mission capabilities without repeated wholesale redesign. The Modular Open Systems Approach (MOSA) is intended to enable that adaptability through modular architectures, severable components, open interfaces, technical data planning, and lifecycle competition. Yet public oversight shows a persistent measurement problem: programs frequently document MOSA intent without consistently verifying conformance or demonstrating the resulting cost, schedule, interoperability, and competition outcomes. This thesis develops and evaluates an evidence driven conformance harness that translates MOSA claims into reproducible technical evidence.

The research combines a targeted evidence synthesis, comparative case analysis, design science development, and a controlled synthetic experiment. The implemented prototype evaluates machine readable interface contracts, manifests, provenance, dependencies, software bills of materials, semantic behavior, determinism, latency, evidence completeness, and non compensable decision gates. Twenty candidate module packages were assessed: two independently implemented conformant controls and eighteen candidates containing a single known nonconformance. Each candidate was evaluated by both a minimal document and happy path baseline and the conformance harness. Twenty repeated executions per condition were used for timing and reproducibility analysis, while primary fault detection inference used the unique candidate packages.

The baseline detected 5 of 18 injected nonconformances, for 27.8 percent sensitivity (Wilson 95 percent confidence interval [CI], 12.5 to 50.9 percent). The harness detected 18 of 18, for 100 percent sensitivity (95 percent CI, 82.4 to 100 percent). The paired correctness difference was statistically significant under an exact McNemar test ($p = 0.000244$). Both methods accepted the two conformant controls, although the small control sample produces a wide specificity interval. Actionable diagnosis increased from 5 of 18 to 18 of 18, and evidence completeness for controls increased from 0.50 to 1.00. The harness executed a median of 177 checks rather than 14 and increased median primary runtime from approximately 0.28 to 5.95 milliseconds, demonstrating a deliberate computation for evidence tradeoff.

Keywords: Modular Open Systems Approach; MOSA; conformance testing; design science; mutation testing; defense acquisition; interoperability; evidence quality; reproducibility

Artifact availability. The executable prototype, synthetic candidates, raw results, statistical analysis, figures, tests, protocol, data dictionary, and integrity hashes are provided as an accompanying digital archive.

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List of Acronyms

Acronym	Meaning	Acronym	Meaning
AAF	Adaptive Acquisition Framework	**API**	Application Programming Interface
A RCI	Acoustic Rapid Commercial Off the Shelf Insertion	**C4ISR**	Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance
CI	Confidence Interval	**CI/CD**	Continuous Integration / Continuous Delivery
CISA	Cybersecurity and Infrastructure Security Agency	**COTS**	Commercial Off the Shelf
DevSecOps	Development, Security, and Operations	**DoD**	Department of Defense
DSR	Design Science Research	**DSRM**	Design Science Research Methodology
EVM	Earned Value Management	**FFRDC**	Federally Funded Research and Development Center
FP	False Positive	**GAO**	Government Accountability Office
HHI	Herfindahl Hirschman Index	**IP**	Intellectual Property
MBSE	Model Based Systems Engineering	**MDAP**	Major Defense Acquisition Program
MOSA	Modular Open Systems Approach	**NIST**	National Institute of Standards and Technology
NTIA	National Telecommunications and Information Administration	**OA**	Open Architecture
OSCAL	Open Security Controls Assessment Language	**OSD**	Office of the Secretary of Defense
OUSD(A&S)	Office of the Under Secretary of Defense for Acquisition and Sustainment	**OUSD(R&E)**	Office of the Under Secretary of Defense for Research and Engineering
p95	95th Percentile	**RACI**	Responsible, Accountable, Consulted, and Informed
RQ	Research Question	**SBIR**	Small Business Innovation Research
SBOM	Software Bill of Materials	**SCRM**	Supply Chain Risk Management
SEP	Systems Engineering Plan	**SOA**	Service Oriented Architecture
TN	True Negative	**TP**	True Positive
UAS	Unmanned Aircraft System	**V&V**	Verification and Validation

1. Executive Summary

The Department of Defense faces a modernization problem defined by repeated change: software updates, alternative sensors, new algorithms, supply chain disruptions, cybersecurity findings, and joint or coalition interface changes must be absorbed without destabilizing entire platforms. MOSA provides the governing business and technical strategy, but the public record demonstrates a gap between documentary compliance and exercised modularity. This thesis addresses that gap by treating openness as a set of claims that must be tied to versioned criteria, configurations, test methods, baselines, provenance, and decision consequences.

The study makes three connected contributions. First, it synthesizes public MOSA evidence and distinguishes realized outcomes from demonstrations, estimates, expectations, and policy assertions. Second, it defines a federated conformance harness architecture spanning structural openness, operational openness, economic openness, and security resilience. Third, it implements a reproducible software prototype and evaluates it against a conventional baseline using twenty synthetic candidate modules with known ground truth.

The controlled evaluation found a large difference in observability. The baseline rejected only 5 of 18 faulted candidates, whereas the harness rejected all 18; both accepted the two conformant controls. The harness also produced the correct actionable diagnosis for all 18 fault classes and complete evidence packages for the controls. An exact McNemar test on paired correctness decisions yielded $p = 0.000244$. These gains required more work by the machine: median evaluation time increased from about 0.28 milliseconds to 5.95 milliseconds and median checks increased from 14 to 177. In this setting, the additional computation was operationally small but materially improved evidence quality.

The experiment also revealed why a MOSA decision should not depend on a weighted score alone. Across 10,000 random dimension weight sets, several faulted candidates could pass when high performance in other dimensions compensated for a decisive defect. Once high and critical severity gates were applied, both controls passed under every weight set and every faulted candidate failed under every weight set. The result supports a governance design in which scores aid prioritization but cannot override missing interfaces, invalid provenance, semantic failure, security violations, or other non compensable conditions.

The public evidence and the prototype results support a conditional conclusion rather than a universal claim. MOSA can improve integration, technology refresh, and supplier participation when module boundaries are real, interface semantics are executable, data and rights are sufficient, and change is exercised under representative conditions. The harness cannot create market depth or erase certification constraints, but it can make those limitations visible and create comparable evidence for acquisition, engineering, test, cybersecurity, and sustainment decisions. The accompanying digital artifact contains the source code, synthetic candidates, raw observations, statistical analysis, generated figures, tests, data dictionary, and integrity hashes required to reproduce the controlled evaluation.

2. Introduction and Background

2.1 Strategic Context

The operational environment facing the U.S. defense enterprise is increasingly dynamic and contested: adversaries iterate rapidly, blend cyber and kinetic effects, and exploit seams between systems, organizations, and authorities. Joint and allied operations now require higher levels of interoperability at the data, interface, and mission thread levels, and systems that cannot be adapted quickly create operational risk.

Traditional defense acquisition approaches have often produced highly capable but tightly coupled systems that are difficult to adapt at the speed of operational change. In software intensive contexts this rigidity shows up as slow integration cycles, costly upgrades, and limited ability to introduce alternative vendors without extensive rework. These failure modes are precisely what MOSA was designed to address by shifting design emphasis from monolithic stacks to modular, interoperable compositions.

2.2 Problem Statement

A persistent gap continues to exist between the policy intent of the Modular Open Systems Approach (MOSA) and its execution in practice. This gap manifests in three recurring forms. First, hidden coupling behind compliant artifacts allows programs to meet documentation requirements while retaining tight dependencies in interfaces, data models, deployment assumptions, or tool chains. Second, unvalidated modularity claims mean that assertions of openness and modularity are rarely tested through rigorous third party integration, substitution, or competitive prototyping. Third, acquisition incentives remain misaligned with interchangeability, as contract and program structures often reward bespoke integration and short term delivery rather than long term competition, sustainment agility, and vendor independence. Collectively, these issues create a compliance posture that appears satisfactory at formal milestones but fails under modernization stress when new capabilities, vendors, or mission threads must be integrated rapidly and securely. Closing this gap requires operationalized conformance mechanisms that make openness measurable, testable, and contractually reinforced

(DoD, 2025a; GAO, 2025; 10 U.S.C. §§ 4401 to 4403, 2026).

2.3 Research Basis and Source Hierarchy

The research is grounded in authoritative public sources and a transparent source hierarchy. Statute establishes mandatory MOSA requirements; DoD instructions and guidebooks define implementation expectations; GAO reports provide independent oversight and program level evidence; NIST publications provide conformance, cybersecurity, and machine readable assessment concepts; and peer reviewed systems engineering literature supplies theoretical and comparative context. Claims are traced to named sources and are classified as realized, demonstrated, estimated, expected, or policy based so that evidence strength remains visible.

The proposed artifact is developed as an unclassified reference design rather than as an assessment of a specific acquisition program. Local prototype materials, source code, criteria files, and test fixtures are treated as reproducibility artifacts and are described in Appendix D. They support demonstration of the research design but do not substitute for external evidence of MOSA effectiveness.

2.4 Thesis Objective

The objective of this thesis is to design, implement, and evaluate a practical conformance harness for verifiable MOSA outcomes in defense acquisition. The research translates policy intent into versioned criteria, machine readable evidence, automated and human checks, substitution oriented tests, transparent scoring, and governance rules. It then tests a minimum viable implementation in a controlled synthetic environment to determine whether the harness detects and diagnoses known integration nonconformances more reliably than a minimal document and happy path baseline. The broader objective is to make MOSA actionable for engineers, auditable for assessors, useful for decision authorities, and measurable across the lifecycle (DoD, 2025a; GAO, 2025).

2.5 Research Questions and Analytical Proposition

The primary research question is: To what extent does publicly available evidence demonstrate that MOSA produces measurable improvements, and can an evidence driven conformance harness make modularity and openness more observable, diagnosable, and reproducible?

The question is decomposed into five supporting questions: (1) What measurable MOSA outcomes have been reported in defense programs? (2) How credible, independent, and

reproducible is the supporting evidence? (3) What technical, contractual, and organizational conditions appear necessary for benefits to emerge? (4) Does the implemented harness detect and diagnose known module nonconformances more reliably than a minimal baseline? and (5) What computation, evidence, reliability, and governance tradeoffs accompany the stronger assessment method?

The central proposition is that MOSA effectiveness is conditional: structural openness creates the possibility of change, operational openness demonstrates that change can be executed, and economic openness shows whether credible alternatives and lifecycle value emerge. A conformance harness should therefore improve the quality of evidence without claiming that a passing laboratory result alone proves program level cost savings or mission effectiveness (GAO, 2013; GAO, 2025; Zimmerman et al., 2019).

2.6 Scope

The scope of this thesis is intentionally focused on program and enterprise level MOSA conformance mechanisms, rather than prescribing a single technical stack or mandating uniform implementation technologies across the Department of Defense. This distinction is critical: MOSA is fundamentally an architectural and governance philosophy, not a fixed set of tools or platforms. As emphasized in DoD guidance, effective MOSA implementation requires a balance between enterprise coherence and program specific flexibility, ensuring that shared standards and evidence expectations provide a common foundation while still allowing tailoring based on mission needs, operational constraints, and risk environments (DoD, 2025a; DoD, 2020b). To support this balance, the thesis adopts a federated architectural model, in which a core set of enterprise standards, interface patterns, and conformance evidence types anchor interoperability, while domain level variations accommodate unique technical ecosystems such as C4ISR, autonomy, space systems, or logistics platforms.

Within this federated framing, the thesis examines a broad set of interdependent focus areas that collectively shape MOSA conformance. These include system architecture and modularity patterns, which define how components are decomposed, bounded, and governed; implementation concepts and reference workflows for a conformance harness that can be embedded into engineering pipelines; and cybersecurity and supplychain assurance mechanisms that ensure openness does not introduce vulnerabilities or compromise mission assurance. The scope also

encompasses governance structures and escalation pathways, recognizing that MOSA conformance is sustained not only through technical artifacts but also through decision authorities, configuration management processes, and crossprogram coordination bodies. Additionally, the thesis addresses metrics and feedback loops that enable continuous improvement, workforce adoption and tooling strategies that support practical implementation, and acquisition implications, including solicitation language, deliverables, and contractual enforcement mechanisms that translate technical expectations into binding program obligations.

These focus areas reflect the integrated nature of MOSA as articulated in DoD implementation guidance, which stresses that modularity, openness, and interchangeability must be reinforced through both technical standards and acquisition and sustainment practices (DoD, 2025a). By defining the scope at the intersection of engineering, governance, and contracting, the thesis positions MOSA conformance not as a static compliance activity but as a multidisciplinary, lifecyclespanning discipline. This approach ensures that the resulting conformance harness is not only technically sound but also organizationally viable, contractually enforceable, and adaptable to evolving mission and technology landscapes.

2.7 Research Design Overview

The study combines a targeted evidence synthesis, comparative case analysis, design science development, and a controlled mutation based feasibility evaluation. Public cases identify reported outcomes and evidence gaps. Those findings define the artifact requirements. The prototype is then evaluated against known conformant and nonconformant candidate modules under paired conditions, with unique packages used for detection inference and repeated runs reserved for timing and reproducibility.

Chapter 5 presents the evidence review and academic foundations. Chapter 6 specifies the research method and prototype implementation. Chapter 7 reports the controlled results. The remaining chapters generalize the artifact into a defense acquisition architecture, concept of operations, governance model, verification framework, metric system, contracting approach, implementation roadmap, and research agenda.

Table 1. Research questions, evidence inputs, and analytical outputs.

Research question	Principal evidence inputs	Analytical output
RQ1: What measurable outcomes have been reported?	Named programs; quantified cost, schedule, integration, interoperability, refresh, and supplier outcomes	Outcome inventory and evidence taxonomy

Research question	Principal evidence inputs	Analytical output
RQ2: How credible and reproducible is the evidence?	Source independence, baseline quality, measurement transparency, time horizon, and reproducibility	Evidence strength rating and confidence narrative
RQ3: What conditions enable or constrain benefits?	Architecture, data rights, governance, funding, workforce, security, and market depth	Cross case propositions and contextual moderators
RQ4: Does the prototype outperform the baseline?	Paired decisions for 20 candidate packages with known ground truth	Sensitivity, control acceptance, diagnosis accuracy, and exact paired test
RQ5: What tradeoffs accompany stronger assessment?	Runtime, test volume, evidence completeness, repeatability, digest stability, and weight sensitivity	Operational overhead, reliability, and gate design findings

Note. The questions link the evidence review to the design science artifact and prevent the thesis from treating MOSA policy statements as outcome evidence.

3. MOSA Principles and Design Logic

3.1 Modularity as a Mission Enabler

Modularity is the foundational architectural principle that enables systems to evolve at the pace of mission needs rather than the pace of monolithic acquisition cycles. In defense systems, where platforms often remain in service for decades, the ability to upgrade subsystems independently is essential for maintaining operational relevance in contested environments. By decomposing systems into modules with **coherent responsibilities**, **bounded dependencies**, and **explicit lifecycle ownership**, programs can isolate technical risk, accelerate modernization, and reduce the cascading impacts of subsystem changes (DoD, 2025a). This decomposition also supports more predictable integration pathways, as each module becomes a unit of capability that can be tested, validated, and evolved without destabilizing the broader architecture.

DoD MOSA guidance emphasizes that modularity is not solely a technical design choice but a **mission enabler** that must be deliberately planned and governed across the acquisition lifecycle (DoD, 2022a). Programs must define module boundaries early, establish interface contracts, and ensure that integration and substitution pathways remain viable as systems evolve. This requires governance mechanisms that maintain architectural integrity, enforce interface discipline, and prevent the reintroduction of hidden coupling. When executed effectively, modularity enables rapid insertion of emerging technologies such as autonomy algorithms, sensor payloads, or cyber defense components without requiring wholesale system redesign. This agility is critical for maintaining overmatch in dynamic operational environments where adversaries adapt quickly.

Modularity also enhances resilience by enabling targeted remediation of failures. When modules are loosely coupled and responsibilities are well defined, failures can be isolated and corrected without system wide disruption. This reduces downtime, improves maintainability, and supports mission continuity. In this sense, modularity becomes a strategic asset that aligns engineering practices with operational imperatives, enabling the Department of Defense to field adaptable, resilient, and competitive systems over long lifecycles.

3.2 Open Interfaces and Open Semantics

Openness is a multi layered property that extends far beyond the publication of interface documentation. While **syntactic openness** the availability of documented protocols, schemas, and

message formats is necessary, it is insufficient without **semantic openness**, which ensures that third parties can interpret and implement the interface correctly. Semantic openness includes consistent meaning, timing rules, state transitions, error behaviors, and assumptions about operational context (DoD, 2025a). Without these semantic details, an interface may be technically "open" but practically unusable, as third party developers cannot reliably implement compatible behavior.

DoD and Defense Standardization Program guidance increasingly emphasize the need for **machine readable interface artifacts**, such as interface definition languages (IDLs), schema files, timing constraints, and executable test suites (DoD, 2025a). These artifacts reduce ambiguity, support automated validation, and enable independent implementation without privileged knowledge. A MOSA conformance harness must therefore include both syntactic and semantic validation assets such as schema validators, timing tests, state machine conformance checks, and edge case test suites to ensure that interfaces behave consistently across vendors and platforms.

Open semantics also support long term sustainability. When interfaces are fully specified, programs can evolve underlying implementations without breaking interoperability. This reduces integration risk, supports competition, and ensures that interfaces remain stable even as technologies change. In this way, openness becomes a strategic enabler of modularity, interoperability, and lifecycle affordability.

3.3 Interoperability as a System Property

Interoperability is not a static achievement but a dynamic system property that must be continuously validated as systems evolve. Defense systems operate in complex, multi program environments where components from different vendors, services, and mission domains must interact reliably. Achieving durable interoperability requires governance structures that align architectural decisions across programs, disciplined release management that controls interface evolution, and continuous integration practices that validate cross program interactions as changes occur (DoD, 2021a).

A MOSA conformance harness must therefore incorporate mechanisms for **ongoing regression testing**, **substitution testing**, and **cross program integration validation**. These tests ensure that new releases do not break existing integrations and that modules remain replaceable throughout the lifecycle. Interface repositories play a critical role by recording version

relationships, compatibility constraints, and deprecation timelines. Configuration management practices must make downstream impacts visible before fielding, enabling programs to anticipate integration risks and coordinate interface changes across stakeholders (DoD, 2025a).

Interoperability also depends on organizational alignment. Programs must coordinate interface governance, share authoritative interface artifacts, and participate in enterprise level architecture boards. Without this alignment, technical interoperability can be undermined by inconsistent implementation practices, divergent interface interpretations, or uncoordinated changes. By treating interoperability as a sustained system property rather than a milestone event, MOSA implementation ensures that systems remain compatible, adaptable, and mission ready over time.

3.4 Portability and Replaceability

Portability and replaceability are central to MOSA's goal of enabling competition and reducing vendor lock in, but they must be decomposed into measurable dimensions to support meaningful conformance assessment. **Compute portability** refers to the ability of a module to run across different hardware or hosting environments, validated through container formats, runtime requirements, and platform abstraction layers. This ensures that modules can be deployed on alternative computer platforms without extensive rework (DoD, 2020b).

Data portability ensures that data can be moved, exported, and reused without proprietary encumbrances. This requires stable schemas, documented semantics, and accessible import/export tools. Without data portability, even a technically modular system can become practically locked in, as alternate vendors cannot access or interpret the data needed to integrate or replace components.

Integration portability focuses on the ability of modules to connect to different systems using stable, well documented interfaces and machine readable contracts. This dimension ensures that modules can be integrated into new environments without privileged knowledge or proprietary tooling.

A module may be technically replaceable but practically irreplaceable if critical artifacts such as test data, toolchains, or data rights are missing. DoD MOSA contracting guidance emphasizes the need to secure the data rights, interface artifacts, and test fixtures necessary for

alternate vendors to succeed (DoD, 2020c). A conformance harness must therefore validate the **entire replacement pathway**, ensuring that a third party can realistically integrate or substitute a module without privileged access. This includes verifying the availability of interface documentation, test fixtures, configuration scripts, and data rights required for independent implementation.

3.5 Competition Across the Lifecycle

Sustained competition is a core objective of MOSA, but it does not emerge automatically from modular or open architecture. Instead, competition must be deliberately cultivated through transparent conformance thresholds, complete documentation, and acquisition strategies that reward openness and interchangeability. High quality interfaces, reusable evidence artifacts, and accessible integration knowledge are essential to preventing de facto lock in, even when architectures are nominally modular (GAO, 2022).

Programs must structure solicitations and incentives to reward vendors who provide modular, well documented, and interoperable components. This includes requiring vendors to deliver machine readable interface artifacts, substitution test results, and evidence of portability and replaceability. Interface repositories and governance processes must ensure equal access to integration knowledge, preventing incumbents from gaining unfair advantages through privileged information.

MOSA guidance emphasizes that competition must be **observable and enforceable**, not assumed. This requires acquisition strategies that integrate technical conformance with contractual mechanisms, ensuring that openness is maintained across decades of sustainment (DoD, 2025a). When executed effectively, competition drives innovation, reduces lifecycle costs, and ensures that the Department of Defense can leverage the full breadth of the industrial base.

3.6 MOSA and Resilience

Modularity enhances resilience by enabling targeted updates, rapid patching, and substitution of compromised components. However, it also introduces new security considerations by expanding the number of interfaces, suppliers, and integration pathways. A balanced conformance approach must therefore assess both **adaptability** and **security**, ensuring that openness does not inadvertently increase mission risk (DoD, 2022b).

Security oriented conformance criteria should include:

- **Boundary controls** that enforce isolation between modules and prevent unintended interactions.
- **Supply chain assurance** mechanisms that validate the provenance, integrity, and trustworthiness of components.
- **Authenticated and authorized interfaces** that prevent unauthorized access or manipulation.
- **Configuration management** that tracks component versions, dependencies, and security posture.

A MOSA conformance harness must integrate these security tests alongside substitution, portability, and interoperability checks. This ensures that modularity strengthens rather than weakens mission assurance. DoD guidance increasingly links MOSA implementation to cybersecurity and supply chain resilience, emphasizing that modular architecture must be both adaptable and secure (DoD, 2025a).

Table 2. Conformance dimensions and representative measurable indicators.

Dimension	Core question	Representative indicators
Structural openness	Whether the architecture and rights posture make change technically and legally possible	Module cohesion and coupling; interface completeness; semantic coverage; data rights sufficiency; portability assets
Operational openness	Whether independent change can be executed safely under representative conditions	Integration time; substitution and rollback success; mission thread performance; regression rate; second integrator reproducibility
Economic openness	Whether the ecosystem supports credible alternatives and affordable change	Qualified supplier count; onboarding effort; competition events; HHI; cost of change; switching cost
Security and resilience	Whether openness preserves mission assurance and supply chain integrity	Provenance coverage; vulnerability response time; boundary control results; dependency exposure; recovery performance

4. DoD Policy and Institutional Context

Table 3. Policy drivers and implications for the conformance harness.

Policy driver	Primary source	Implication for the harness
Statutory MOSA requirement	10 U.S.C. §§ 4401 to 4403	Use MOSA to the maximum extent practicable; define and verify interfaces; address IP, technical data, integration, and configuration management.
DoD MOSA implementation guidance	DoD MOSA Implementation Guidebook (2025)	Translate objectives into measurable criteria, verification plans, cost benefit analysis, and acquisition deliverables.
Adaptive Acquisition Framework	DoDI 5000.02 and pathway instructions	Tailor evidence rigor to pathway and risk while retaining comparable core artifacts.
Digital engineering	DoDI 5000.97	Use configuration controlled models and authoritative data to generate traceable evidence.
Software and DevSecOps	DoDI 5000.87; DoD software guidance	Execute interface, dependency, and security checks continuously in delivery pipelines.
Supply chain and cybersecurity	NIST SP 800 to 161 Rev. 1 Update 1; DoD Zero Trust Strategy	Include SBOMs, provenance, vulnerability response, identity, authorization, and boundary controls.

Note. The harness does not replace these authorities; it organizes their required and recommended evidence into a repeatable assessment process.

4.1 Policy Drivers

DoD guidance frames MOSA as an integrated business and technical strategy that drives acquisition, engineering, and sustainment decisions rather than an optional best practice. Programs are expected to adopt modular designs, publish interface artifacts, and plan for severable modules so that competition, technology refresh, and reuse are achievable across long lifecycles.

A practical conformance harness therefore must translate those policy expectations into concrete evidence types (machine readable interface definitions, interface test suites, and documented modularity decisions) and make them discoverable to program offices, assessors, and acquisition authorities. This aligns MOSA's stated benefits competition, technology refresh, and interoperability with verifiable deliverables.

The December 2024 Tri Service memorandum further characterized MOSA as a warfighting imperative, raising the standard from an optional architecture preference to an operational modernization necessity (DoD, 2024).

4.2 Statutory and Regulatory Pressure

DoD's MOSA posture is grounded in statutory direction that requires major defense acquisition programs to *consider* and, where appropriate, adopt modular open systems practices;

this statutory baseline is reflected in DoD MOSA materials and the explicit references to U.S. Code on the DoD MOSA site. Programs therefore cannot treat MOSA as optional guidance alone program managers must document architectural choices, datarights strategies, and modularity tradeoffs as part of their acquisition justification and decision records. (DoD, 2025a)

Regulatory instruments and implementing guidance create discrete checkpoints where MOSA evidence should be produced and evaluated. The MOSA community of practice and associated DoD guidebooks identify deliverables (e.g., interface definitions, modularity rationales, and datarights plans) that should accompany milestone submissions so reviewers can assess whether modularity claims are actionable rather than aspirational. In practice, this means a conformance harness must produce auditable, machinereadable artifacts that map to regulatory decision points so that documentary compliance is backed by executable evidence rather than freetext assertions. (DoD, 2025a)

4.3 Alignment with the Adaptive Acquisition Framework

The Adaptive Acquisition Framework (AAF) explicitly endorses tailoring acquisition approaches to program context and risk, offering multiple pathways (e.g., Major Capability, Middle Tier, Software) that programs may combine or transition between as needed. Because MOSA is a lifecycle posture rather than a single technical prescription, its evidence and verification artifacts must be pathway aware: the same underlying conformance evidence (interface schemas, substitution test results, SBOMs, etc.) should be expressible in formats and levels of rigor appropriate to the pathway in use.

A practical conformance harness therefore serves two roles within the AAF: (1) it standardizes evidence so reviewers across pathways see consistent, comparable artifacts; and (2) it preserves tailoring by allowing programs to select the subset and rigor of checks that match mission urgency and risk tolerance. By mapping harness outputs to AAF checkpoints and tailoring decisions, program offices and Milestone Decision Authorities can reduce interpretive drift while retaining the AAF's flexibility to adapt acquisition strategy to operational needs. (DoD, 2026)

4.4 Software Pathway and DevSecOps Implications

Software acquisition pathways emphasize rapid, iterative delivery through DevSecOps practices software factories, CI/CD pipelines, and automated testing are now central to how

capability is produced and sustained. In this environment, MOSA verification cannot remain an episodic, milestoneonly activity; instead, conformance checks must be integrated into the software delivery pipeline so that interface conformance, portability validators, and semantic tests run continuously and produce signed, machinereadable evidence. (DoD, 2026)

Embedding MOSA checks in DevSecOps pipelines yields two operational benefits: it converts compliance into continuous telemetry that surfaces architectural drift early, and it produces reusable artifacts (test results, schema validation logs, SBOM snapshots) that acquisition reviewers can ingest without bespoke manual review. For programs using the Software Acquisition pathway or Middle Tier approaches, this alignment reduces review latency, supports incremental capability insertion, and makes MOSA a routine part of software quality rather than an external audit after the fact. (DoD, 2026)

4.5 Joint and Coalition Demands

Modern operations increasingly require cross Service and multinational integration data sharing, common situational awareness, and coordinated effects across partners are routine mission requirements. MOSA conformance therefore must be evaluated not only within single program boundaries but against mission thread scenarios that reflect joint and coalition interactions.

A conformance harness should support scenariobased test suites and sanitized mission threads that exercise timing, security, and semantic interoperability across participating systems while respecting classification and partnersharing constraints. By capturing missionlevel evidence (endtoend traces, compatibility matrices, and sanitized logs) and making those artifacts discoverable, the harness enables programs to demonstrate that modular interfaces and semantics hold under realistic joint conditions rather than only in isolated lab tests. (DoD, 2025a)

4.6 Institutional Friction Points

Institutional friction misaligned incentives, uneven assessor expertise, and fragmented authority undermines MOSA outcomes when documentary compliance substitutes for executable evidence. Standardized evidence formats, assessor playbooks, and clear role definitions embedded in the harness reduce duplication and clarify accountability across contracting, engineering, test, and cybersecurity communities.

By making MOSA expectations explicit, repeatable, and traceable, the harness helps align institutional behavior with policy intent and provides program offices with the tools to demonstrate real modularity, not just paperwork.

The July 2025 DoD software engineering guide reinforces continuous delivery, automated evidence, and modular software practices. GAO's June 2026 acquisition synthesis likewise cautions that systems not designed with modular open architectures become harder and more expensive to upgrade (DoD, 2025c; GAO, 2026).

5. Evidence of MOSA Effectiveness in Practice

5.1 Review Scope and Source Hierarchy

The review is a targeted, reproducible evidence synthesis rather than a claim of exhaustive systematic coverage. Searches prioritized the U.S. Code, DoD issuances and guidebooks, GAO reports, official program material, NIST conformance and security publications, peer reviewed software and systems engineering literature, and publicly accessible thesis repositories. Sources were included when they identified a named program, demonstration, dataset, or implemented artifact and described an intervention, baseline, or observable outcome. Expected benefits and policy statements were retained for context but were not coded as realized evidence. Appendix G records the repositories, representative search strings, screening rules, and search date, following transparency principles from software engineering evidence reviews (Kitchenham & Charters, 2007).

The review is structured as an evidence synthesis rather than a conventional meta analysis because the public record does not provide a sufficiently homogeneous set of effect estimates, baselines, or samples. Reported results use different units, observation windows, system boundaries, and counterfactual assumptions. The analytical task is therefore to classify the type and strength of each claim, preserve uncertainty, and identify recurring mechanisms and boundary conditions.

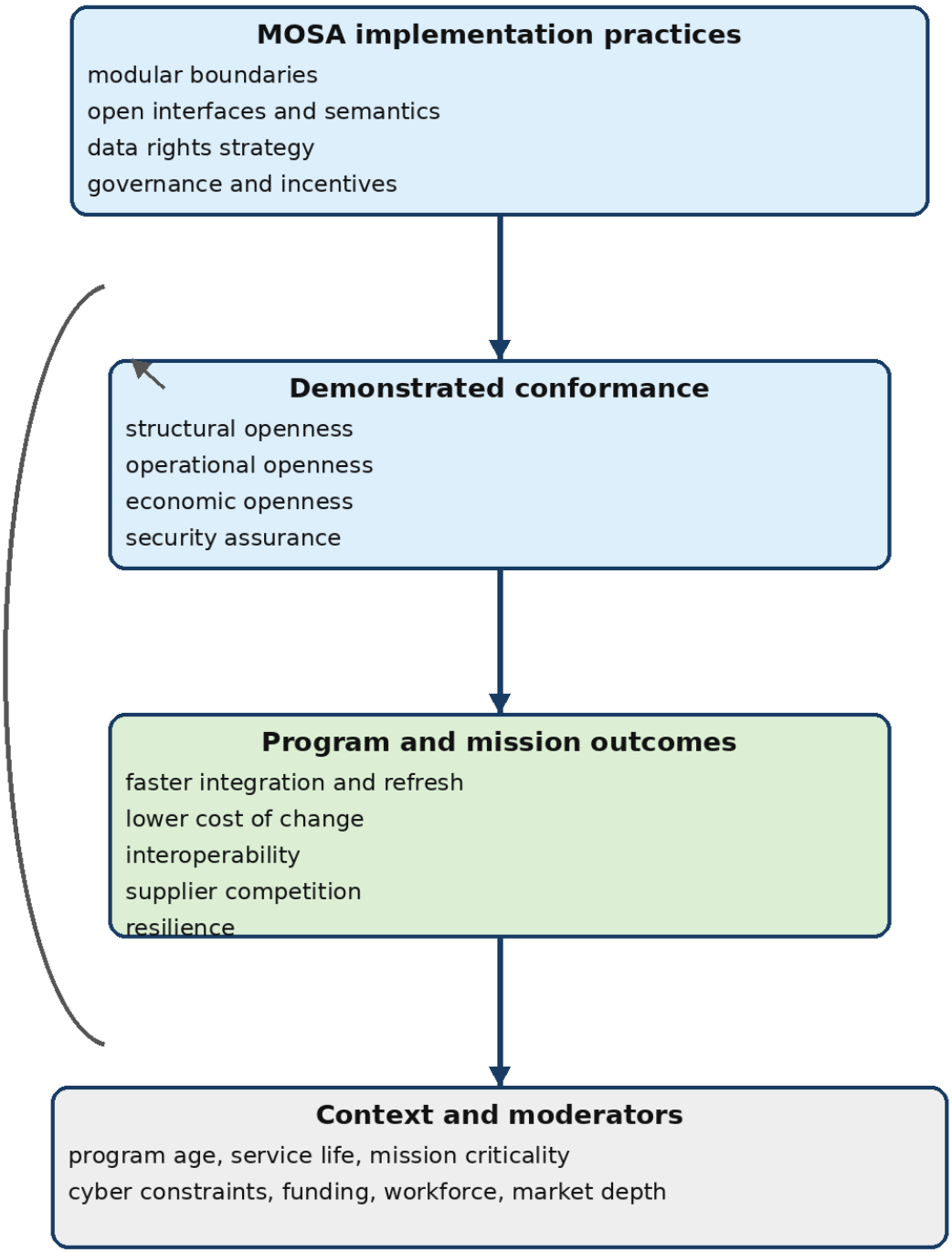


Figure 1. Conceptual logic linking MOSA practices, demonstrated conformance, and program outcomes.

Note. The framework treats context as a moderator and does not assume that conformance indicators alone prove causal program outcomes.

Table 4. Evidence source hierarchy and admissibility rules.

ier	Source class	Examples	Admissibility rule
ier 1	Independent oversight and statute	GAO reports; U.S. Code	Admit as high authority evidence of policy, reported program behavior, and documented outcomes; distinguish GAO analysis from program reported figures.
ier 2	Official DoD and Service sources	Instructions, guidebooks, program reports	Admit for requirements, implementation practices, and program data; flag self reporting and estimated counterfactuals.
ier 3	Peer reviewed and research center studies	Journal articles; systems engineering research	Admit for theory, synthesis, and methods; assess case provenance and empirical transparency.
ier 4	Industry and practitioner material	White papers; presentations; vendor reports	Use for implementation insight only when claims are traceable; do not treat marketing assertions as outcome evidence.

Note. Evidence is classified by authority and transparency, not merely by publication format.

5.2 Academic Foundations and Comparable Applied Theses

The artifact is grounded in three academic traditions. Information hiding modularity treats a module boundary as a decision boundary that shields likely change from the rest of the system (Parnas, 1972). Design science research requires an artifact to be motivated, designed, demonstrated, evaluated, and communicated rather than presented only as a conceptual recommendation (Hevner et al., 2004; Peffers et al., 2007). Empirical software engineering contributes executable conformance, architecture reflexion, and fault injection methods for determining whether implementation behavior matches declared design intent (Murphy et al., 2001; Jia & Harman, 2011). Together, these traditions support the thesis claim that documentary openness is weaker evidence than reproducible substitution and behavioral conformance.

Comparable master's theses also establish a useful quality benchmark. Vogler (2023) tested open architecture claims against programmatic cost and schedule data and reported a result that did not simply confirm the expected benefit. Cole (2011) used acquisition professional interviews to examine risk and open architecture. Ryburn (2012) combined archival data, interviews, and cases to study small business participation in open system initiatives. Nyman (2022) and Truong (2025) implemented and evaluated software testing frameworks rather than stopping at a proposed design. The present thesis follows the same build and evaluate expectation while retaining a broader acquisition and governance scope.

Table 5. Comparable applied theses and implications for this research

Study	Primary method	Implication for the present thesis
Cole (2011)	Practitioner interviews on risk, uncertainty, and open architecture	Treat organizational incentives and practitioner judgment as part of the system.

Study	Primary method	Implication for the present thesis
Ryburn (2012)	Archival SBIR analysis, interviews, and company cases	Connect openness to supplier participation using more than policy assertions.
Vogler (2023)	Empirical comparison of program management metrics	Allow negative or mixed findings and disclose the counterfactual.
Nyman (2022)	Framework construction and pilot evaluation	Demonstrate that an interface testing artifact can be used, not merely specified.
Truong (2025)	Implemented automated framework with repeated operational runs	Report execution time, reliability, and gating performance from observed runs.

Note. The comparison is methodological. The studies address different systems, samples, and outcome domains and are not treated as direct effect size comparators.

5.3 Evidence Classification

Every claim is coded as a realized outcome, demonstrated outcome, estimated benefit, expected benefit, or policy assertion. A realized outcome records an event or result that occurred in a program and was measured. A demonstrated outcome was observed in a prototype, laboratory, or integration event but may not generalize to fleet scale operations. An estimated benefit relies on a modeled or official counterfactual. An expected benefit is prospective. A policy assertion explains why MOSA should be used but does not contain an observed result. This classification prevents projected savings from being presented as actual savings and keeps technical demonstrations distinct from longitudinal lifecycle evidence.

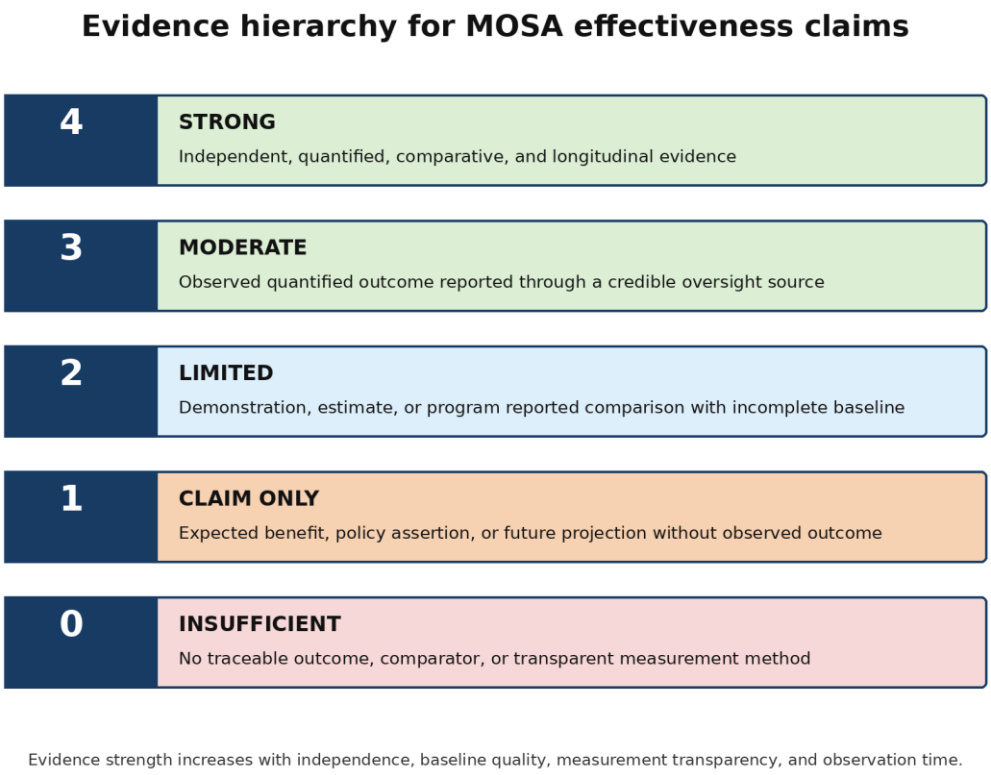


Figure 2. Evidence hierarchy used to evaluate public claims of MOSA effectiveness.

Note. Author developed synthesis. Higher ratings require stronger independence, baseline quality, transparency, and longitudinal coverage.

5.4 Evidence Strength Rubric

Evidence strength is evaluated on a 0 to 4 scale using five factors: outcome realization, source independence, baseline quality, measurement transparency, and longitudinal coverage. The score is an analytical aid, not a claim that complex evidence can be reduced without judgment. Each rating is accompanied by a narrative explaining why the evidence is strong or limited.

Table 6. Evidence strength scoring rubric.

Score	Outcome	Independence	Baseline	Transparency	Coverage
4, Strong	Observed and quantified	Independent or auditable	Documented comparator or counterfactual	Method and data sufficiently transparent	Longitudinal or repeated
3, Moderate	Observed and quantified	Credible oversight source, often relaying program data	Plausible documented baseline	Method partially available	Adequate but bounded
2, Limited	Demonstration, estimate, or self report	Primarily program reported	Estimated or weak comparator	Partial method	Short term or one event
1, Claim only	Expected or asserted	Interested source or policy text	No comparator	No usable method	No observation window
0, Insufficient	Outcome unclear	Untraceable	None	None	None

5.5 Comparative Case Evidence

Public evidence supports a qualified conclusion that MOSA can be effective in practice. The strongest reported results involve technical integration, development time, selected cost avoidance, and participation by alternate suppliers. The underlying evidence is heterogeneous: some outcomes were observed in operational programs, others occurred in demonstrations, and several baselines were estimates supplied by program officials rather than independently reconstructed counterfactuals (GAO, 2013; Zimmerman et al., 2019).

Table 7. Selected public evidence of MOSA effectiveness.

Program or study	Reported result	Interpretation	ating
Acoustic Rapid COTS Insertion (A RCI)	Navy reported 17% lower development and production cost, 13% lower operating and support cost, and 80% less development time; first ship installation occurred 2 years after program start rather than a decade or more.	Realized, quantified, but program reported through GAO; public counterfactual detail is incomplete.	oderate (3/4)

Program or study	Reported result	Interpretation	ating
Virginia class submarine technology refresh	Navy reported $96 million in cost avoidance from 2001 through 2010.	Realized cost avoidance claim with limited public calculation detail.	imited moderate (2/4)
UAS Control Segment architecture	A demonstration integrated 10 applications in less than 2 weeks versus an official comparison of about 1 year for traditional integration.	Direct technical demonstration; not a fleet wide longitudinal result.	oderate (3/4)
Fire Scout radar integration	Radar integrated in 18 months versus an estimated 3 years using the prior approach.	Observed event with estimated rather than controlled baseline.	imited moderate (2/4)
Shadow universal payload pod	Four third party vendors used standardized power and data connections to design payloads.	Direct evidence of supplier enablement; no quantified cost effect.	oderate (3/4)
GAO review of 20 newer programs	Fourteen reported at least some MOSA implementation; none documented a formal MOSA cost benefit analysis.	Strong oversight evidence about planning and measurement gaps, not technical conformance or causal effects.	trong for process gap (4/4)

Note. Program results in the first five rows are reported in GAO (2013). GAO (2025) did not independently rate these cases, and the ratings shown are the author's application of the rubric in Table 5.

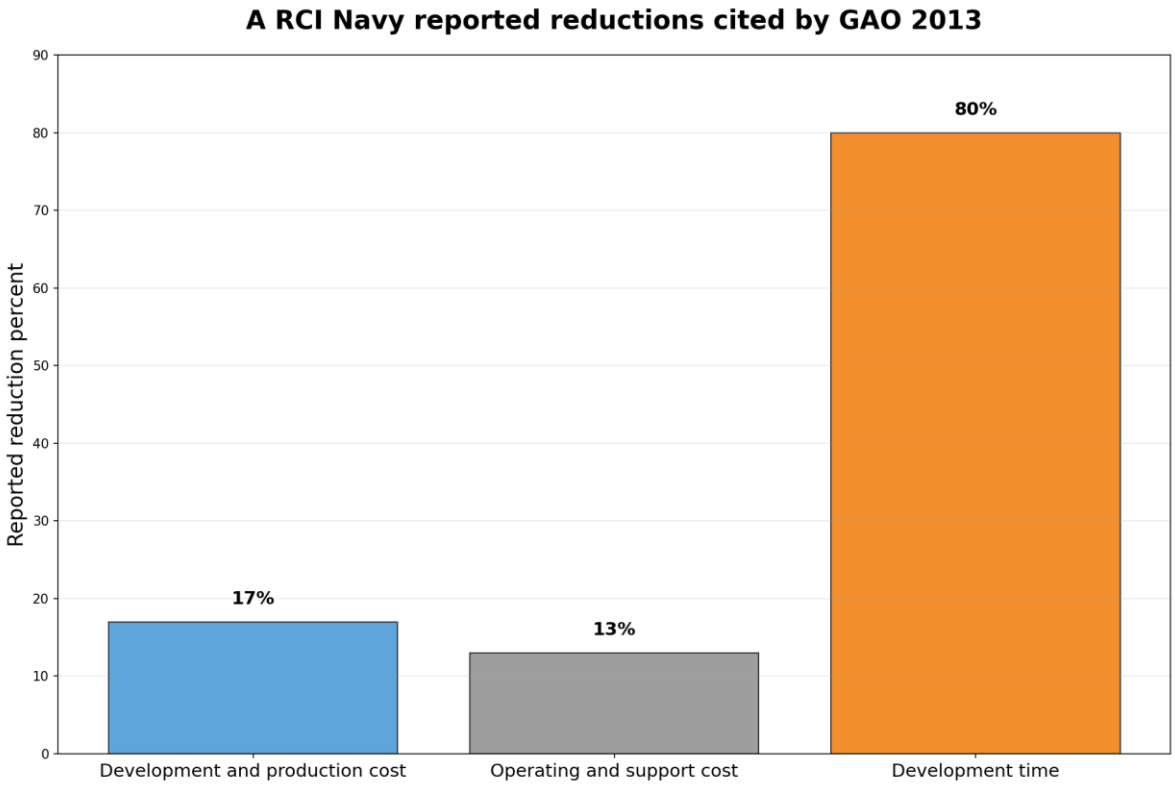


Figure 3. Navy reported A RCI reductions cited by GAO (2013).

Note. Values are Navy reported results reproduced by GAO (2013); they should not be interpreted as controlled causal estimates.

5.6 Cross Case Findings

The cases support several cross case propositions. First, MOSA appears most valuable for long lived systems that require repeated technology refresh. Second, stable interface syntax is insufficient without usable semantics, test assets, integration knowledge, and technical data rights. Third, the most persuasive evidence occurs when an alternate component or supplier is actually integrated, not when a program merely publishes an interface document. Fourth, near term architecture and test costs are visible immediately, whereas sustainment and competition benefits emerge over longer periods and are therefore easier for programs to discount. Fifth, leadership, funding, government technical expertise, and portfolio coordination materially influence whether technical openness becomes operational and economic openness.

These propositions are mechanisms and conditions, not universal laws. A mature proprietary system with few anticipated upgrades may not produce the same benefit profile as a software intensive mission system with frequent refresh cycles. Similarly, an open interface may create little competition when the qualified supplier market is thin or when cybersecurity, safety, certification, or classified data constraints make substitution inherently expensive.

5.7 Evidence Gaps and Interpretive Limits

GAO's 2025 review provides the most important counterweight to positive case evidence. Fourteen of 20 selected programs reported implementing MOSA to some extent, but none provided documentation of a formal MOSA cost benefit analysis. Only two of nine program executive offices had a formal process specifically intended to coordinate MOSA opportunities across programs. GAO also did not assess technical conformance; its findings concern program decisions, planning documents, resources, and policy gaps (GAO, 2025).

The public record is affected by self reporting bias, survivorship bias, inconsistent definitions, unavailable classified data, weak counterfactuals, and the difficulty of separating MOSA effects from management quality, funding stability, workforce skill, and concurrent modernization. Consequently, the evidence supports the statement that MOSA can create substantial value under appropriate conditions, but it does not support a universal percentage savings claim or the assumption that documentary compliance automatically causes improved outcomes.

5.8 Implications for the Conformance Harness

The central evidence gap is not an absence of successful examples; it is an absence of standardized, reproducible measurement. The proposed harness addresses that gap by binding each claim to a criterion, system boundary, configuration, baseline, observation window, test method, provenance record, and confidence assessment. Structural evidence establishes whether change is possible; operational evidence demonstrates whether change can be executed; economic evidence establishes whether credible alternatives and lifecycle value actually emerge.

The evidence review therefore changes the role of the harness. It is not merely a compliance checker. It is a measurement infrastructure that enables programs to generate comparable evidence over time, supports more defensible cost benefit analysis, and makes it possible to distinguish realized outcomes from expectations. This logic directly informs the research design in Chapter 6 and the scoring, confidence, and evaluation framework developed later in the thesis.

6. Research Methodology and Prototype Implementation

6.1 Design Science Strategy

The study follows a design science research methodology because it seeks both understanding and an evaluated artifact. The problem identification stage synthesizes evidence of MOSA outcomes and recurring measurement gaps. Objectives are derived from statute, DoD guidance, conformance testing principles, modularity theory, and comparable implemented theses. The design and development stage produces machine readable contracts, candidate modules, baseline and harness evaluators, evidence packages, analytics, and governance rules. Demonstration and evaluation occur through a controlled synthetic experiment, and the final artifact is communicated through the thesis and companion archive (Hevner et al., 2004; Peffers et al., 2007).

6.2 Research Hypotheses and Unit of Analysis

The primary unit of analysis is a unique candidate module package. A package contains an implementation and, when applicable, a manifest, SBOM, documentation, and known ground truth status. The paired design exposes every package to both assessment methods. Repeated executions are not treated as additional fault cases; they are used only to characterize runtime and reproducibility. This distinction prevents pseudo replication and keeps the inferential sample tied to the twenty independently constructed packages.

- H1: The conformance harness will produce more correct accept/reject decisions than the minimal baseline on the paired candidate set.
- H2: The conformance harness will produce more complete and actionable evidence than the minimal baseline.
- H3: The conformance harness will require more computation and execute more checks than the minimal baseline.
- H4: Non compensable gates will make accept/reject decisions less sensitive to alternative dimension weights than score only decisions.

Table 8. Research variables and operational definitions

Construct	Operational measure	Interpretation
Fault detection sensitivity	Faulted candidates rejected / 18	Ability to expose known nonconformance.
Control acceptance (specificity proxy)	Conformant controls accepted / 2	Ability to avoid rejecting the two valid controls.
Diagnostic accuracy	Correct root cause labels / 18	Actionability of the evidence.
Evidence completeness	Observed expected evidence items / total expected items	Coverage of manifest, provenance, SBOM, documentation, and executable results.
Assessment effort proxy	Checks or tests executed and runtime in milliseconds	Machine effort required to produce the decision.

Construct	Operational measure	Interpretation
Reliability	Decision and evidence digest stability over 20 repeated runs	Repeatability under the recorded environment.
Weight robustness	Pass probability across 10,000 random weight sets	Sensitivity of score only and gate adjusted decisions.

6.3 Prototype Requirements and Architecture

The minimum viable artifact was required to provide executable interface contracts, a conventional comparator, deterministic ground truth, multiple fault classes, per candidate evidence, transparent decision rules, statistical outputs, automated tests, and a complete reproduction path. The implementation is intentionally small enough to audit while exercising the same evidence patterns proposed for a larger federated defense harness: criteria, ingestion, provenance, static analysis, dynamic tests, scoring, gates, reports, and integrity hashes.

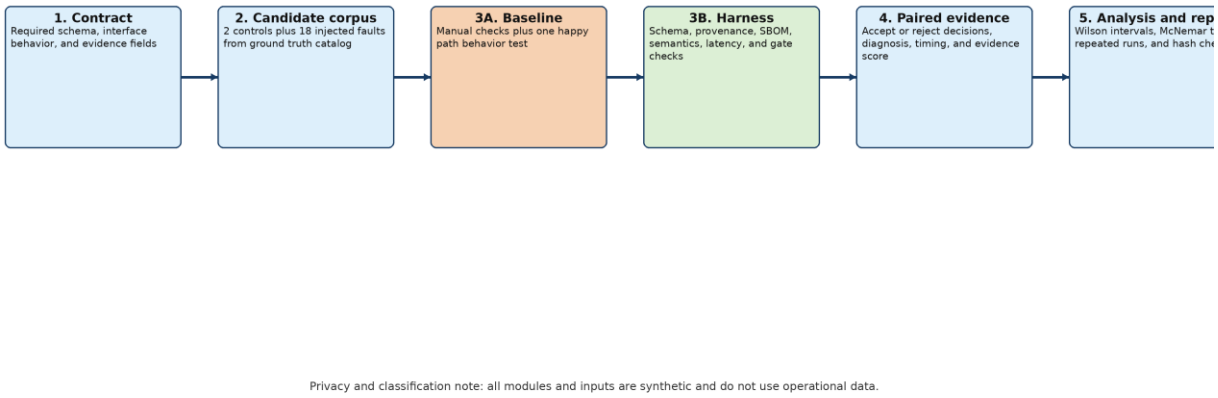


Figure 4. Controlled study workflow from interface contract and candidate construction through paired evaluation and evidence analysis.

Table 9. Implemented prototype components

Component	Implementation	Research role
Interface contracts	JSON Schema for sensor input, processor output, and module manifest	Defines machine checkable syntax and version expectations.
Candidate packages	Two independent controls and eighteen single fault implementations	Provides known ground truth and diverse failure mechanisms.
Baseline evaluator	Manifest presence, import, one happy path execution, and basic output checks	Represents a low cost documentary and smoke test condition.
Harness evaluator	Provenance, schema, dependency, security, semantic, determinism, latency, evidence, scoring, and gate checks	Represents evidence driven conformance assessment.
Experiment runner	Paired primary run plus twenty repeated runs in randomized candidate order	Produces decisions, timing, and reproducibility observations.
Analysis and reporting	Wilson intervals, exact McNemar, Wilcoxon, effect description, weight sensitivity, figures, and CSV/JSON outputs	Supports transparent interpretation and independent re analysis.

Component	Implementation	Research role
Reproducibility controls	Pinned dependencies, tests, seed, data dictionary, protocol, evidence digests, and SHA 256 manifest	Supports reconstruction and tamper detection.

6.4 Synthetic Mission Thread and Interface Contract

The testbed models a stateless processing module that receives a sensor event and returns a classification, confidence, and risk score while preserving the event identifier and schema version. The contract specifies required fields, numerical ranges, enumeration values, and deterministic semantics. Eight fixed vectors exercise nominal, watch, alert, and boundary cases. Forty eight additional vectors are generated from the recorded seed during each harness evaluation. The synthetic mission thread is non sensitive and deliberately excludes hardware, classified data, safety certification, and human operator behavior.

6.5 Fault Model and Ground Truth

Mutation based fault injection provides explicit ground truth for evaluating detection methods. Each nonconformant candidate contains one intended fault class, and experiment metadata records the expected status and diagnosis. The fault model spans documentary omissions, structural version defects, provenance failures, hidden dependencies, security boundary violations, semantic incompatibilities, state leakage, nondeterminism, edge behavior, and performance. Mutation testing is not a substitute for field defects, but it is a recognized empirical strategy for determining whether a test or analysis method can expose seeded faults (Jia & Harman, 2011).

Table 10. Injected fault catalog and primary detection result

Fault class	Domain	Injected condition	Baseline	Harness
Configuration Coupling	Operational	Requires an undocumented runtime configuration secret.	Missed	Detected
Edge Case Exception	Operational	Fails on a boundary quality input omitted from the happy path.	Missed	Detected
Event Id Not Preserved	Semantic	Returns a different event identifier than the input contract requires.	Missed	Detected
Forbidden Network Import	Security	Imports a prohibited network capable dependency.	Missed	Detected
Hidden File Dependency	Competition	Reads an undeclared local file at runtime.	Missed	Detected
Latency Violation	Performance	Exceeds the pre specified 10 ms p95 event latency criterion.	Missed	Detected
Missing Manifest	Structural	Omits the machine readable module manifest.	Detected	Detected
Missing Output Field	Syntactic	Omits a required output field.	Detected	Detected
Missing Sbom	Evidence	Omits the referenced software bill of materials.	Missed	Detected
Nondeterministic Output	Operational	Produces different results for repeated identical inputs.	Missed	Detected
Provenance Hash Mismatch	Provenance	Source digest does not match the manifest attestation.	Missed	Detected
Semantic Threshold Shift	Semantic	Changes the classification threshold while preserving output shape.	Missed	Detected
State Leakage	Operational	Uses mutable global state that contaminates subsequent calls.	Missed	Detected

Fault class	Domain	Injected condition	Baseline	Harness
Undeclared Dependency	Competition	Imports a dependency not declared in the manifest.	Missed	Detected
Units Error	Semantic	Computes the risk score using an incompatible unit transformation.	Missed	Detected
Wrong Interface Version	Structural	Declares an unsupported interface version.	Detected	Detected
Wrong Output Type	Syntactic	Returns a required numeric field using an invalid type.	Detected	Detected
Wrong Schema Version	Syntactic	Returns an unsupported output schema version.	Detected	Detected

Note. Every faulted package contains one intended ground truth nonconformance. Secondary findings may also appear when a primary defect blocks downstream checks.

6.6 Baseline and Harness Conditions

The baseline checks for a manifest, attempts to import the module, executes one happy path vector, and inspects the output for required keys, supported versions, numerical types, and enumeration values. It approximates a low friction integration process in which documentation and a smoke test carry most of the decision burden. The baseline was intentionally credible but limited; it was not designed as a strawman that performs no validation.

The harness executes the same candidate in a richer evidence context. It validates the manifest schema and source digest, confirms the SBOM, inspects imports and file access, checks data rights metadata, executes fixed and generated vectors, validates output schema, compares observable semantics, tests determinism and latency, calculates dimension scores, records findings, and applies high and critical severity gates. The methods therefore differ in evidence depth, not in access to the candidate ground truth during execution.

Table 11. Controlled evaluation conditions

Design element	Recorded condition
Candidates	20 unique packages: 2 conformant controls and 18 single fault candidates.
Paired exposure	Every package evaluated once by each method for primary inference.
Repeated execution	20 repetitions per package and method for timing and reliability only.
Test data	8 fixed vectors; harness adds 48 seeded generated vectors per evaluation.
Environment	Python 3.13.5 on Linux 4.4.0 x86_64 with glibc2.41.
Random seed	20260623
Harness decision	Composite score at least 80 and no high or critical severity gate failure.
Baseline decision	No finding from manifest presence, import, happy path execution, or basic output checks.

6.7 Data Collection and Reproducibility

The experiment runner writes one primary record per candidate method pair, repeated run timing records, per candidate JSON evidence packages, experiment metadata, statistical summaries, figures, and a machine readable method table. Canonicalized harness evidence is hashed with SHA 256. The archive also includes source, schemas, fixed vectors, generated vector logic, pinned dependencies, a data dictionary, protocol, automated tests, and a checksum manifest.

Three executable tests verify control passage, fault rejection, and evidence package stability requirements.

6.8 Statistical Analysis

The primary outcome is fault detection sensitivity. Wilson score intervals are used because the number of fault classes is small and extreme observed proportions make simple normal intervals unreliable. Control acceptance is reported with the same interval but interpreted cautiously because only two controls were available. Paired correctness is assessed with an exact McNemar test across the twenty unique packages. Candidate level median runtimes are compared with a paired Wilcoxon signed rank test; distributional dominance is described using Cliff's delta. Ten thousand random dimension weight vectors drawn from a symmetric Dirichlet distribution evaluate how scoring choices affect passage with and without critical gates (McNemar, 1947; Wilcoxon, 1945; Cliff, 1993; Lakens, 2013).

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN}) \tag{1}$$

Equation 1. Fault detection sensitivity.

TP is a faulted candidate correctly rejected; FN is a faulted candidate incorrectly accepted.

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) \tag{2}$$

Equation 2. Control acceptance specificity.

TN is a conformant control correctly accepted; FP is a conformant control incorrectly rejected.

$$\text{Diagnostic accuracy} = \text{Dcorrect} / \text{Nfault} \tag{3}$$

Equation 3. Actionable root cause diagnostic accuracy.

Dcorrect is the number of faulted candidates assigned the injected root cause class; Nfault = 18 in this experiment.

6.9 Validity and Ethical Boundaries

Internal validity is strengthened by a paired design, explicit ground truth, identical candidate packages, fixed acceptance semantics, a recorded seed, and machine generated results. Construct validity is bounded by the selected fault model: the study measures whether the methods expose these eighteen nonconformances, not every form of modularity failure. External validity is deliberately limited to a synthetic software module. The study contains no DoD program data and does not estimate field labor, certification effort, cost avoidance, schedule reduction, or mission

effectiveness. Statistical precision is limited by eighteen fault classes and two controls, and runtime reflects a shared container environment rather than production infrastructure. These limits are carried into Chapter 7 and the final conclusions.

7. Controlled Evaluation Results

7.1 Primary Detection and Control Acceptance Results

The baseline detected 5 of 18 injected nonconformances, yielding 27.8% sensitivity with a Wilson 95 percent CI of 12.5% to 50.9%. The harness detected 18 of 18, yielding 100.0% sensitivity with a 95 percent CI of 82.4% to 100.0%. Both methods accepted both controls. Because the control denominator is two, the 100 percent point estimate has a wide Wilson interval of 34.2% to 100.0% and should not be generalized as a mature false positive estimate.

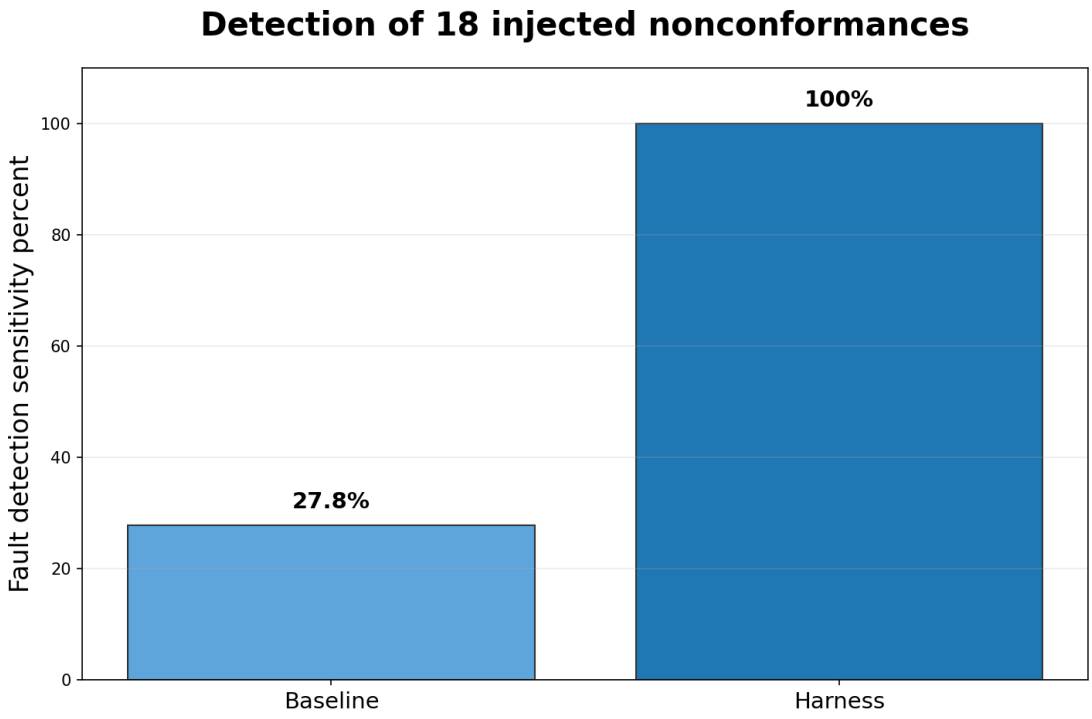


Figure 5. Fault detection sensitivity with Wilson 95 percent confidence intervals for the baseline and conformance harness.

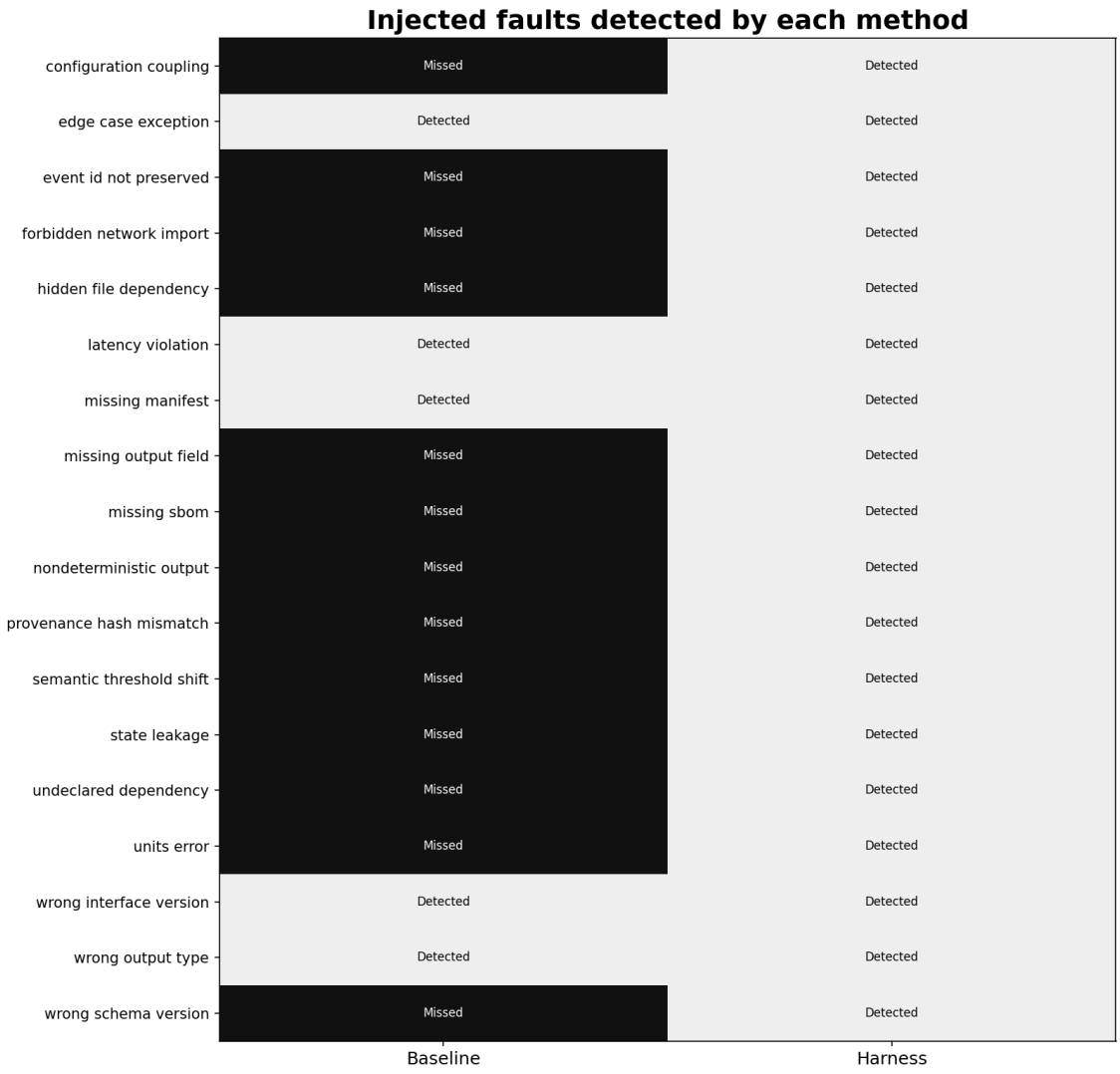


Figure 6. Detection matrix for the eighteen injected nonconformance classes.

Table 12. Primary method results

Metric	Baseline	Harness	Interpretation
Faults detected	5/18	18/18	Harness exposed all injected fault classes.
Sensitivity	27.8%	100.0%	Difference is large, but intervals reflect the 18 fault denominator.
Wilson 95% CI	12.5% 50.9%	82.4% 100.0%	Exact boundaries are not claimed beyond the fault model.
Controls accepted	2/2	2/2	No false rejection among two controls.
Correct diagnosis	5/18	18/18	Harness findings were actionable for every injected fault.
Control evidence completeness	0.50	1.00	Harness produced the full expected evidence set for controls.
Median checks/tests	14	177	Improved observability required a deeper assessment.

7.2 Paired Correctness Test

Across the twenty candidate packages, both methods were correct for seven packages, the harness alone was correct for thirteen, the baseline alone was correct for none, and both were wrong for none. The exact two sided McNemar p value was 0.000244. Within this constructed candidate set, the result rejects equality of paired correctness and supports H1. It does not estimate the effect size expected for an operational DoD portfolio, because the candidate distribution was designed for mechanism coverage rather than population sampling.

Table 13. Paired and reliability statistics

Analysis	Result	Interpretation
McNemar discordant pairs	Harness only = 13; baseline only = 0	All disagreements favored the harness.
Exact McNemar p value	0.000244	Evidence against equal paired correctness in this candidate set.
Paired runtime Wilcoxon p value	9.537e 07	Harness runtime was consistently greater.
Cliff's delta on candidate medians	0.905	Large distributional dominance toward longer harness runtime.
Decision stability	Baseline 100.0%; harness 100.0%	No accept/reject changes across twenty repeated executions.
Harness evidence digest stability	Controls 100.0%; all candidates 90.0%	The nondeterministic fault appropriately caused evidence variation.

7.3 Diagnostic and Evidence Quality

The baseline generated the injected root cause diagnosis for 5 of 18 faulted candidates; the harness generated the correct diagnosis for all 18. The difference matters because a rejection without mechanism level evidence can still impose significant engineering burden. The harness evidence packages record criterion codes, severity, category, message, dimension scores, evidence confidence, gate status, and a digest. For the two controls, mean evidence completeness increased from 0.50 to 1.00, supporting H2.

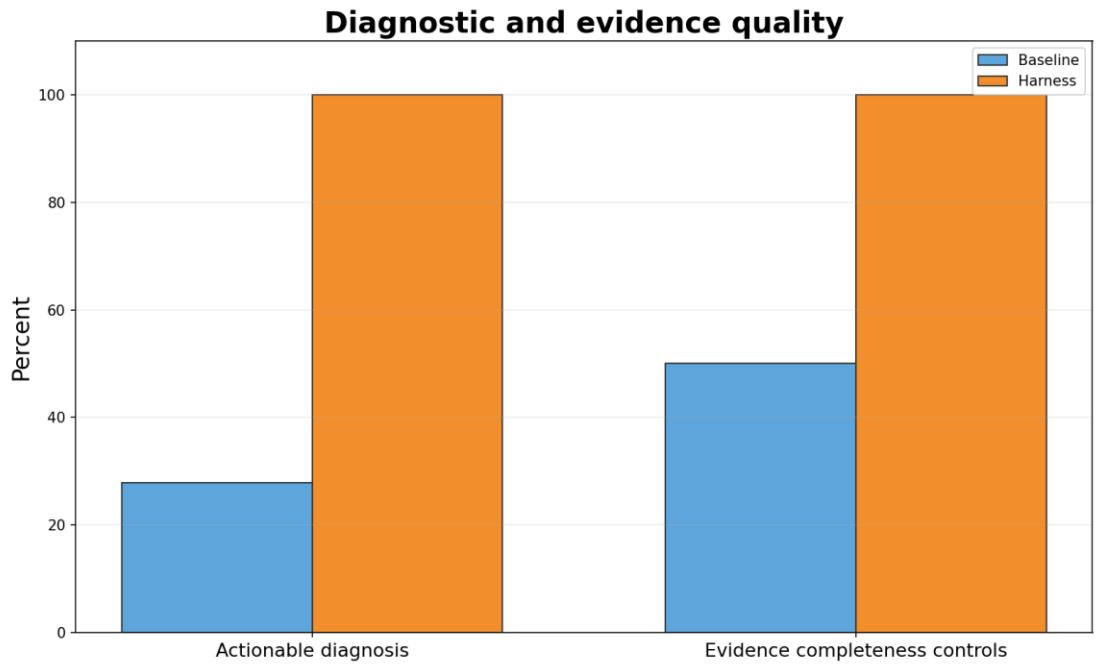


Figure 7. Actionable diagnosis accuracy and evidence completeness for conformant controls.

7.4 Computational Overhead

The stronger method was not free. Median primary runtime increased from 0.348 ms to 7.042 ms, a ratio of approximately 20.2 to 1. Across candidate level medians from repeated runs, the median paired increase was 5.554 ms and the median ratio was 20.0 to 1. The harness also executed a median of 177 checks or tests rather than 14. The paired Wilcoxon test confirmed greater runtime ($p = 9.537e\ 07$). The magnitude should be interpreted in context: the repeated run median harness runtime was 5.863 ms and the primary median was approximately 7.0 ms, while the deliberately slow latency mutant created a long tail case. H3 is supported as a computation for evidence tradeoff, not as evidence of burdensome human integration cost.

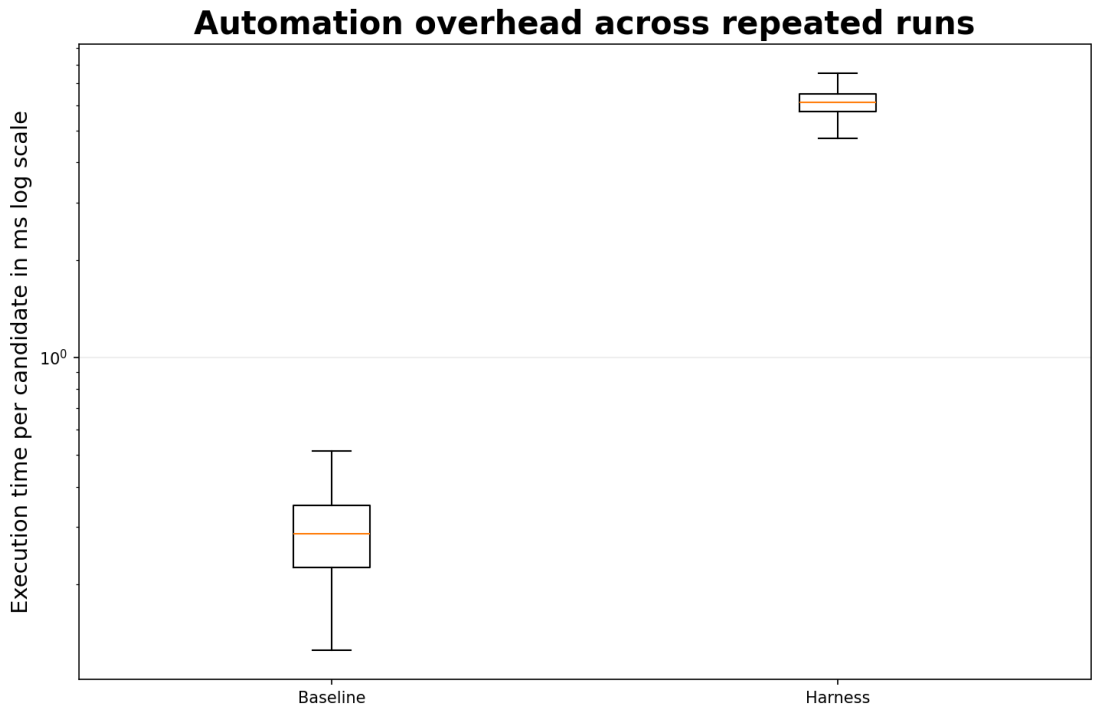


Figure 8. Repeated run runtime distributions on a logarithmic scale; outliers are suppressed in the boxplot but retained in the statistical data.

7.5 Conformance Scores and Non Compensable Gates

Composite scores were useful for ranking evidence and identifying affected dimensions, but a threshold alone was unsafe. Several faulted packages retained scores above or near the nominal 80 point threshold because correct behavior in unaffected dimensions compensated for a decisive defect. A missing SBOM, undeclared dependency, provenance mismatch, semantic threshold shift, edge case failure, or latency violation could therefore pass under some weight configurations if the decision relied only on arithmetic aggregation.

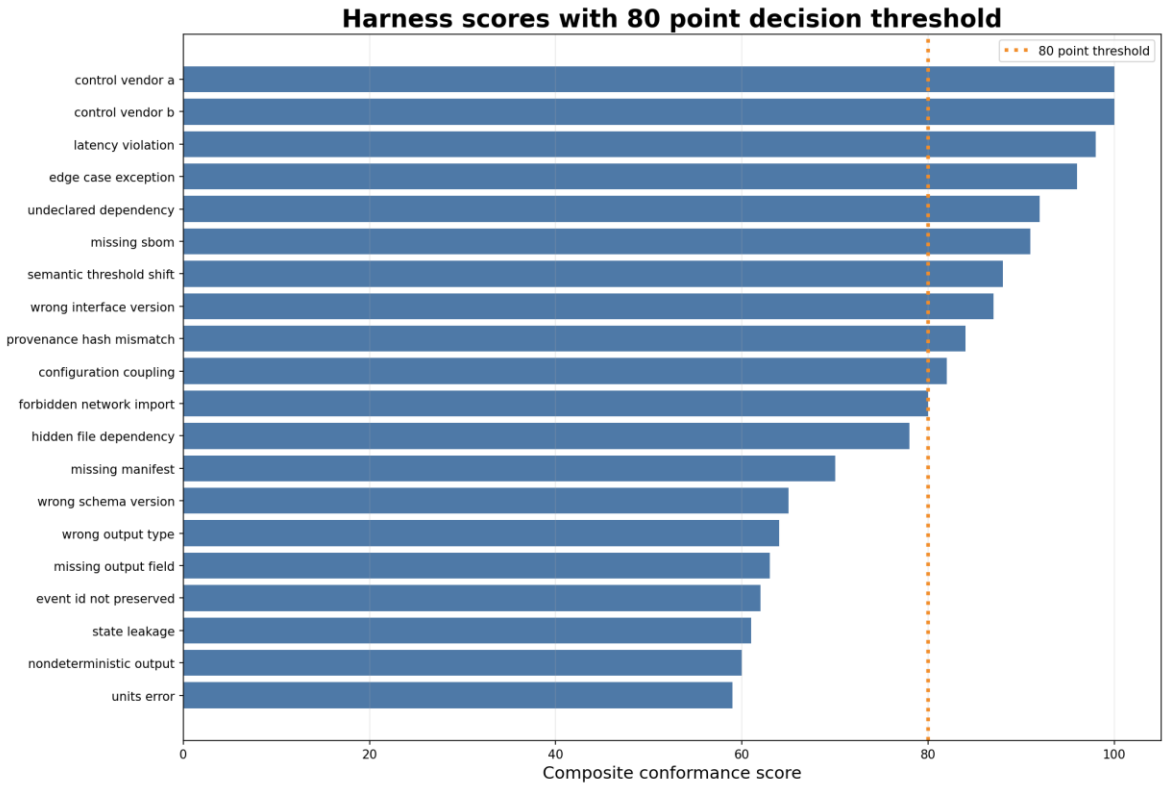


Figure 9. Harness composite scores by candidate with the nominal 80 point score threshold.

The weight sensitivity analysis drew 10,000 random four dimension weight sets. The maximum score only pass probability among faulted candidates was 100.0%. After high and critical severity gates were applied, the maximum pass probability among faulted candidates was 0.0%, while the minimum pass probability among controls remained 100.0%. This result supports H4 and justifies a governance rule: scores summarize and prioritize, but non compensable defects determine eligibility.

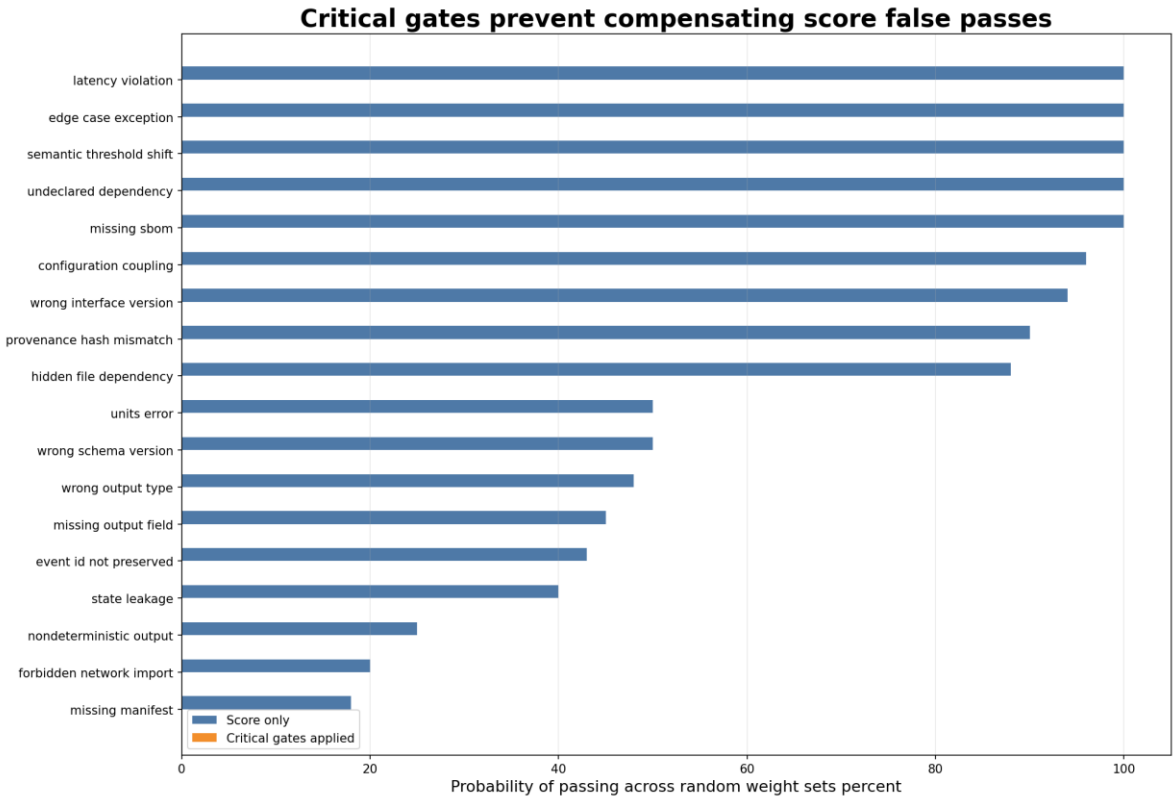


Figure 10. Pass probability for faulted candidates across 10,000 random dimension weight sets, with and without critical gates.

Table 14. Weight sensitivity and decision gate results

Measure	Observed result	Design implication
Random weight sets	10,000 symmetric Dirichlet draws	Tests sensitivity without privileging one dimension.
Maximum fault pass probability, score only	100.0%	A weighted average can compensate for a decisive defect.
Maximum fault pass probability, gates applied	0.0%	All injected high/critical defects remained ineligible.
Minimum control pass probability, gates applied	100.0%	Controls remained eligible across all draws.

7.6 Reliability and Reproducibility

Accept/reject decisions were stable across all repeated executions for both methods. Harness evidence digests were stable for both conformant controls and for 90 percent of all candidates. The two unstable candidates included the deliberately nondeterministic behavior and conditions whose findings contain variable execution evidence, which is an expected signal rather than an unexplained reproducibility failure. The complete archive allows the reader to distinguish deterministic decision stability from byte for byte evidence stability.

7.7 Findings by Research Question

Table 15. Research question findings

Question	Finding	Strength and boundary
RQ1	Public cases report selected cost, schedule, integration, refresh, and supplier participation benefits.	Moderate case level support; heterogeneous baselines and limited lifecycle analysis.
RQ2	Evidence quality varies substantially and often depends on program reporting or estimated counterfactuals.	Strong support for the measurement gap; weak support for universal effect magnitude.
RQ3	Stable interfaces, semantics, data rights, test assets, governance, funding, and market depth recur as enabling conditions.	Cross case propositions, not causal population estimates.
RQ4	The implemented harness detected and diagnosed all 18 injected faults; the baseline detected and diagnosed five.	Strong internal evidence for the supplied synthetic fault model.
RQ5	The harness increased runtime and test volume but improved evidence completeness, repeatability, and decision robustness.	Observed computational tradeoff; human and field costs remain unmeasured.

7.8 Interpretation and Boundary of Claim

The controlled results close the most important gap in the earlier thesis version: the artifact is no longer only proposed. The implementation demonstrates that machine readable contracts, provenance, static analysis, generated semantic tests, evidence packaging, and critical gates can expose failure mechanisms that a plausible happy path process misses. The results do not show that every DoD system should use the exact prototype, that the observed percentages will replicate in a field population, or that automation alone creates economic openness. They show that the measurement logic is executable and produces materially different evidence under controlled conditions. The appropriate next step is external validation on real, releasable module change events with independent integrators and observed labor, schedule, certification, and sustainment outcomes.

8. Conformance Harness Concept and Value Proposition

8.1 Definition

A **MOSA DoD Conformance Harness** is a federated ecosystem of criteria, tools, test artifacts, scoring methods, governance workflows, and reporting interfaces that together assess the quality and fidelity of MOSA implementation across program lifecycles. The harness is both technical and organizational: it codifies what "good" looks like in architectural and interface terms, and it embeds those expectations into repeatable processes that shape decisions by program offices, testers, and acquisition authorities. By design the harness treats conformance as an outcome to be demonstrated with evidence rather than a checklist to be asserted in prose.

The harness is intentionally federated to reflect DoD's distributed acquisition and engineering environment. Local programs retain responsibility for producing evidence and remediate findings, while enterprise services provide shared criteria, ingestion pipelines, and assessment tooling. This separation of responsibilities preserves program agility while ensuring that conformance evidence is comparable, discoverable, and auditable across the enterprise.

8.2 Why a Harness, Not Just Guidance

Guidance documents communicate expectations and recommended practices; a conformance harness operationalizes those expectations into executable, repeatable mechanisms. Where guidance leaves room for interpretation, the harness reduces variance by embedding criteria into tools, workflows, and standardized artifacts so that assessments are consistent across reviewers, programs, and acquisition pathways. This shift from narrative guidance to instrumented practice makes assessments repeatable rather than ad hoc, enables meaningful comparison across programs and vendors, and establishes auditable evidence chains that survive personnel turnover and organizational change.

Operationalizing MOSA through a harness also changes incentives. When conformance criteria are machinereadable, testable, and contractually referenced, vendors and program offices can plan to meet measurable outcomes rather than optimize for minimal paperwork. The harness therefore strengthens institutional memory and reduces reliance on individual reviewer judgment, turning MOSA conformance into a structured engineering discipline rather than a subjective policy exercise.

8.3 Primary Outcomes

The harness is designed to produce five primary outcomes that together close the gap between policy intent and fielded capability. First, it yields **clear and testable conformance expectations** so programs and vendors understand what is required, how it will be measured, and what evidence is sufficient. Second, it enables **faster identification of architecture and integration risks** by surfacing systemic issues such as hidden coupling, brittle interfaces, or weak portability earlier and more consistently. Third, it improves the **quality of milestone and contracting decisions** by providing decisionmakers with structured, traceable evidence about MOSA implementation.

Fourth, the harness enhances **lifecycle competition viability** by making openness and replaceability visible and measurable, thereby sustaining supplier diversity during sustainment and modernization. Fifth, it strengthens **mission confidence in adaptation readiness** by enabling stakeholders to assess whether systems can integrate new capabilities, partners, and threat responses at operational tempo. Collectively these outcomes reduce integration cycle time, lower sustainment risk, and increase the defensibility of acquisition choices.

8.4 Stakeholder Benefits

Stakeholder communities derive distinct, mutually reinforcing benefits from a MOSA conformance harness. **Program managers** gain clearer decision support, prioritized remediation recommendations, and a defensible basis for trade offs among cost, schedule, and openness. **Systems engineers** receive predictable quality gates, reusable test and validation assets, and clearer feedback loops linking architecture choices to conformance outcomes. **Contracting officers** obtain objective evidence for evaluating claims about openness, modularity, and replaceability and for structuring incentives and remedies in contract language.

Independent assessors benefit from a common framework, shared criteria, and standardized evidence formats that improve consistency and transparency in adjudication. Industry partners gain better visibility into conformance expectations, onboarding pathways, and reuse opportunities, reducing ambiguity in proposal development and implementation planning. These aligned benefits reduce friction across acquisition, engineering, test, and sustainment communities and make MOSA outcomes more durable.

8.5 Prototype Repository and Stakeholder Portal

The prototype repository is now a completed companion research artifact rather than a future concept. It contains the executable harness, baseline comparator, contracts, synthetic candidates, fixed vectors, analysis code, raw observations, generated figures, tests, experiment protocol, data dictionary, license, citation metadata, and checksum manifest. The package is deliberately independent of informal workspace files so that a committee member or external reviewer can reproduce the controlled results from the submitted archive.

Public transparency should be separated from protected program evidence. Releasable criteria and examples can accelerate industry participation, while controlled repositories retain proprietary, classified, export controlled, or security sensitive artifacts. Versioned links between criteria, test suites, evidence schemas, and governance decisions allow reviewers to reconstruct how a judgment was reached and allow programs to demonstrate improvement over time.

9. Architecture and Design of the Conformance Ecosystem

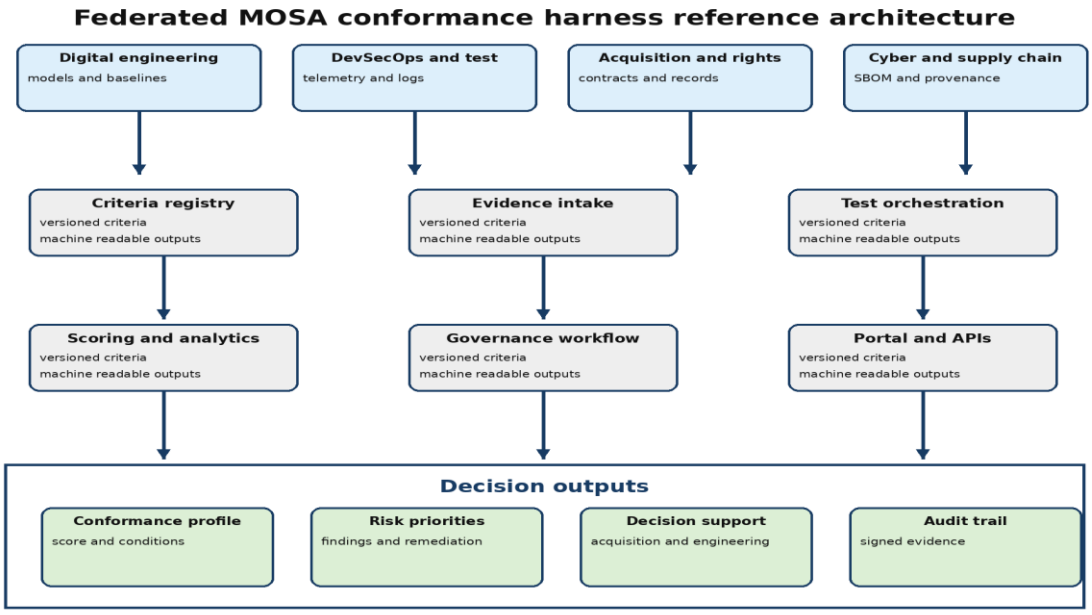


Figure 11. Reference architecture for the federated MOSA Conformance Harness.

Note. Author developed reference architecture. Local tools may vary; shared services standardize criteria, evidence, and decision interfaces.

9.1 Reference Architecture

The reference architecture for a conformance ecosystem should be implemented as a modular set of services that cleanly separates (1) the authoritative specification of conformance criteria, (2) evidence ingestion and normalization, (3) test orchestration, (4) scoring/analytics and reporting, and (5) governance decision workflows, rather than as a monolithic "compliance platform." This separation is consistent with DoD engineering policy that mandates a modular open systems approach (MOSA) to the maximum extent practicable, including severable components for incremental addition/replacement and explicit interface definition supported by the acquisition of appropriate technical data rights (DoD, 2020c). It is also consistent with DoD digital engineering policy that makes digital models and their underlying data central to engineering communication and life cycle decision making, establishing and maintaining technically accurate models as an authoritative source of truth for stakeholders (DoD, 2023). As an enabling context, a defense.gov hosted advisory assessment stresses that an interoperable digital ecosystem requires common standards and requirements, shared accurate real time data, and a

foundational architecture supporting tooling, communication, and interoperability across the weapon system lifecycle (Defense Business Board, 2024). Within this frame, "conformance ecosystem" services should be designed to be independently deployable and evolvable, with explicit interface contracts and data interchange standards, so that conformance logic can change (e.g., new standards, new thresholds, new evidence types) without requiring wholesale replacement of tooling an outcome that is directly aligned with MOSA's intent to support upgrades and evolution via modularity (DoD, 2020c).

9.2 Data Model and Evidence Schema

The data model should treat conformance as an evidence backed claim that can be reproduced from recorded digital artifacts, rather than as an opaque score or manual certification. NIST emphasizes that conformance testing is only possible when the requirements/criteria for conformance are specified (typically in a conformance clause) and can be decomposed into testable assertions; without specified criteria, "there can be no conformance testing" (NIST, 2012). Accordingly, the evidence schema should (at minimum) bind each conformance claim to: the criterion/assertion identifier and version, the system boundary and scope of applicability (what component/interface/configuration is covered), the lifecycle stage and technical milestone context, the method of evaluation (automated test, inspection, analysis), and the provenance and configuration controlled identity of each evidence artifact. This is structurally consistent with DoD digital engineering requirements that digital models and twins are configuration controlled and traceable from operational capabilities through requirements, design constructs, production, test, training, and sustainment, and that baselines be verified and validated before technical milestones (DoD, 2023). For interoperability and automation, the evidence schema should be compatible with structured assessment representations; NIST's OSCAL 1.0.0 release illustrates a practical approach by defining stable models for catalogs/profiles, component definitions, system security plans, assessment plans, assessment results, and plans of action/milestones explicitly aimed at automating assessment, auditing, and continuous monitoring and reducing non interoperable proprietary solutions (NIST, 2021b). Finally, because supply chain and software transparency increasingly influences system assurance, the evidence schema should anticipate machine readable attestations and component inventories; NIST's EO 14028 supply chain guidance describes SBOMs as formal records of component details and relationships and specifically links the value of SBOMs to transparency, provenance, and faster vulnerability identification/remediation, while

warning that acquirers must be able to ingest/analyze SBOM data for it to improve posture (NIST, 2024).

9.3 Conformance Levels

Conformance levels should be defined as nested, testable subsets of requirements that support incremental adoption and risk based decision making, rather than as aspirational maturity labels without verification anchors. NIST's conformance testing guidance explicitly describes "levels" as nested subsets where level 1 is a basic core that must be implemented by all products and higher levels include the prior level plus additional functionality; it also explains how profiles and level's structure conformance claims and testing scope (NIST, 2012). In an engineering context, a defensible level design should therefore (a) define the minimal, mandatory interface/behavior requirements and mandatory evidence artifacts, (b) specify the verification method(s) and timing (e.g., before milestone decision points), and (c) articulate what a passing result means and does not mean. This caveat matters because NIST notes that conformance testing provides assurance but not proof: passing all tests is not a guarantee of conformance, because exhaustive testing is infeasible and untested behaviors may still violate the specification (NIST, 2024). DoD acquisition oversight findings reinforce the need to link "levels" to explicit verification timing: GAO reports that programs do not consistently plan to verify MOSA implementation before key points such as production, and that neither systems engineering policy nor MOSA guidance (as observed by GAO) clearly addresses when verification testing should be completed or documented (GAO, 2023). Therefore, a rigorous conformance level framework should explicitly define which conformance level must be achieved (and evidenced) at which lifecycle gate to prevent "paper MOSA" claims and to support incremental upgrades, interoperability, and competitive insertion the intended benefits of MOSA (DoD, 2026; GAO, 2025).

Table 16. Conformance levels and exit criteria.

Level	Required capability	Illustrative exit criteria
Level 1, Foundational	Boundaries and major interfaces identified; minimum technical data and rights plan; criteria profile approved	Complete required artifacts; no unresolved critical documentation or rights blockers
Level 2, Operational	Interfaces validated; representative interoperability, substitution, rollback, and security tests executed	Required mission thread tests pass; high severity findings mitigated or time bounded by approved waiver

Level	Required capability	Illustrative exit criteria
Level 3, Advanced	Continuous pipeline verification; demonstrated alternate supplier onboarding; portfolio reuse and outcome measurement	Sustained performance across releases; evidence confidence and economic openness thresholds achieved

9.4 Tailoring Framework

A tailoring framework should enable mission and domain specific adjustments to conformance criteria while preserving comparability, auditability, and decision authority. DoD engineering policy explicitly allows tailoring: Systems Engineering Plan (SEP) content can be tailored with approval by the SEP approval authority, and the issuance itself can be tailored as necessary for each acquisition pathway (DoD, 2026). In conformance testing terms, tailoring is best modeled as an explicit, versioned "profile" that selects and constrains requirements from a base catalog: NIST describes profiles as subsets of specifications needed for a particular user community and emphasizes that the conformance clause can specify sets of functions as profiles and/or levels, along with how extensions and options are handled (NIST, 2012). A thesis grade tailoring design should therefore (1) establish a non negotiable core baseline (mandatory criteria) that preserves enterprise interoperability and audit comparability, (2) define domain profiles that add or refine criteria (e.g., platform class, mission environment), and (3) define controlled waiver mechanisms that record rationale, affected scope, compensating measures, and an expiry/review trigger. Critically, tailoring decisions should be "evidence bearing" i.e., the system should require the same type of traceable artifact and decision record that DoD digital engineering expects for technical baselines and life cycle decision support (DoD, 2023). Where tailoring modifies or relaxes MOSA centered interface expectations, the framework should require explicit justification and governance because DoD policy mandates consensus based standards for interfaces unless unavailable/unsuitable and requires justification if such standards are not used (DoD, 2026).

9.5 Interoperability of the Harness Itself

The conformance harness itself must be interoperable, or it risks becoming a new enterprise silo that undermines MOSA and digital engineering goals. DoD engineering policy requires that MOSA implementation includes clearly defined major system interfaces and the use of consensus based standards for interfaces (unless unsuitable), along with open sharing of definitions to interdependent programs (DoD, 2020a). Complementing this, DoD digital engineering policy requires maximizing interoperability by using standard and best practice model representations,

methods, and underlying data structures, and it positions digital models as the primary means of communicating system information (DoD, 2023). Translating these requirements into harness design implies: (a) the harness must expose stable APIs and event interfaces to integrate with program toolchains (model repositories, test infrastructure, CI/CD, vulnerability scanning) and (b) it must produce standardized, machine readable outputs that can be consumed by multiple governance and reporting systems, avoiding vendor lock in. NIST's OSCAL release provides a concrete interoperability pattern: by publishing stable models for catalogs/profiles and assessment results, OSCAL is explicitly intended to automate assessments and reduce proprietary, non interoperable processes (NIST, 2021b). Strategically, the defense.gov hosted digital ecosystem assessment argues that an interoperable digital ecosystem requires common standards and requirements and real time data exchange across platforms and partners, with architecture strong enough to support tooling and interoperability throughout lifecycles (Defense Business Board, 2024). Therefore, interoperability of the harness is not a "nice to have"; it is an architectural obligation derived from MOSA and digital engineering policy objectives, and it should be treated as a first order conformance requirement of the ecosystem itself (DoD, 2020a; DoD, 2023).

9.6 Reliability and Operational Considerations

Reliability and operational considerations should treat the conformance ecosystem as a mission relevant decision system: if conformance results inform milestone decisions, upgrade eligibility, and system integration readiness, then availability, integrity, and auditability become engineering requirements rather than operational afterthoughts. At the policy level, DoD digital engineering calls for technically accurate digital models supported by configuration management and for infrastructures that may need secure networks supporting multilevel classification environments, implying the harness must be engineered for secure, reliable operation under realistic constraints (DoD, 2023). From a risk governance standpoint, NIST's Risk Management Framework provides a disciplined, life cycle process for managing security and privacy risk that includes control selection, implementation, assessment, authorization, and continuous monitoring, explicitly aiming to support near real time risk management and decision making (NIST, 2018). NIST's SP 800 to 53 control catalog complements this by framing security and privacy controls as flexible and customizable to mission needs and emphasizing assurance the "measure of confidence" in capabilities that protects organizational operations and assets from diverse threats and failures (NIST, 2020). Practically, this implies that operational requirements for the harness

should include (at minimum) immutable audit logging of evidence and decision changes, role based access/segregation of duties for criteria modification and waiver approval, backups and disaster recovery, and continuous monitoring of the harness itself consistent with NIST's continuous monitoring guidance that stresses visibility into assets and the effectiveness of deployed controls to provide ongoing assurance aligned with risk tolerance (NIST, 2011). Finally, because evidence artifacts such as SBOMs only improve posture when acquirers can ingest and act on them, operational design should include data pipelines, validation gates, and analytics capacity sufficient to avoid "collecting evidence without decision utility" (NIST, 2024). Where specific numeric reliability targets (e.g., availability SLOs, recovery time objectives) are not defined, they are unspecified and should be derived from mission impact analysis and acquisition decision criticality

10. Conformance Harness Concept of Operations

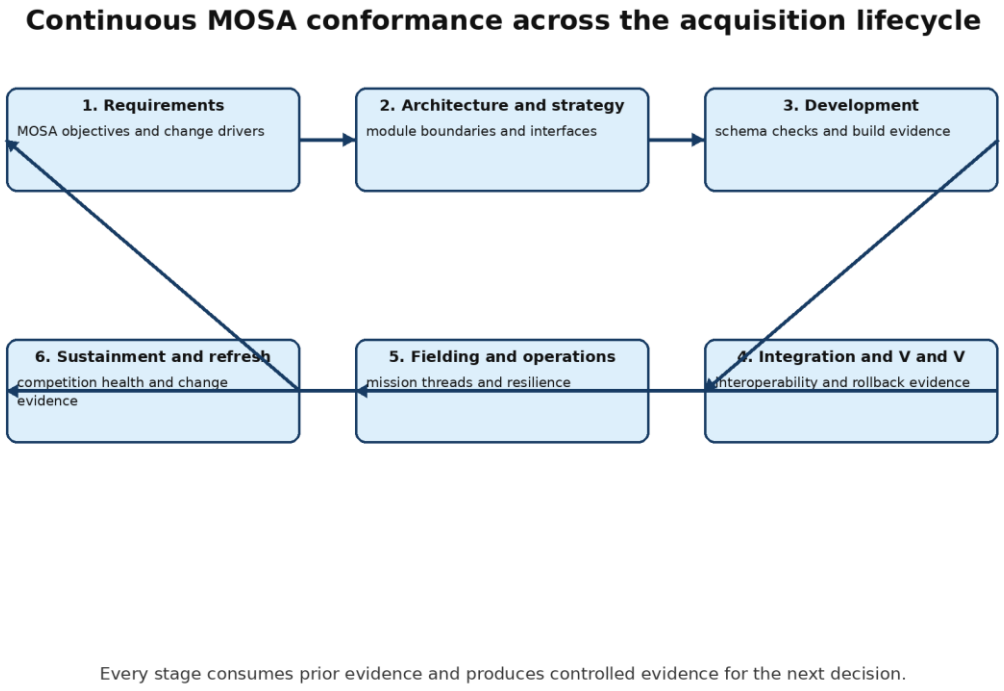


Figure 12. Lifecycle touchpoints for continuous MOSA conformance.

Note. Author developed lifecycle model showing continuous evidence generation from requirements through sustainment.

10.1 Lifecycle Touchpoints

A MOSA Conformance Harness must integrate with the full defense acquisition lifecycle rather than operate as a standalone assessment event. Conformance expectations begin during early requirements and architecture formulation, where programs define module boundaries, interface contracts, datarights strategies, and initial openness objectives. At this stage, the harness provides criteria and exemplars that guide architectural decomposition and ensure that modularity and openness are treated as design drivers rather than late stage compliance artifacts. Early touchpoints also include preliminary interface definitions, initial SBOM generation, and early substitutionrisk assessments, all of which establish the baseline against which future conformance will be measured.

As programs progress into development, integration, and test phases, the harness becomes more operational. Evidence pipelines ingest machinereadable interface artifacts, CI/CD generated

conformance telemetry, and integration test results. These touchpoints allow the harness to detect architectural drift, hidden coupling, or portability regressions as they occur rather than at milestone reviews. During operational test and fielding, the harness evaluates missionthread interoperability, substitution performance, and security posture under realistic conditions. Finally, in sustainment, the harness supports continuous verification by monitoring interface evolution, supplychain changes, vulnerability response performance, and the viability of competitive alternatives. In this way, the harness becomes a persistent lifecycle companion rather than a onetime audit mechanism.

10.2 Assessment Workflow

The assessment workflow begins with **criteria selection and tailoring**, where programs identify the relevant conformance dimensions structural, operational, and economic openness and apply riskweighted adjustments based on mission context, acquisition pathway, and system criticality. Once criteria are selected, the harness orchestrates evidence ingestion from engineering repositories, DevSecOps pipelines, test environments, and acquisition documentation. Evidence is normalized into machinereadable formats and linked to specific criteria through traceability metadata, ensuring that reviewers can understand not only what evidence exists but how it supports conformance claims.

Next, the harness executes automated and manual assessments. Automated checks validate interface schemas, semantic rules, SBOM completeness, dependency boundaries, and portability indicators. Manual assessments focus on architectural rationale, datarights sufficiency, and missionthread interoperability. The workflow culminates in a structured scoring and analytics phase, where findings are aggregated, weighted, and contextualized. The output is a conformance profile that includes scores, risk indicators, recommended remediation actions, and evidence provenance. This profile is then routed through governance workflows for adjudication, milestone preparation, or contract action, ensuring that conformance results directly inform program decisions.

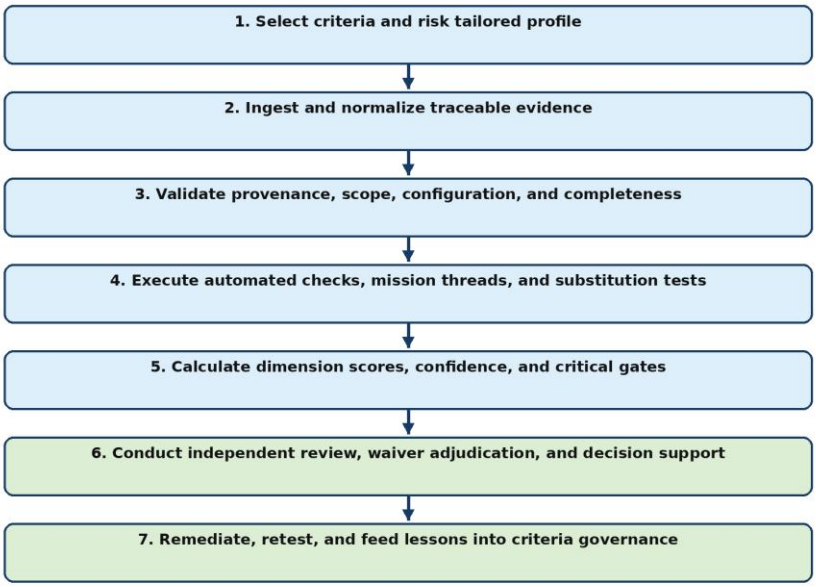


Figure 13. End to end assessment and adjudication workflow.

Note. Author developed workflow. Each result remains linked to its criterion, configuration, evidence, reviewer, finding, and disposition.

10.3 Roles and Authorities

Effective MOSA conformance requires clear delineation of roles and authorities across the acquisition ecosystem. Program managers serve as the primary stewards of conformance, responsible for ensuring that architectural decisions, contract structures, and engineering practices align with MOSA objectives. They own tailoring decisions, remediation planning, and integration of conformance results into milestone documentation. Systems engineers and software architects provide the technical foundation for conformance by producing interface artifacts, modularity rationales, SBOMs, and integration evidence. Their work forms the core of the evidence pipeline.

Independent assessors, including engineering review boards, test organizations, and enterprise MOSA authorities, provide objective evaluation and adjudication of findings. Their authority ensures that conformance judgments are consistent across programs and not influenced by local incentives. **Contracting officers** play a critical role by translating conformance expectations into enforceable contract language, incentives, and remedies. Finally, **enterprise governance bodies**such as servicelevel MOSA offices or crossprogram architecture boards maintain the criteria catalog, manage versioning, and ensure alignment with DoDwide policy.

Together, these roles create a balanced governance structure that supports both program autonomy and enterprise consistency.

10.4 Findings and Severity Model

A structured findings and severity model is essential for translating conformance assessments into actionable program decisions. Findings are categorized based on the dimension of openness they affect structural, operational, or economic and are assigned severity levels that reflect their potential impact on interoperability, mission readiness, or lifecycle affordability. Highseverity findings typically indicate architectural defects such as hidden coupling, proprietary interface dependencies, or missing data rights that could prevent module replacement or integration with external systems. These findings require immediate remediation and may block milestone approval.

Mediumseverity findings often relate to incomplete evidence, partial semantic specification, or insufficient test coverage. While not immediately missionthreatening, they represent risks that could escalate if left unaddressed. Lowseverity findings capture minor documentation gaps, noncritical deviations from best practices, or areas where evidence is adequate but could be strengthened. The severity model also incorporates temporal factors such as whether a finding is trending toward improvement or degradation and links each finding to recommended remediation actions. This structured approach ensures that conformance results are not merely descriptive but drive targeted, riskinformed program behavior.

10.5 Waiver Control

Waiver control is a critical governance mechanism that ensures deviations from MOSA criteria are justified, documented, and bounded. The harness establishes a formal waiver process that requires programs to articulate the operational, technical, or economic rationale for non conformance, supported by evidence and risk analysis. Waivers are reviewed by independent assessors and approved by authorities commensurate with the severity and scope of the deviation. This prevents programs from bypassing MOSA expectations without accountability and ensures that waivers are granted only when they are necessary and defensible.

Waivers are time bounded and subject to periodic review. The harness tracks waiver expiration dates, required mitigation actions, and dependencies on future system changes. This

prevents waivers from becoming permanent exceptions and encourages programs to revisit architectural or contractual decisions as technology and mission needs evolve. By embedding waiver control into the conformance workflow, the harness balances flexibility with discipline, allowing programs to tailor MOSA implementation without undermining enterprise level objectives.

10.6 Continuous Improvement Loop

The continuous improvement loop ensures that the conformance harness evolves alongside technology, policy, and operational needs. Each assessment cycle generates data on common failure modes, recurring architectural issues, and systemic barriers to modularity or interoperability. This data feeds into analytics that identify trends across programs, enabling enterprise governance bodies to refine criteria, update guidance, and adjust scoring models. Over time, the harness becomes more predictive, highlighting emerging risks before they manifest in fielded systems.

Continuous improvement also applies to tooling and workflows. As DevSecOps pipelines mature, new automated checks can be integrated into the evidence ingestion process. As interface standards evolve, the criteria catalog can incorporate updated schemas and semantic rules. Feedback from programs, vendors, and assessors informs usability enhancements, clearer documentation, and more effective remediation pathways. Ultimately, the continuous improvement loop ensures that the harness remains relevant, scalable, and aligned with DoD's modernization priorities, transforming MOSA from a static policy requirement into a dynamic, learningdriven enterprise capability.

11. Implementation Approach and Delivery Plan

11.1 Phased Implementation Strategy

A phased implementation strategy is necessary to balance the urgency of improving MOSA execution with the institutional stability required for enterprise adoption. The four phase model Define, Pilot, Operationalize, and Scale aligns with DoD's emphasis on iterative acquisition and continuous modernization (DoD, 2020b). **Phase 0 (Define)** establishes foundational governance structures, creates the initial criteria baseline, and deploys a minimal publicfacing portal. This phase ensures that MOSA expectations are transparent and consistent with the principles outlined in the DoD MOSA Implementation Guidebook (DoD, 2025a).

Phase 1 (Pilot) introduces the harness into a controlled set of programs to validate assumptions, test workflows, and calibrate criteria under real engineering conditions. Pilot feedback informs refinements to evidence schemas, scoring logic, and governance processes, consistent with the Adaptive Acquisition Framework's emphasis on tailoring and learning (DoD, 2026). **Phase 2 (Operationalize)** expands automation, strengthens analytics, and formalizes integration with digital engineering repositories and DevSecOps pipelines. **Phase 3 (Scale)** federates governance across Components and embeds the harness into enterprise level oversight flows, aligning with GAO recommendations for improving MOSA planning and verification across programs (GAO, 2025).

11.2 Pilot Program Selection

Pilot program selection must be intentional and diverse to ensure that the harness generalizes across mission and technology contexts. A **softwarecentric program** is essential for validating integration with CI/CD pipelines, interface schemas, and software supplychain evidence, consistent with DoD's DevSecOps Strategy (DoD, 2021a). This ensures that the harness can operate at the tempo required for modern software delivery.

A **platformcentric program** such as an aircraft, ship, or groundintroduces challenges related to hardware software integration, long lifecycle timelines, and complex interface dependencies. Finally, a **joint or coalition integration context** is necessary to validate crosssystem interoperability and missionthread performance, reflecting DoD's emphasis on joint all domain operations and coalition interoperability (DoD, 2025a). This diversity ensures that the

harness can accommodate the full spectrum of MOSA implementation challenges, from data constraints to semantic interoperability.

11.3 Minimum Viable Harness

The Minimum Viable Harness (MVH) represents the smallest coherent set of capabilities that delivers tangible value while maintaining tight scope control. At MVH, the harness should include a **criteria registry** with versioned, accessible conformance expectations aligned with MOSA statutory requirements (10 U.S.C. § 4401). It should also provide **evidence submission mechanisms** aligned to a common schema, enabling programs to submit interface artifacts, SBOMs, test results, and modularity rationales in a consistent, machinereadable format.

Early automation should focus on **highleverage, lowrisk checks**, such as schema validation and interface conformance, consistent with digital engineering best practices (DoD, 2025a). A defined **governance board process** is necessary to review findings and adjudicate waivers. Standardized **scorecard reporting** supports milestone and contracting decisions, aligning with GAO's call for more structured MOSA evidence (GAO, 2025). During this phase, the existing public site can serve as the onboarding portal, hosting criteria documentation, process descriptions, and FAQs.

11.4 Automation Roadmap

Automation is a critical enabler of scalable, continuous MOSA conformance, but it must be introduced in deliberate stages. Early stage automation focuses on **schema validation, dependency checks, and traceability completeness**, which align with DevSecOps principles of automated quality gates (DoD, 2021a). These checks provide immediate benefits by reducing manual review burden and surfacing structural issues early in the lifecycle.

Later stage automation can introduce more advanced capabilities, including **semantic interoperability tests**, **synthetic missionthread scenarios**, and **crossprogram pattern analysis**. These capabilities align with DoD's push toward digital engineering ecosystems and modelbased integration (DoD, 2025a). Advanced anomaly detection can highlight deviations from expected conformance trajectories. This staged roadmap ensures that automation strengthens existing engineering practices and that each new capability is informed by pilot and operational experience.

11.5 Organizational Readiness Measures

Organizational readiness is as important as technical capability in ensuring successful adoption of the Conformance Harness. Effective implementation requires **training and education** that provide practical playbooks, worked examples, and templates, not just policy statements. This aligns with DoD's emphasis on workforce development in software intensive acquisition (DoD, 2020a). **Role clarity** is essential to avoid duplication of effort and gaps in accountability, particularly between engineering, contracting, and test communities.

Leadership support is critical: senior leaders must signal that MOSA conformance and the harness as its mechanism is a priority that influences milestone decisions, contract actions, and resource allocation. This reflects GAO's findings that leadership engagement is a key factor in successful MOSA implementation (GAO, 2025). **Assessor calibration** ensures consistency across evaluations, reducing variance in conformance judgments and strengthening institutional trust in the harness.

11.6 Funding and Sustainment

A sustainable conformance ecosystem requires a deliberate funding and product management model. **Core platform capabilities**, including shared services, the criteria registry, the public portal, and foundational automation should be centrally funded to ensure consistency, interoperability, and longterm viability. Central funding also supports enterpriselevel governance and version control, consistent with DoD's approach to shared digital engineering infrastructure (DoD, 2025a).

Program specific assessment activities such as test execution, specialized tailoring, and remediation support can be funded proportionally by participating programs or portfolios. Sustained **product management** is essential to govern criteria evolution, maintain release quality, and provide user support. Without ongoing investment, the harness risks stagnation and divergence from evolving mission and technology needs. A wellfunded, continuously managed harness becomes a durable enterprise asset that improves MOSA execution across the DoD.

12. Governance Model and Decision Architecture

Table 17. Governance RACI matrix for principal harness decisions.

Decision/activity	Program manager	Lead engineer	Independent assessor	Contracting officer	Enterprise authority
Approve base criteria	C	C	C	C	A/R
Tailor program profile	A	R	C	C	C
Produce and maintain evidence	A	R	C	C	I
Execute independent assessment	C	C	A/R	I	I
Approve major waiver	R	C	C	C	A
Use results in contract action	A	C	C	R	I
Update enterprise criteria	C	C	R	C	A

Note. R = Responsible; A = Accountable; C = Consulted; I = Informed. Local governance may tailor roles but should not combine evidence production and final independent adjudication without compensating controls.

12.1 Governance Principles

The governance model for the MOSA Conformance Harness must be grounded in principles that promote clarity, transparency, and accountability. Because MOSA implementation spans engineering, contracting, cybersecurity, and operational communities, ambiguity in decision rights or assessment authority can lead to inconsistent interpretations and delayed decisions. Clear articulation of responsibilities and escalation paths aligns with DoD guidance emphasizing disciplined governance and traceability in acquisition processes (DoD, 2026). This clarity is essential for building trust in the harness and ensuring that conformance outcomes are defensible during milestone reviews and external oversight.

Governance must also remain proportionate to risk. Highimpact decisions such as modifications to core criteria or acceptance of major waivers require more rigorous review and documentation, consistent with GAO's findings that MOSA governance must be strengthened to ensure consistent outcomes across programs (GAO, 2025). Lower risk decisions should be streamlined to avoid unnecessary bureaucracy. The goal is to be a governance structure strong enough to enforce standards yet flexible enough to adapt to diverse mission contexts, acquisition pathways, and system architectures (DoD, 2025a).

12.2 Federated Governance Structure

A federated governance structure provides the most effective balance between enterprise coherence and domain specific autonomy. At the highest level, an **enterprise council** maintains the authoritative MOSA criteria, adjudication policies, and conformance tier definitions. This aligns with DoD's emphasis on enterpriselevel standardization and crossportfolio alignment (DoD, 2025a). The council ensures that all programs operate from a consistent baseline and that changes to core expectations are coordinated across Components.

Beneath the enterprise council, **domain working groups** manage missions specific tailoring and profile extensions. These groups translate enterprise criteria into domain appropriate interpretations and develop specialized test cases that reflect operational realities, consistent with the Adaptive Acquisition Framework's tailoring principles (DoD, 2026). **Program conformance leads** serve as the interface between program teams and governance bodies, coordinating evidence collection, assessment planning, and remediation actions. This federated model ensures consistent standards while allowing tailored implementations aligned with mission needs.

12.3 Decision Right Matrix

A clearly defined decision rights matrix is essential for avoiding overlaps, gaps, and conflicts in governance. **Core criteria changes** fall under the authority of the enterprise council because such changes affect all programs and domains. This approach aligns with DoD's guidance on maintaining authoritative technical baselines and configuration control (DoD, 2020b). **Domain profile changes** originate within domain working groups but require enterpriselevel review to prevent fragmentation and ensure alignment with MOSA principles (DoD, 2025a).

At the program level, **waivers** from either core or domain criteria require approval by designated senior officials, with visibility provided to both the enterprise council and relevant domain working groups. This reflects GAO's recommendation that MOSA waivers and deviations be documented and justified to maintain transparency and accountability (GAO, 2025). **Dispute resolution** should be handled by an independent adjudication panel empowered to interpret criteria and issue binding recommendations. A documented decisionrights matrix ensures that every decision type has a clearly identified owner, participants, and escalation path.

12.4 Auditability

Audit ability is a foundational requirement for governance credibility. The conformance ecosystem must maintain immutable or tamperevident records of criteria versions, assessment findings, waiver decisions, and remediation commitments. This requirement aligns with federal expectations for traceability and configuration control in digital engineering environments (DoD, 2025a). These records enable retrospective analysis of decisions, allowing stakeholders to reconstruct what criteria were in force at a given time and how risks were accepted or mitigated.

Audit ability also supports accountability. Transparent records of waiver expiration, remediation progress, and assessment outcomes ensure that programs follow through on commitments and that deviations from criteria do not persist indefinitely. This capability is consistent with GAO's emphasis on improving MOSA oversight and documentation quality across programs (GAO, 2025). Ultimately, robust audit ability ensures that governance decisions are defensible, reproducible, and aligned with enterprise objectives.

12.5 Governance Performance Metrics

Governance performance must be monitored through a focused set of metrics that assess responsiveness, consistency, and stakeholder trust. **Assessment cycle time** provides insight into whether governance processes are enabling or delaying program progress. **Scoring consistency** across assessors and programs helps identify whether criteria are being interpreted uniformly, supporting GAO's call for improved MOSA assessment consistency (GAO, 2025).

Additional metrics include **waiver closure rates**, **criteria update latency**, and **stakeholder trust indicators**, which align with DoD's broader emphasis on datadriven acquisition performance management (DoD, 2020b). Together, these metrics provide a holistic view of governance health and highlight areas where adjustments may be necessary to maintain effectiveness and credibility.

12.6 Cultural Adoption

Even the most technically sound governance model will fail if it is not culturally embraced. Cultural adoption requires leadership to consistently frame MOSA governance as a mechanism for improving mission outcomes, reducing longterm risk, and enabling innovation not as a punitive or bureaucratic burden. This aligns with DoD's emphasis on cultivating a culture of transparency and continuous improvement in softwareintensive acquisition (DoD, 2021b).

Messaging and practice should emphasize that early risk disclosure, candid findings, and thoughtful waiver requests are indicators of professional rigor rather than weakness. Governance forums should model constructive behavior by using conformance assessments to identify systemic issues and share good practices across programs. This approach reflects the DoD's digital engineering and DevSecOps principles, which emphasize learning, iteration, and crossprogram collaboration (DoD, 2025a). Over time, governance becomes a catalyst for resilience and continuous improvement rather than a barrier to progress.

13. Risk, Cybersecurity, and Supply Chain Security

13.1 Risk Taxonomy

A comprehensive risk taxonomy is essential for treating MOSA conformance as a substantive assurance activity rather than a narrow technical check. DoD policy emphasizes that modularity and openness must be evaluated through a lifecycleoriented, risk based lens that accounts for both technical and organizational factors (DoD, 2025a). **Lockin risk** remains one of the most significant threats to MOSA outcomes, as proprietary interfaces, restrictive datarights strategies, or tightly coupled architectures can prevent vendor substitution and undermine long term affordability. **Interoperability fragility** arises when interfaces degrade under change, often due to insufficient semantic specification or inadequate regression testing issues repeatedly highlighted in GAO assessments of MOSA implementation (GAO, 2025).

Cyber risk is equally critical. **Cyber exploitability** includes vulnerabilities at module boundaries, insecure integration patterns, and weaknesses in update pathways. These risks are amplified in modular systems because each component introduces its own attack surface, consistent with NIST's supply chain risk management guidance (NIST, 2024). **Supplier concentration risk** reflects overreliance on a small number of vendors or sub tier suppliers, which can create single points of failure. **Sustainment debt** captures the accumulation of technical deficiencies, outdated dependencies, and insufficient documentation that increase long term support burden. Finally, **governance noncompliance**failure to follow established criteria, workflows, or waiver processes can mask systemic issues and erode enterprise trust. Assessing each risk in terms of severity, likelihood, detectability, and mission impact enables a structured, defensible approach to prioritizing mitigation.

13.2 Cybersecurity by Design

Cybersecurity must be embedded into MOSA conformance criteria from the outset rather than treated as a separate overlay. This approach aligns with DoD's DevSecOps Strategy, which emphasizes integrating security controls into early design, development, and integration activities (DoD, 2021a). **Boundary authentication and identity verification** ensure that module interfaces enforce strong identity controls rather than relying on implicit trust. **Leastprivilege access control** ensures that data flows and API permissions are constrained to the minimum necessary for mission execution, reducing lateral movement opportunities for adversaries.

Secure update pathways including **code signing**, **provenance checks**, and **rollback protections**are essential for preventing supplychain compromise and unauthorized modifications. These expectations mirror federal guidance on secure software development and supply chain assurance (NIST, 2024). Finally, **telemetry and logging requirements** ensure that modules generate sufficient data for threat detection, anomaly analysis, and forensic reconstruction. By embedding these elements as baseline conformance criteria, the harness reinforces cybersecuritybydesign practices and reduces the likelihood that security controls are deferred or inconsistently applied across modules.

13.3 Zero Trust Mapping

MOSA module boundaries align naturally with Zero Trust principles, which require explicit identity, authorization, and policy enforcement at every boundary. DoD's Zero Trust Strategy emphasizes minimizing implicit trust zones and enforcing continuous authentication and authorization (DoD, 2025a). In modular architecture, each module boundary becomes a potential enforcement point for Zero Trust controls. Conformance checks should therefore verify that **trust assumptions are minimized**, **policies are enforced as close to the asset as possible**, and **authentication and authorization decisions are continuously evaluated** rather than statically configured.

This mapping ensures that modularity does not inadvertently create new trust zones or expand attack surfaces. Instead, MOSA boundaries become opportunities to strengthen security posture by enforcing granular, contextaware controls. Aligning MOSA with Zero Trust principles also ensures consistency with DoD's broader cybersecurity modernization efforts and reduces the risk of architectural drift between mission systems and enterprise security expectations.

13.4 Supply Chain Integrity

Supply chain integrity is a critical dimension of MOSA risk, particularly as modern systems rely on complex software and hardware dependency chains. Federal supply chain guidance emphasizes the need for **provenance**, **transparency**, and **vulnerability management** across all components (NIST, 2024; CISA, 2023). Programs should provide **provenance attestations** for critical components, demonstrating that suppliers follow secure development practices and that artifacts have not been tampered with. **SBOMs and dependency mappings**

provide transparency into thirdparty libraries, firmware, and hardware dependencies, enabling more accurate vulnerability assessments.

Vulnerability management evidence including timelines for identifying, triaging, and remediating issues demonstrates operational readiness and responsiveness. GAO has repeatedly noted that DoD programs often lack sufficient supplychain visibility and timely remediation processes, contributing to long term sustainment risk (GAO, 2025). **Contingency plans** for supplier disruption or compromise ensure that programs can maintain operational continuity even when vendors fail or components are compromised. Conformance scoring should reflect both current posture and remediation velocity, encouraging continuous improvement rather than static compliance.

13.5 Data Protection and Cross Domain Controls

Data protection and crossdomain safeguards are essential components of MOSA conformance. DoD cybersecurity policy requires correct handling of classification markings, encryption for data at rest and in transit, and strict enforcement of needto know principles (DoD, 2020b). Modular architectures often involve data flowing across multiple components, increasing the risk of misclassification, unauthorized access, or improper dissemination. Conformance checks should therefore verify that **classification rules are consistently applied**, **encryption is implemented using approved algorithms**, and **access controls align with mission roles**.

Crossdomain transfer controls are particularly important in systems that operate across security enclaves. These controls must follow approved mechanisms and policy constraints, consistent with DoD's crossdomain solution (CDS) guidance. Failure to implement proper crossdomain safeguards can result in mission degradation, data compromise, or regulatory violations. As such, data protection and crossdomain controls should be treated as firstclass conformance dimensions.

13.6 Incident Learning and Criteria Updates

Major cyber, supplychain, or interoperability incidents should trigger structured review of relevant conformance criteria and test assets. DoD's digital engineering and cybersecurity strategies emphasize continuous improvement and rapid incorporation of lessons learned (DoD, 2025a). When systemic patterns are identified such as recurring interface vulnerabilities or

repeated supplychain weaknesses governance bodies should propose criteria updates, revise test cases, and issue transition guidance specifying effective dates and expected program responses.

This structured approach ensures that the harness becomes a mechanism for institutional learning rather than a static compliance tool. By linking incident learning to criteria evolution, the harness ensures that lessons from realworld failures are systematically captured, versioned, and propagated across the MOSA ecosystem. This strengthens enterprise resilience and reduces the likelihood of repeated failures.

13.7 Quantitative Risk Prioritization

A simple risk priority score can support consistent triage, provided it is not treated as a substitute for mission judgment. Profiles should define ordinal scales, decision thresholds, and override conditions for catastrophic or non compensable risks.

$$RPS = P \times I \times D \tag{4}$$

Equation 4. Risk priority score for findings triage.

P is likelihood, I is mission or lifecycle impact, and D is detectability or latency before discovery. The score supports prioritization; it is not a probabilistic loss estimate.

14. Testing, Verification, and Validation Framework

Table 18. Verification and validation test classes.

Test class	Purpose	Principal evidence
Interface syntax	Schema, message, API, and IDL conformance	Machine readable specifications and validator logs
Semantic behavior	State transitions, timing, errors, units, and edge cases	Executable assertions, test vectors, behavioral traces
Interoperability	Cross vendor and cross program operation	Compatibility matrix and mission thread results
Substitution and rollback	Replacement without privileged knowledge; restoration after failure	Integration package, logs, defect record, rollback evidence
Portability	Deployment across approved compute, runtime, or hosting environments	Environment manifests and repeatable deployment results
Security and supply chain	Identity, authorization, isolation, SBOM, provenance, vulnerability response	Security test results, attestations, and incident response telemetry
Sustainment reproducibility	Build, test, release, and recovery by an alternate team	Reproduction report and configuration controlled artifacts

14.1 Multi Method Verification Model

No single testing mode is sufficient to establish robust MOSA conformance. DoD testing guidance emphasizes the need for multimethod verification that integrates automated checks, integration demonstrations, expert review, and missionthread validation (DoD, 2020c). Automation supports scale and repeatability, integration events expose emergent behavior, expert review interprets architectural nuance, and missionthread validation connects technical properties to operational effectiveness.

This multimethod approach ensures that MOSA conformance is evaluated holistically rather than through narrow or isolated tests.

14.2 Core Test Classes

To support consistent and comprehensive assessment, the framework should include core test classes aligned with MOSA principles and DoD's digital engineering strategy (DoD, 2025a):

- **Interface syntax and semantics tests**, ensuring schema and behavioral conformance
- **Multivendor interoperability tests**, demonstrating crosssupplier compatibility
- **Component substitution and rollback tests**, validating replaceability
- **Performance and reliability under reconfiguration**, assessing stability during modular changes

- **Security control verification and dependency checks**, consistent with NIST SP 800161 (NIST, 2024)
- **Sustainment reproducibility and release recovery tests**, ensuring longterm maintainability

Together, these test classes provide a structured basis for evaluating modularity, openness, and operational robustness.

14.3 Mission Thread Validation

Missionthread validation is essential for linking engineering claims to warfighter relevance. DoD operational test guidance emphasizes validating system behavior in realistic mission scenarios (DoD, 2020c). Missionthread tests should verify that modular substitutions preserve or enhance operational outcomes and should be grounded in representative use cases defined with operators and mission owners.

By demonstrating that modular changes do not degrade mission effectiveness, missionthread validation provides a critical bridge between MOSA conformance and operational assurance.

14.4 Synthetic and Live Environments

Both synthetic and live environments play essential roles in the conformance framework. Digital twins and simulation environments support rapid iteration and costeffective exploration of configuration space, consistent with DoD's digital engineering strategy (DoD, 2025a). However, synthetic evidence must be labeled with appropriate confidence levels.

Transition criteria should define when live validation is required for example, for safetycritical functions or final predeployment assurance. Synthetic results should inform, not replace, live testing.

14.5 Evidence Confidence Scoring

Not all evidence carries equal weight. The framework should include an evidence confidence scoring model that accounts for source trustworthiness, recency, reproducibility, and assessor independence. This aligns with federal guidance on evidencebased risk management and supplychain assurance (NIST, 2024).

Low confidence evidence may still be useful for identifying concerns but should not be sufficient to close critical findings. Confidence scoring helps decisionmakers interpret conformance results in a nuanced and defensible way.

14.6 Continuous Verification Integration

Where feasible, conformance checks should be integrated into continuous integration and delivery pipelines. This aligns with DoD's DevSecOps Strategy, which emphasizes automated quality gates and continuous security validation (DoD, 2021a). Embedding interface validation, dependency analysis, and selected interoperability tests into routine workflows produces nearrealtime quality telemetry and reduces late discovery of significant issues.

Continuous verification shifts MOSA conformance from an episodic activity to a routine part of development and sustainment, improving release reliability and decisionmaking confidence.

14.7 Completed Controlled Feasibility Demonstration

The controlled feasibility demonstration described in Chapters 6 and 7 was executed as an executable, non sensitive study of a simulated sensor processing interface. The implemented corpus contained twenty independently packaged processing modules: two conformant controls and eighteen candidates, each containing one deliberately injected nonconformance with known ground truth. Every package was evaluated once by a minimal conventional baseline and once by the conformance harness under the same schemas, test vectors, runtime environment, acceptance rules, and recorded configuration. The resulting artifact includes the source code, candidate packages, evidence schemas, raw CSV and JSON outputs, statistical summaries, generated figures, automated tests, and integrity hashes required for independent replay.

The baseline detected 5 of 18 injected faults, whereas the harness detected all 18; both methods accepted the two conformant controls. The harness also produced an actionable root cause diagnosis for all 18 faulted candidates, compared with 5 for the baseline. Thirteen paired fault cases were discordant in favor of the harness, yielding an exact McNemar p value of.00024. Across twenty randomized order repetitions, accept/reject decisions remained stable. The stronger assessment increased the median number of checks from 14 to 177 and increased median primary

runtime from approximately 0.348 milliseconds to 7.042 milliseconds, a 20.2 fold ratio whose absolute cost remained in the millisecond range.

These findings demonstrate mechanism feasibility rather than a universal defense acquisition effect size. The study shows that versioned evidence, provenance checks, static dependency analysis, semantic and edge case tests, and critical decision gates can expose known nonconformances that a nominal smoke test baseline misses. Because the corpus was synthetic, deliberately fault seeded, and evaluated in one software environment, the results do not establish DoD wide cost or schedule savings and do not replace operational pilots, independent assessor calibration, or evaluation against safety critical and classified mission systems. Their value is to close the thesis research cycle with an auditable artifact and controlled evidence while clearly defining the boundary of the claim.

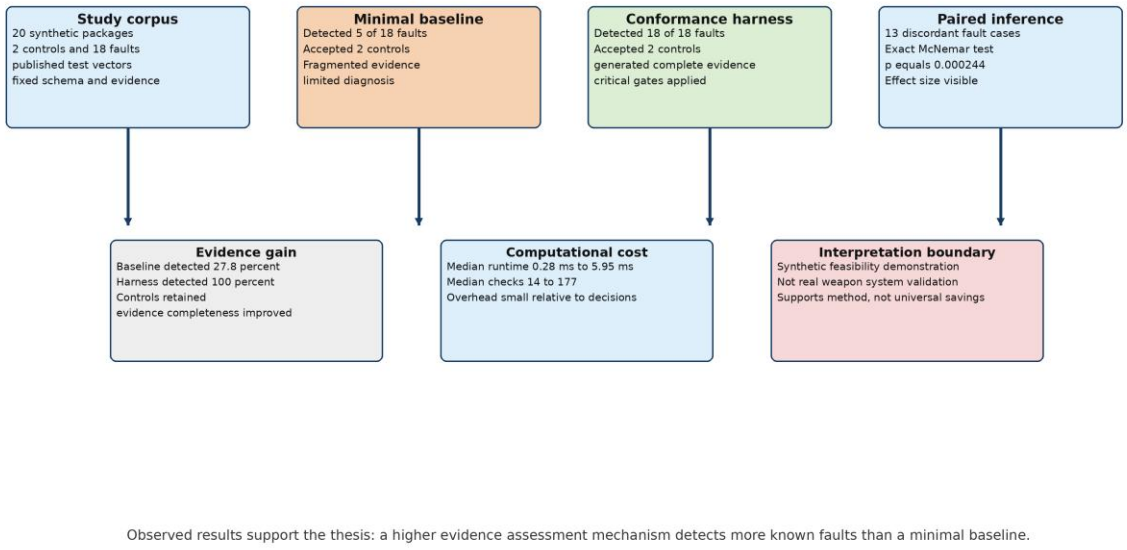


Figure 14. Completed controlled feasibility demonstration and principal results.

Note. Author developed summary of the executable synthetic study. Ground truth was deliberately seeded; the results establish assessment mechanism feasibility under controlled conditions, not a universal DoD cost or schedule effect.

15. Metrics, Analytics, and Performance Management

15.1 Metric Design Principles

Effective MOSA conformance metrics must be meaningful, comparable across programs, and resistant to gaming. DoD acquisition policy emphasizes that metrics should support decision making rather than serve as compliance artifacts (DoD, 2020b). This requires a balanced mix of **leading indicators**, such as evidence quality, interface completeness, and integration cadence, and **lagging indicators**, such as realized delays, cost growth, and mission impacts. Leading indicators help programs detect emerging risks early, while lagging indicators provide insight into whether MOSA practices are producing the intended lifecycle outcomes.

Metrics must also be designed to avoid perverse incentives. GAO has repeatedly warned that poorly designed metrics can encourage superficial compliance rather than substantive improvement (GAO, 2025). Therefore, MOSA metrics should emphasize **trend analysis**, **contextual interpretation**, and **evidencebased scoring** rather than binary pass/fail outcomes. The overarching goal is to support better engineering, acquisition, and governance decisions, not simply to populate dashboards.

15.2 Structural Openness Metrics

Structural openness metrics assess whether architecture and interfaces support modularity, interchangeability, and long term adaptability. These measures align with MOSA's statutory emphasis on modular design and open interfaces (10 U.S.C. § 4401). Representative indicators include **critical interface documentation completeness**, which reflects whether programs have produced the artifacts necessary for independent development and integration. **Schema conformance pass rates** measure how consistently modules adhere to published interface definitions, supporting automated validation and reducing integration risk.

Additional indicators include **coupling index trends**, which quantify the degree of interdependence between modules. High coupling increases lock in risk and reduces replaceability, a concern highlighted in the MOSA Implementation Guidebook (DoD, 2025a). **Version compatibility stability** measures how often interface changes break backward compatibility, signaling whether the architecture is evolving in a controlled, modular manner. Together, these metrics reveal whether the underlying design is moving toward or away from vendor locking and brittle integration patterns.

15.3 Operational Openness Metrics

Operational openness metrics capture how systems behave under real integration and change conditions. These measures reflect DoD's emphasis on rapid, iterative delivery and integration in software intensive systems (DoD, 2021b). **Median integration cycle time** for new or updated modules indicates how quickly programs can absorb change a key determinant of modernization responsiveness. **Thirdparty integration success rates** measure how often external vendors can successfully integrate components using published interfaces, providing direct evidence of openness.

Regression rates following module updates reveal how often changes introduce unintended side effects, indicating the maturity of interface contracts and test coverage. **Missionthread substitution success** measures whether alternative implementations can be inserted without degrading operational outcomes, aligning with DoD's focus on missionlevel performance validation (DoD, 2020c). Collectively, these metrics reflect how safely and efficiently a program can incorporate new capabilities. As illustrated in Figure 1, reductions in integration cycle time after harness adoption demonstrate how standardized criteria and reusable test assets reduce integration friction.

15.4 Economic Openness Metrics

Economic openness metrics assess whether MOSA is delivering lifecycle competition and cost benefits. GAO has emphasized that MOSA's value depends on whether programs achieve competitive options over time (GAO, 2025). **Supplier diversity counts** in key module categories indicate whether multiple vendors can realistically compete for upgrades or replacements. **Time to onboard alternate vendors** measure how quickly new suppliers can enter the ecosystem, reflecting the clarity of interface documentation and the maturity of onboarding processes.

Module replacement cost deltas compare the cost of replacing a module under MOSA conditions with traditional monolithic approaches, providing evidence of affordability gains. **Concentration risk indicators** measure reliance on single vendors for critical capabilities, aligning with federal supplychain risk management guidance (NIST, 2024). These metrics help determine whether openness is translating into practical competitive options rather than remaining a policy aspiration.

15.5 Security and Resilience Metrics

Security and resilience metrics ensure that modularity and openness are balanced with adequate protection and recovery. DoD's Zero Trust Strategy and DevSecOps guidance emphasize continuous security validation and resilience under change (DoD, 2021a; DoD, 2025a). **Vulnerability patch latency** measures how quickly modules are updated after vulnerabilities are identified, reflecting operational readiness and supply chain hygiene. **Coverage of verified provenance and supplychain attestations** aligns with NIST SP 800161 requirements for transparency and integrity (NIST, 2024).

Boundary control conformance assesses whether module interfaces enforce authentication, authorization, and policy controls consistent with Zero Trust principles. **Resilience exercise performance** evaluates how systems behave during reconfiguration, failover, and recovery scenarios, demonstrating whether modularity enhances or undermines mission assurance. These metrics link MOSA conformance directly to cyber risk management and operational resilience.

15.6 Dashboard and Decision Integration

Metrics must be presented through dashboards tailored to different decision levels. DoD's digital engineering strategy emphasizes multilayered visualization to support enterprise, portfolio, and program decision making (DoD, 2025a). An **executive layer** should present a small set of aggregate indicators and trends, enabling senior leaders to assess enterprise health at a glance. **Domainlevel views** allow Component and portfolio leaders to compare programs, identify systemic issues, and prioritize interventions.

Program level drill downs expose specific findings, trends, and remediation status, enabling engineering teams to take targeted action. Importantly, scores should always be accompanied by narrative context and confidence information to avoid false precision a concern highlighted in GAO's assessments of DoD performance management (GAO, 2025). Figure 2 illustrates a notional multidomain conformance trend, showing progressive improvement in structural openness, security and resilience, and economic openness over multiple reporting periods.

15.7 Evidence to Outcome Measurement Logic

Performance management should distinguish conformance indicators from realized program outcomes. Structural indicators such as interface completeness or lower coupling are leading measures; integration time, substitution success, and regression rates are operational outcomes; supplier entry, competitive awards, and cost of change trends are economic outcomes. The dashboard should preserve these levels rather than combine them into a single score that implies causation.

Each outcome claim should include its baseline, observation window, system configuration, data source, confidence rating, and contextual variables. For cost claims, the harness should record whether the comparison is an estimate, a realized expenditure, or cost avoidance, and it should retain the underlying counterfactual assumptions. For schedule claims, it should distinguish calendar time from engineering labor and identify whether integration was performed by the incumbent, the government, or an independent supplier. These requirements convert MOSA benefit claims into auditable evidence packages.

15.8 Quantitative Scoring and Measurement Equations

The equations below organize evidence without implying that a single score can replace engineering judgment. Dimension weights, criterion weights, confidence factors, penalties, and thresholds must be versioned and approved in the applicable profile. Critical safety, cybersecurity, rights, or interoperability gates can override an otherwise high average.

$$C = 100 \sum_{d=1}^{D} w_d \, S_d, \qquad \sum_{d=1}^{D} w_d = 1 \tag{5}$$

Equation 5. Composite conformance score and normalized dimension weights.

C is the composite score, S_d is the normalized score for dimension d, and w_d is its approved weight. The 0 to 100 result must be reported with dimension scores and confidence rather than as a stand alone certification.

$$S_d = \frac{\sum_{i=1}^{n_d} q_i \, e_i x_i}{\sum_{i=1}^{n_d} q_i \, e_i} \tag{6}$$

Equation 6. Evidence weighted score for a conformance dimension.

For criterion i, x_i is the observed 0 to 1 conformance value, q_i is the criterion importance weight, and e_i is evidence confidence. Missing or weak evidence therefore reduces the influence of an asserted result.

$$e_i = (r_i b_i t_i v_i)^{1/4} \tag{7}$$

Equation 7. Geometric evidence confidence factor.

The confidence factor combines source reliability r_i, baseline quality b_i, traceability t_i, and independent verification v_i. A geometric mean penalizes a severely weak component more than a simple arithmetic average.

$$C^* = \max\left(0,\ C \sum_{j=1}^{m} p_j\right) \tag{8}$$

Equation 8. Gate adjusted conformance score after critical penalties.

C* is the decision score after approved penalties p_j. A profile may also define non compensable gates that prevent certification regardless of the arithmetic score.

$$ICR = \frac{T_{baseline}\, T_{harness}}{T_{baseline}} \times 100\% \tag{9}$$

Equation 9. Integration cycle reduction.

Integration cycle reduction should be paired with labor, defect, mission performance, and sample information so speed is not mistaken for successful integration.

$$ECR = \frac{\sum_{k=1}^{K} a_k\, z_k}{\sum_{k=1}^{K} a_k} \tag{10}$$

Equation 10. Weighted evidence completeness ratio.

The evidence completeness ratio uses artifact importance a_k and availability/acceptability z_k. It measures whether required evidence exists and is usable, not whether the underlying system conforms.

$$MCI = 1\ \frac{E_{cross}}{E_{total}} \tag{11}$$

Equation 11. Modularity coupling index.

The modularity coupling index is an illustrative structural indicator in which E_cross represents dependencies that cross declared module boundaries. Its interpretation depends on the dependency model and should be calibrated by domain.

Table 19. Metric dictionary for structural, operational, economic, and security openness.

Dimension	Metric	Definition	Cadence	Type
Structural	Interface completeness	Required syntactic and semantic elements accepted / required elements	Release or architecture baseline	Leading indicator
Structural	Modularity coupling index	1, cross boundary dependencies / total dependencies	Release	Leading indicator
Operational	Integration cycle time	Elapsed time from accepted package to passing acceptance tests	Change event	Outcome
Operational	Substitution success	Successful substitutions / attempted substitutions	Quarter or release	Outcome
Operational	Regression rate	New material defects / change events	Release	Outcome
Economic	Qualified supplier count	Suppliers that can meet technical, security, and rights thresholds	Annual	Outcome/enabler
Economic	Cost of change	Total engineering, test, certification, and contract cost per bounded change	Change event	Outcome
Security	Vulnerability response time	Time from validated notification to tested and deployable remediation	Incident	Outcome
Evidence	Evidence completeness	Weighted accepted artifacts / weighted required artifacts	Assessment	Confidence input

Table 20. Illustrative conformance score calculation.

Dimension	Weight	Observed score	Confidence	Contribution / result
Structural openness	0.30	0.82	0.90	22.14
Operational openness	0.35	0.74	0.85	22.02
Economic openness	0.20	0.55	0.70	7.70
Security and resilience	0.15	0.88	0.95	12.54
Composite before penalties	1.00			64.40
Critical penalties				8.00
Gate adjusted score				56.40

Note. Illustrative values only. The example shows why a composite score must remain subordinate to critical gates and confidence information.

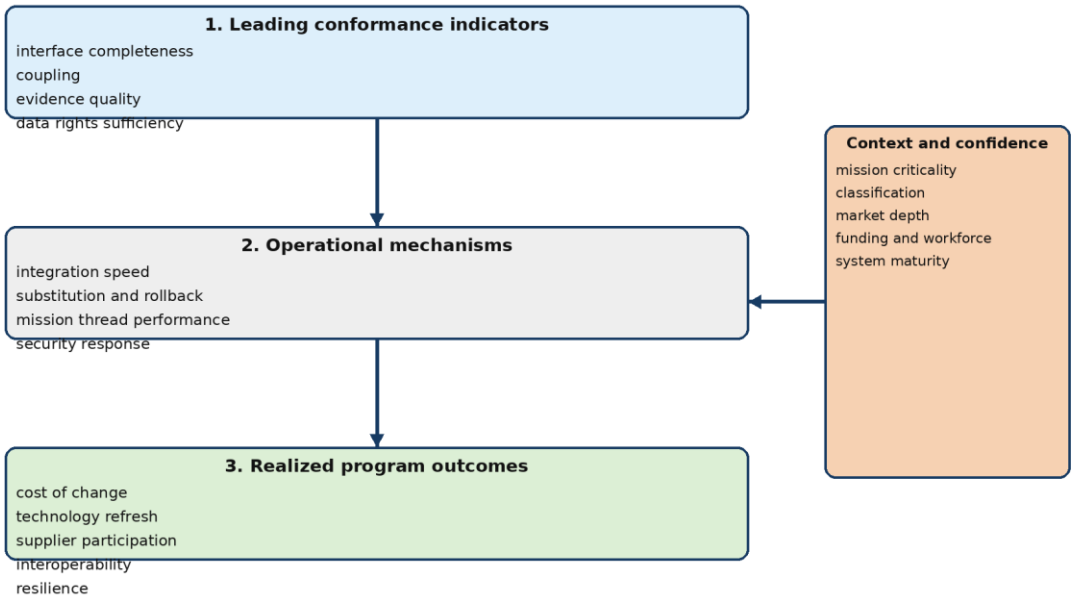


Figure 15. Measurement chain from leading indicators to realized program outcomes.

Note. Author developed measurement model. Outcome attribution requires context, baselines, and confidence information.

16. Acquisition, Contracting, and Business Model Implications

Table 21. Acquisition evidence deliverables by lifecycle decision point.

Decision point	Evidence question	Principal deliverables
Requirements / materiel decision	MOSA objectives; anticipated change drivers; candidate major components and interfaces	Capability document; preliminary profile; change frequency assumptions
Acquisition strategy / solicitation	Modularity strategy; interface standards; IP and technical data plan; verification approach	RFP requirements; data item descriptions; evaluation factors; rights assertions
Preliminary and critical design reviews	Boundary integrity; interface maturity; semantic completeness; test readiness	Models, schemas, interface control artifacts, test plans, traceability
Milestone / production decision	Verified conformance; unresolved findings; waiver posture; cost benefit update	Conformance profile; confidence; substitution evidence; decision brief
Fielding and sustainment	Mission thread performance; supplier health; cost of change; vulnerability response	Continuous telemetry; competition readiness assessment; refresh and obsolescence plan

16.1 Acquisition Integration Logic

MOSA conformance evidence must be treated as a core acquisition artifact rather than an auxiliary technical report. DoD acquisition policy emphasizes that modularity, openness, and data rights are statutory requirements that must inform milestone decisions, tailoring strategies, and risk assessments (DoD, 2026; 10 U.S.C. § 4401). Accordingly, conformance scorecards and risk summaries should be integrated directly into **Milestone A/B/C reviews**, **Acquisition Strategy approvals**, **source selection evaluations**, **contract option exercises**, and **sustainment planning gates**.

At each decision point, acquisition authorities should view MOSA conformance evidence alongside traditional cost, schedule, and performance data. This alignment ensures that openness and modularity are not treated as aspirational goals but as measurable, enforceable acquisition outcomes. GAO has repeatedly found that MOSA is often referenced in acquisition strategies but not meaningfully evaluated during decision reviews, leading to inconsistent implementation and unrealized benefits (GAO, 2025). Integrating conformance evidence into acquisition governance corrects this gap and ensures that MOSA expectations influence real program decisions.

16.2 Source Selection Factors

Source selection criteria should explicitly evaluate MOSA implementation quality. DoD's MOSA Guidebook emphasizes that openness, interface quality, and data rights must be considered

during competitive evaluations to ensure longterm adaptability and competition (DoD, 2025a). Beyond technical merit and management approach, evaluations should assess:

- **Interface openness and testability**, including schema completeness and semantic clarity
- **Maturity of evidence automation**, such as CI/CDintegrated conformance checks
- **Credible thirdparty integration and substitution plans**, demonstrating replaceability
- **Sufficiency of data rights** to support future competition and sustainment
- **Approaches to enabling multivendor participation**, including onboarding support

These factors reward vendors who design longterm interchangeability rather than optimizing them solely for near term delivery. This aligns with GAO's recommendation that MOSA must be embedded in source selection to avoid vendor locking and ensure lifecycle competition (GAO, 2025).

16.3 Incentive Structures

Contract incentives can reinforce desired MOSA outcomes by aligning vendor behavior with openness and modularity objectives. DoD acquisition guidance encourages the use of incentive structures that reward measurable performance outcomes rather than compliance artifacts (DoD, 2020b). Examples include:

- Incentive fees tied to achieving specific conformance levels by defined dates
- Rewards for reducing integration cycle time while maintaining quality
- Incentives for successful multivendor integration or substitution events

Conversely, persistent major nonconformance, especially in interoperability, openness, or data rights should trigger corrective action plans and, if unresolved, contractual consequences such as withheld fees or negative past performance assessments. This approach aligns with federal acquisition principles that emphasize accountability and performancebased contracting.

16.4 IP and Data Rights Posture

Intellectual property (IP) and data rights are foundational to MOSA success. Even the most modular architecture cannot support competition or adaptability if the government lacks sufficient rights to technical data, interface definitions, or test assets. DoD policy stresses that data rights strategies must be aligned with long term sustainment and recompete objectives (DoD, 2020c). Conformance assessments should therefore examine whether:

- The government holds adequate rights in technical data and software

- Key interfaces and data models can be disclosed to alternate suppliers
- License terms enable modular substitution and multivendor participation

GAO has noted that insufficient data rights remain a major barrier to MOSA implementation, often locking programs into incumbent vendors despite nominally modular architectures (GAO, 2025). Treating IP and data rights as firstclass conformance dimensions ensures that legaltechnical coupling supports, rather than undermines, MOSA objectives.

16.5 Competition Readiness

Competition should be treated as an engineering capability, not merely a contracting aspiration. Periodic conformance assessments can estimate **competition readiness** by evaluating:

- The expected effort to onboard alternate vendors
- The completeness and usability of technical data, test assets, and interface evidence
- Integration risk associated with replacing critical modules

These assessments provide an evidencebased view of how quickly and safely a program could pivot to new suppliers if mission needs or technology changes require it. This aligns with DoD's emphasis on maintaining competitive options throughout the lifecycle to reduce cost and improve innovation (DoD, 2025a).

16.6 Ecosystem Participation

Predictable conformance pathways and transparent expectations lower barriers to entry for nontraditional vendors and small businesses. Clear criteria, reusable test packs, and onboarding guides reduce ambiguity about what is required to participate in a modular ecosystem. This aligns with DoD's push to expand the defense industrial base and increase participation from emerging technology suppliers (DoD, 2025a).

As more participants align with a shared conformance model, programs gain access to a broader supplier base, strengthening competition, innovation, and resilience. Supplier diversity patterns such as those illustrated in Figure 3 provide quantitative insight into ecosystem health and economic openness.

16.7 Supplier Concentration and Competition Readiness

Supplier counts alone can conceal concentration. The Herfindahl Hirschman Index (HHI) summarizes the concentration of qualified supplier capacity or award share, while the reciprocal

gives an intuitive effective supplier count. The measure should be calculated at the module and interface level because a program wide count can mask a single critical bottleneck.

$$HHI = \sum_{k=1}^{K} s_k^2, \qquad N_{eff} = \frac{1}{HHI} \tag{12}$$

Equation 12. Supplier concentration and effective supplier count.

s_k is supplier k's share expressed as a proportion. A single supplier yields HHI = 1 and N_eff = 1; evenly distributed shares reduce concentration. The metric does not capture qualification barriers, rights, or onboarding time and must be interpreted with those factors.

17. Workforce, Change Management, and Adoption Strategy

17.1 Workforce Readiness as a Limiting Factor

The durability of a MOSA conformance harness depends primarily on workforce capability. DoD modernization strategies consistently emphasize that workforce readiness is often the limiting factor in implementing new engineering and acquisition practices (DoD, 2021b; DoD, 2025a). Engineers, contracting officers, test professionals, cybersecurity teams, and program managers all interact with the harness in different ways, yet they require a shared vocabulary, common mental models, and aligned expectations.

If workforce readiness lags tool and policy development, MOSA conformance risks devolving into a boxchecking exercise. Therefore, workforce readiness must be treated as a critical constraint and a deliberate focus of early implementation efforts.

17.2 Training Architecture

A layered training architecture can build capability in a structured, scalable manner. **Foundation training** introduces MOSA principles, conformance logic, and mission relevance to a broad audience. **Rolebased training** provides tailored instruction for responsibilities such as evidence preparation, test execution, governance participation, and contracting support, using concrete examples and templates.

Advanced assessor calibration training is essential for those responsible for scoring, interpreting criteria, and adjudicating complex cases. This training emphasizes consistency, evidence confidence models, and cross program alignment. Finally, **leadership briefings** focus on interpreting conformance outputs, understanding risk posture, and using evidence in tradeoffs among cost, schedule, performance, and openness. This layered approach ensures that each role understands both its own tasks and its place in the broader ecosystem.

17.3 Change Management Mechanics

Successful adoption requires intentional management change, not just technical deployment. DoD organizational change literature emphasizes the importance of executive sponsorship, visible early wins, and transparent issue resolution (DoD, 2020b). Key mechanics include:

- **Executive sponsorship** that visibly endorses the harness and models its use

- **Visible pilot wins** demonstrating reduced integration time or clearer risk visibility
- **Transparent issue resolution**, where pilot challenges are acknowledged and addressed
- **Practical support channels**, such as office hours, help desks, and peer networks

Resistance is most likely when new processes are perceived as overhead without mission benefit. Early communication should therefore emphasize time savings, risk reduction, and improved negotiating position with vendors not just compliance.

17.4 Industry Engagement

Industry engagement is essential for realistic and implementable conformance expectations. Vendors should be included in **draft criteria feedback cycles**, **pilot design workshops**, and **interoperability or plugfest events**. This aligns with DoD's emphasis on collaborative development of standards and open architectures (DoD, 2025a).

Early and repeated industry involvement surfaces practical constraints, reduces implementation surprises, and increases the likelihood that the harness will be seen as a predictable framework rather than a shifting compliance burden.

17.5 Communication Strategy and Portal Evolution

The current minimal public site serves as a communication seed and should evolve into a persistent communication platform. Over time, it should host:

- Release notes and criteria change logs
- Frequently asked questions and interpretation notes
- Onboarding guides and exemplary artifacts
- Clear contact paths for programs, assessors, and industry stakeholders

Predictable, authoritative communication reduces rumors, minimizes interpretation drift, and provides a single reference point for "what the harness really requires." This aligns with DoD's digital engineering strategy, which emphasizes authoritative sources of truth (DoD, 2025a).

17.6 Adoption Risks and Countermeasures

Key adoption risks include **assessor shortages**, **uneven leadership enforcement**, and **metric gaming**, where teams optimize scores rather than underlying outcomes. Countermeasures include:

- Phased rollout with realistic capacity planning

- Independent quality audits of assessments and metrics
- Incentive alignment with mission outcomes, such as rewarding early risk disclosure

These countermeasures reflect GAO's findings that governance consistency and workforce capability are essential for successful MOSA implementation (GAO, 2025).

18. Roadmap, Milestones, and Execution Timeline

Table 22. Roadmap phases, outputs, and exit conditions.

Phase	Core outputs	Exit condition
0 to 6 months: Define	Enterprise sponsor, governance charter, criteria v0.1, evidence schema, pilot selection	Approved baseline and two or more pilot agreements
6 to 12 months: Pilot	Manual and semi automated assessments, assessor calibration, defect and burden tracking	Pilot lessons incorporated; inter rater agreement and workflow targets met
12 to 18 months: Operationalize	Pipeline integration, APIs, analytics, domain profiles, evidence signing	Repeatable automated checks and auditable evidence packages
18 to 24 months: Embed	Solicitation language, milestone mappings, waiver controls, training	Harness artifacts accepted in acquisition decisions and contracts
24 to 36 months: Scale	Federated service governance, portfolio analytics, cross program reuse, sustainment funding	Multi Component adoption with stable operating model and outcome reporting

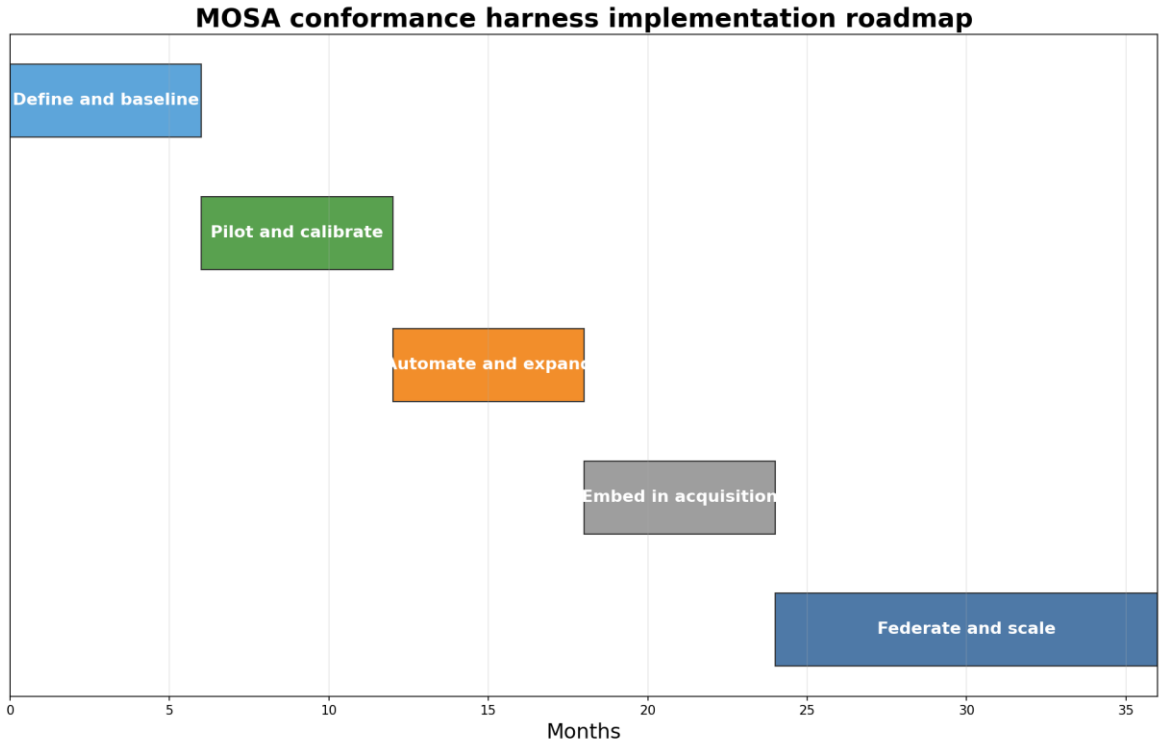


Figure 16. Thirty six month implementation roadmap.

Note. Author developed roadmap. Programs should tailor sequence and rigor to pathway, mission urgency, and risk.

With governance, engineering, acquisition, and adoption elements defined, the roadmap provides a structured sequence for implementing the MOSA Conformance Harness in a way that balances urgency with institutional stability. DoD modernization guidance emphasizes that

enterpriselevel transformation requires phased capability maturation, iterative learning, and deliberate integration into acquisition and governance processes (DoD, 2020b; DoD, 2025a). The roadmap therefore focuses not only on deploying tools but on embedding new behaviors, governance rhythms, and evidencebased decision practices across the defense ecosystem.

18.1 36 Month Roadmap Overview

A 36month roadmap provides sufficient time to stand up core capabilities, validate them in pilots, and embed them into enterprise acquisition and governance processes. This timeline aligns with DoD's typical acquisition planning cycles and allows for multiple iterations of criteria refinement, automation enhancement, and workforce calibration (DoD, 2020b). The roadmap is structured around two intertwined dimensions:

- Technical capability maturity including criteria development, automation, analytics, and evidence pipelines
- Institutional embedment including governance practices, acquisition integration, and workforce adoption

Each phase concludes with tangible artifacts and observable behavior changes, ensuring that progress is measured not by documentation alone but by demonstrated improvements in conformance quality, integration speed, and decisionmaking rigor.

18.2 Months 0 to 6: Stand Up and Baseline

The initial phase focuses on establishing the minimum foundations required for meaningful pilot use. This includes approval of a governance charter defining roles, decision rights, and federation logic an essential step for ensuring clarity and accountability (DoD, 2025a). Programs publish criteria v1.0 and evidence schema v1.0, establishing the baseline against which early assessments will be conducted.

An initial set of pilot programs is selected to ensure diversity across mission domains, consistent with GAO's recommendation that MOSA implementation be validated across heterogeneous environments (GAO, 2025). The existing public site is expanded into a practical onboarding portal with basic documentation, FAQs, and contact information. The goal is to move quickly to a usable baseline, accepting that refinement will be needed after pilot feedback.

18.3 Months 6 to 12: Pilot Assessments and Calibration

The second phase uses pilots to stress, test, and refine the harness. Pilot assessments are executed in selected programs, generating the first wave of realworld evidence. This phase focuses heavily on scoring calibration, ensuring that assessors interpret criteria consistently across domains an issue repeatedly highlighted in DoD oversight reviews (GAO, 2025).

Workflow bottlenecks and unclear criteria are identified and corrected, leading to publication of criteria v1.1 with explicit change rationale. Basic analytics dashboards are introduced to track early metrics such as assessment cycle time, common findings, and evidence completeness. This phase establishes the feedback loops necessary for continuous improvement.

18.4 Months 12 to 18: Automation and Domain Expansion

The third phase expands both technical depth and domain coverage. Additional automated checks are introduced, including extended schema validation, interface analysis, and traceability checks. These are aligned with DoD's DevSecOps Strategy, which emphasizes automated quality gates and continuous verification (DoD, 2021a).

Mission thread scenario packs are developed and deployed to support representative integration and substitution tests. Integration pathways are built to connect the harness with program toolchains such as CI/CD systems and model repositories. Assessor training courses are expanded to support additional programs and domains. By the end of this phase, the harness begins to function as part of normal engineering practice rather than a standalone activity.

18.5 Months 18 to 24: Acquisition Embedding

The fourth phase embeds conformance more deeply into acquisition and contracting processes. Conformance checkpoints are formalized in acquisition strategies, RFP templates, and milestone review artifacts, aligning with statutory MOSA requirements (10 U.S.C. § 4401). Pilot incentive structures are introduced, linking contract fees to conformance quality, remediation performance, and integration outcomes.

Source selection language is refined to incorporate harness outputs, ensuring that MOSA implementation quality influences competitive evaluations. By the end of this phase, conformance results routinely influence real acquisition decisions, not just technical reviews.

18.6 Months 24 to 36: Federated Scale

The final phase focuses on federated scaling and enterpriselevel visibility. Governance federation is extended across additional Services, Components, and mission domains, consistent with DoD's push for enterprisewide digital engineering alignment (DoD, 2025a). Enterpriselevel trend reporting is operationalized, highlighting crossdomain patterns in openness, security, and competition.

An annual conformance posture report is published, summarizing progress, systemic risks, and planned criteria evolutions. By month 36, the MOSA Conformance Harness functions as a stable, evolving institutional capability embedded in engineering, acquisition, and governance rhythms rather than a timelimited initiative.

19. Domain Specific Application Profiles

Table 23. Comparison of domain specific application profiles.

Profile	Typical modules	Dominant constraints	Priority evidence
Air systems	Mission computers, payloads, avionics applications	Airworthiness, real time timing, safety, platform certification	Compute and payload substitution; deterministic timing; certified rollback
Maritime systems	Combat system services, sensors, weapons interfaces	Long service life, ship availability, cyber physical integration	Incremental refresh; shore based integration; configuration reproducibility
Ground systems	Vehicle electronics, mission command applications, sensors	SWaP constraints, harsh environments, fleet diversity	Hardware abstraction; ruggedized portability; field maintenance
Space systems	Ground segments, payload processing, mission data	Launch irreversibility, radiation, cross domain and latency constraints	Ground system modularity; interface simulation; fault containment
Enterprise IT	Cloud services, data platforms, business systems	Rapid release cadence, identity, data governance, commercial dependency	API conformance; portability; zero trust enforcement; exit planning
Joint C2	Data and mission services across Services and partners	Semantic interoperability, releasability, coalition constraints	Mission thread validation; data mediation; profile version governance

Domainspecific profiles illustrate how a single conformance framework can preserve core rigor while adapting evidence depth, test emphasis, and governance practices to distinct operational realities. DoD's digital engineering and MOSA guidance emphasize the need for shared criteria with domainappropriate tailoring rather than separate, incompatible standards (DoD, 2025a). Programs across domains therefore work from a common baseline for modularity, openness, and security, but tailor how those expectations are demonstrated, tested, and governed based on domain constraints such as safety certification, connectivity patterns, environmental conditions, and sustainment models.

19.1 Air Systems Profile

Air systems combine long lived hardware platforms with rapidly evolving software payloads, mission data requirements, and stringent safety and airworthiness constraints. MOSA conformance in this domain must emphasize **interface stability**, **mission computer abstraction layers**, and **robust configuration management** for safetycritical updates. DoD aviation authorities require disciplined version governance to prevent unintended ripple effects into flightcritical systems (DoD, 2020a).

Simulationbacked regression evidence is essential, allowing programs to demonstrate that modular substitutions or updates have been exercised across realistic flight profiles, environmental

conditions, and failure modes. Supplier onboarding packages must include sufficient technical, test, and configuration information to enable alternate vendors to implement and certify modules without relying on incumbentcontrolled toolchains. Missionthread tests should span sensor ingest, track fusion, and decisionaid services under degraded communication conditions, demonstrating that modular changes preserve operational resilience.

19.2 Maritime Systems Profile

Maritime systems operate in harsh environments with constrained maintenance windows and long deployment cycles. MOSA conformance must therefore emphasize **reliable and recoverable remote updates**, **lowbandwidth interoperability**, and **sustainment mechanisms** that function while underway. Long service lives amplify supplier concentration risks, making openness and replaceability particularly important.

Conformance evidence should demonstrate that update mechanisms are recoverable in the face of partial failures, power interruptions, or intermittent connectivity. Digital twin or shorebased integration environments are essential for rehearsing updates before deployment. Data exchange consistency across command, control, and combat systems must be validated, especially where crosscombat system integration depends on shared modular services. Cybersecurity checks should stress identity and access controls in intermittent connectivity scenarios, consistent with DoD Zero Trust guidance (DoD, 2023).

19.3 Ground Systems Profile

Ground systems modernization often involves integrating new software and electronics into legacy vehicles and tactical networks operating under contested spectrum conditions. Conformance criteria must account for heterogeneous hardware constraints, limited computing resources at the edge, and strict backward compatibility requirements across mixed fleets.

Replaceability tests should emphasize lowfootprint module deployments that can operate under degraded network conditions. Governance should prioritize sustainment outcomes, including whether units can receive, verify, and install updates without depotlevel intervention. Missionthread tests may involve inserting updated mission applications or sensors into older vehicles, demonstrating that critical functions remain available and interoperability is preserved across mixedconfiguration formations.

19.4 Space and Enterprise IT Profiles

Space systems demand high reliability, deterministic behavior, and carefully managed autonomy. MOSA conformance must therefore emphasize **fault management**, **predictable behavior**, and **autonomous recovery**. Test packs may include extendedduration simulations, radiation and fault injection scenarios, and demonstrations that modular substitutions do not introduce unacceptable mission risk.

Enterprise IT environments, by contrast, emphasize rapid deployment, frequent updates, and cloud portability. Conformance evidence should focus on **continuous deployment safeguards**, **rollback mechanisms**, **bluegreen deployments**, and **data portability** across hosting environments. These environments benefit from rapid vulnerability mitigation and scalable access control enforcement, consistent with DoD's DevSecOps Strategy (DoD, 2021a).

19.5 Joint C2 Profile

Joint command and control (C2) environments rely on multi domain data fusion, shared situational awareness, and coordinated effects across Services and coalition partners. MOSA conformance must emphasize **data model harmonization**, **latency tolerance**, and **finegrained, rolebased information sharing**. Missionthread tests should cross Service boundaries and, where policy allows, include coalition interoperability proxies.

Representative threads might involve sensor tracks originating in one Service, processed through joint fusion services, and delivered to decision aids in another Service, all while enforcing crossdomain security constraints. Evidence from these tests demonstrates that the combined system can support joint decisionmaking under realistic operational and policy conditions.

19.6 Cross Profile Lessons

Across all profiles, recurring success factors include disciplined interface governance, automated evidence pipelines, and early alignment between engineering, contracting, and legal communities. Programs that treat interface changes as controlled, welldocumented events supported by automated regression tests and clear data rights strategies are better positioned to exploit modular substitution and multivendor competition.

Recurrent failure factors include ambiguous ownership of key interfaces, underscoped rights planning, and late discovery of hidden dependencies such as undocumented timing

assumptions or proprietary tooling. A federated but coherent conformance framework makes these patterns visible, enabling targeted remediation and incremental strengthening of MOSA practice across domains.

20. Data Rights, Interfaces, and Legal Technical Coupling

Table 24. Data rights and interface artifact matrix.

Artifact	Operational purpose	Minimum content	Rights / access condition
Interface specification	Implement compatible behavior	Definition, semantics, version history, error and timing rules	Rights permit government and authorized third party use and disclosure
Reference implementation / adapter	Accelerate integration and clarify ambiguous behavior	Source or executable, build instructions, dependencies	License supports modification, test, and authorized redistribution
Test suite and fixtures	Verify conformance without incumbent knowledge	Assertions, vectors, expected results, environment manifests	Rights support execution and delivery to alternate integrators
Configuration and deployment artifacts	Reproduce build, integration, and recovery	Scripts, infrastructure definitions, parameter baselines	Government can use and modify throughout sustainment
Data models and representative data	Preserve semantic and data portability	Schemas, dictionaries, transformation rules, sanitized samples	Rights and releasability allow alternate supplier access
SBOM and provenance records	Assess supply chain and vulnerability risk	Component identity, version, supplier, hashes, attestations	Contract requires timely, machine readable delivery and updates

Effective MOSA implementation depends not only on engineering quality but also on alignment between architecture design, data rights, and contractual structures. Programs can design clean, modular interfaces and still fail to achieve openness if they do not hold sufficient rights to technical data, test assets, or software to enable independent implementation, reuse, and sustainment. Conversely, generous data rights without coherent interface definitions may not substantially improve competition or adaptability. This section extends the conformance discussion to legal technical coupling, arguing that MOSA assessment must treat rights posture as a first class dimension of architectural quality.

20.1 Why Legal Technical Coupling Matters

Legal technical coupling matters because many of the practical benefits of MOSA such as multi vendor competition, flexible sustainment, and rapid onboarding of new capabilities depend on the ability of alternate suppliers to build, test, and integrate conformant modules without being blocked by proprietary constraints. A program may publish interface descriptions and even schemas, yet lack rights to share representative test vectors, error semantics, or timing tolerances with potential new vendors. In such cases, the architecture appears open on paper but remains effectively closed in practice. Conformance assessment that focuses only on technical artifacts

risks missing this gap unless it explicitly examines whether rights and access arrangements are sufficient to support the intended level of openness.

20.2 Rights Taxonomy for Conformance

A practical rights taxonomy can give conformance assessments a more structured way to describe sufficiency. At one end of the spectrum, modules can be classified as operationally sufficient, meaning that alternate suppliers can reasonably implement, integrate, and sustain them with acceptable technical, schedule, and legal risk. This typically requires access to interface specifications, test artifacts, and enough technical data to diagnose issues and evolve implementations over time. In the middle, modules may be conditionally sufficient, where additional negotiation, special licenses, or restricted use terms are likely to be necessary before a new supplier can participate. At the far end, modules may be deemed insufficient, where the rights posture materially impedes competition, adaptation, or sustainment transfer.

This taxonomy becomes more meaningful when tied to module criticality and lifecycle timing. For highly critical modules, programs may accept a more constrained posture early in development but should plan to move toward operational sufficiency before key modernization or competition milestones. Conformance assessments can then track whether rights evolution is keeping pace with architecture evolution, rather than treating rights as a static, one time concern.

20.3 Interface Rights versus Interface Access

Another important distinction is between having rights to interface documentation and having practical access to what is needed for independent implementation and verification. An interface specification that omits realistic test vectors, boundary conditions, timing tolerances, or error semantics may satisfy minimal documentation requirements but still leave alternate suppliers dependent on incumbent proprietary tools or trial and error integration. From a MOSA conformance perspective, this creates hidden coupling and undermines replaceability.

Conformance criteria should therefore ask whether a technically independent team, operating with the rights and artifacts available under the contract, could build a conformant implementation and validate its behavior without relying on privileged back channel support. Evidence might include shared test suites, model exports, or simulators that exercise the interface

across nominal and edge conditions. By assessing interface access in this richer sense, the harness can distinguish between nominal openness and genuine, enforceable openness.

20.4 Contract Data Requirement Alignment

Contract data requirements are the primary mechanism by which programs secure the artifacts needed for conformance, competition, and long term sustainment. Ambiguous or overly generic deliverable language can lead to documents that are technically delivered but of limited verification utility, for example, PDFs that cannot be parsed by tools, or descriptions that lack the necessary detail for integration or regression testing. Conformance assessment should examine whether data requirements are explicit about structure, timing, and verification use.

Where possible, deliverables should be machine usable, version controlled, and accompanied by validation artifacts that demonstrate their correctness and relevance. This includes not only interface documents but also model exports, software bills of materials, test harnesses, and configuration data. Programs that treat these data products as integral to MOSA conformance rather than as administrative attachments are better positioned to maintain openness and manage risk over time.

20.5 Negotiation Patterns and Risk Reduction

Rights related risk can be reduced through deliberate negotiation patterns early in the program lifecycle. Early market engagement can clarify what types of rights and data vendors are willing to provide under different pricing and scope assumptions. Modular package scoping, in which work is structured around well defined modules with clear interface and data boundaries, helps both the government and industry reason about what rights are needed where. Rights strategy workshops that bring engineering, legal, and contracting stakeholders together can reduce misalignment between technical aspirations and contractual language.

Conformance reviews should verify not only that such planning activities occurred, but that they are translated into enforceable contractual structures. This includes checking whether key interfaces and modules are covered by appropriate data requirements, whether options or step up rights exist to support future competition, and whether rights strategies are reflected in the way work is partitioned among vendors. Programs that institutionalize this negotiation patterns are less

likely to discover late in the lifecycle that they lack the rights necessary to realize MOSA objectives.

20.6 Lifecycle Health Checks

Finally, rights posture should be reassessed at key lifecycle points rather than assumed static once initial contracts are signed. Architecture and implementation details evolve; suppliers may change through mergers, acquisitions, or performance issues; and mission needs can shift in ways that increase the importance of certain interfaces or modules. An initially adequate rights posture can erode relative to emerging requirements if it is not periodically reviewed.

Lifecycle rights health checks can be integrated with broader conformance assessments at major upgrades, sustainment transitions, and competition planning milestones. These checks should ask whether the current rights and data arrangements still support the intended level of openness, whether any new critical dependencies have emerged without corresponding rights coverage, and what remedial actions may be needed. By making rights posture a recurring, evidence based topic in MOSA conformance, the harness helps prevent latent lock in and ensures that legal technical coupling remains aligned with the program’s long term competition and adaptability goals.

21. Financial and Industrial Base Dynamics

The financial and industrial base implications of MOSA conformance are as important as the technical and governance dimensions. A conformance harness that ignores cost behavior, supplier structure, and market incentives will struggle to deliver sustainable benefits, even if it improves interoperability and openness on paper. This section extends the analysis to show how conformance analytics can quantify the cost of change, highlight industrial base vulnerabilities, and shape healthier market dynamics over time.

21.1 Cost of Change as Core Indicator

The central economic promise of MOSA is not simply lower total cost, but a sustained reduction in the **cost of changing** the marginal cost of replacing, upgrading, or substituting modules under realistic operational and contractual constraints. GAO has repeatedly emphasized that DoD programs often underestimate the cost of change because they lack modular architecture, adequate data rights, or reusable test assets (GAO, 2025). Conformance analytics should therefore track not only aggregate program cost but also the **effort, time, and money required to incorporate new suppliers, implement updates, or respond to emergent requirements**.

If the marginal cost of change remains high measured in extensive reengineering, prolonged regression cycles, or exceptional contracting actions then MOSA outcomes are not being realized, regardless of how modular the architecture appears on paper. This aligns with DoD's digital engineering strategy, which stresses that modularity must reduce integration friction and accelerate modernization cycles (DoD, 2025a). Tracking costofchange metrics over time provides a quantitative signal of whether MOSA is delivering its intended economic benefits.

21.2 Industrial Base Resilience

Industrial base resilience depends heavily on how supplier relationships are distributed across critical modules. When a small number of vendors control large portions of key capabilities, operational and bargaining risks increase even in architectures that are nominally modular. This risk is welldocumented in DoD industrial base assessments, which highlight the dangers of singlesource dependencies and fragile supply chains (DoD, 2025a; NIST, 2024).

The conformance harness can strengthen industrial base resilience by tracking supplier concentration for critical modules and surfacing trends that indicate emerging single points of

failure. By identifying where capabilities are effectively tied to one or two suppliers, conformance metrics can inform targeted diversification strategies, such as:

Prioritizing alternate implementations

Negotiating additional data rights

Structuring future solicitations to encourage new entrants

This approach aligns with DoD's statutory requirement to sustain a healthy, competitive industrial base capable of supporting modernization and surge demands.

21.3 Market Signaling Effects

The financial and industrial base implications of MOSA conformance are as consequential as the technical and governance dimensions. DoD modernization policy emphasizes that modularity and openness must produce measurable economic benefits reduced lifecycle cost, increase competition, and improved industrial base resilience not merely architectural elegance (DoD, 2020b; DoD, 2025a). A conformance harness that ignores cost behavior, supplier structure, and market incentives risks delivering only superficial gains. This section extends the analysis to show how conformance analytics can quantify the cost of change, highlight industrial base vulnerabilities, and shape healthier market dynamics over time. These economic dimensions are essential for ensuring that MOSA delivers sustainable, enterpriselevel value rather than isolated technical improvements.

21.4 Small Business Entry Conditions

Small and innovative vendors are particularly sensitive to ambiguity and hidden complexity. They often lack the resources to navigate bespoke integration processes or reverseengineer unstated assumptions buried in legacy systems. A welldesigned conformance ecosystem can improve their entry conditions by providing:

- Clearly scoped interface profiles
- Publicly accessible subsets of test harnesses
- Structured onboarding guidance
- Transparent evaluation criteria

These elements reduce the cognitive and financial burden of entering defense markets. Programs should actively monitor whether conformance design inadvertently favors incumbents

for example, by relying on proprietary tools, undocumented integration steps, or informal knowledge networks. Where such patterns are found, profiles and processes should be adjusted to make genuine competition feasible. This aligns with DoD's small business strategy, which emphasizes reducing barriers to entry and expanding supplier diversity (DoD, 2025a).

21.5 Budget Planning Implications

Implementing and operating a MOSA conformance harness requires upfront investment in infrastructure, training, and process adaptation. These costs should be treated not as overhead but as **enabling investments** that support modernization and competition over the lifecycle. DoD acquisition guidance stresses that early investment in modularity, automation, and data rights can reduce downstream costs associated with integration delays, emergency rework, and constrained competition (DoD, 2020b).

Conformance analytics can reinforce this view by documenting reductions in:

- Integration cycle time
- Rework volume
- Negotiation friction
- Vendor onboarding effort

Once the harness is in place, these improvements make the economic case for continued funding more concrete. This aligns with GAO's recommendation that DoD track MOSA outcomes using measurable, repeatable indicators rather than narrative assertions (GAO, 2025).

21.6 Portfolio Level Optimization

At the enterprise level, the real power of conformance analytics lies in comparative analysis across programs and portfolios. By examining patterns in openness, supplier diversity, cost of change, and resilience metrics, decisionmakers can direct investments toward areas where risk reduction or competition expansion will yield the greatest benefit. This aligns with DoD's digital engineering and portfolio management guidance, which emphasizes datadriven prioritization (DoD, 2025a).

For example, a portfolio view might reveal that certain capability clusters such as mission computing, communications, or autonomy are consistently associated with high marginal cost of change and low supplier diversity. These insights can inform targeted rearchitecture, rights renegotiation, or competitive prototyping efforts. In this sense, the conformance harness becomes

not only a program level assessment tool but also a **portfoliolevel optimization instrument**, guiding where to focus scarce resources for maximum impact.

21.7 Lifecycle Value and Cost Benefit Analysis

Formal cost benefit analysis should compare upfront architecture, interface, rights, and test investment with discounted future benefits such as avoided redesign, faster refresh, competitive pressure, reuse, reduced integration effort, and lower obsolescence risk. Benefits should be scenario based and should identify which are realized expenditures, cost avoidance, or modeled options.

$$NPV_{MOSA} = \sum_{t=0}^{T} \frac{B_t \, C_t}{(1+r)^t} \tag{13}$$

Equation 13. Net present value of MOSA enabled lifecycle benefits.

B_t and C_t are incremental MOSA enabled benefits and costs in period t, r is the approved discount rate, and T is the analysis horizon. Sensitivity analysis should vary upgrade frequency, supplier depth, interface stability, and service life.

22. Governance Failure Modes and Recovery Patterns

No governance model is immune to drift, overload, or misinterpretation. DoD oversight bodies have repeatedly observed that even welldesigned governance frameworks degrade over time if they are not actively maintained, calibrated, and reinforced (GAO, 2025). As organizations evolve, mission demands shift, and personnel rotate, governance mechanisms can become bureaucratic rituals or lose alignment with operational realities. Anticipating common failure modes and defining recovery patterns in advance reduces the risk of governance collapse and helps restore credibility when problems arise. This section examines characteristic failure modes and outlines elements of a recovery playbook consistent with DoD governance best practices.

22.1 Failure Mode 1: Checklist Compliance Drift

One of the most common governance failures is the drift from **evidencebased assessment** to **checklist compliance**. This occurs when governance bodies emphasize the presence of artifacts rather than their quality or operational relevance. GAO has documented similar patterns in acquisition oversight, where programs optimize for producing required documents rather than demonstrating real capability (GAO, 2025).

Checklist drift leads to a system that appears compliant but provides weak assurance. Recovery requires re coupling conformance to **actual tests and outcomes**. This includes strengthening the role of integration demonstrations, missionthread evaluations, and independent evidence audits. Governance must make clear that incomplete or low quality evidence, even if formally present is insufficient to close significant findings. This aligns with DoD's emphasis on "trustworthy evidence" in digital engineering and test strategies (DoD, 2025a).

22.2 Failure Mode 2: Waiver Inflation

Waiver processes are necessary for managing exceptions, but they can undermine conformance if they become too easy to obtain or too difficult to close. **Waiver inflation** occurs when waivers accumulate without meaningful remediation, often due to schedule pressure or leadership turnover. DoD acquisition reviews frequently highlight the risks of unmanaged waivers, which can mask systemic issues and erode accountability (DoD, 2020b).

Recovery requires reestablishing waiver discipline through strict expiration dates, regular review of waiver aging, and verification that compensating controls are implemented and effective.

Executivelevel visibility into persistent waivers especially for highcriticality modules creates the accountability needed to reverse this trend.

22.3 Failure Mode 3: Criteria Volatility

Criteria volatility occurs when conformance criteria change too frequently or unpredictably, destabilizing planning and eroding trust. Programs may struggle to keep up, leading to redundant work, inconsistent implementation, and frustration. This failure mode is common in largescale modernization efforts where governance bodies iterate too quickly without structured transition windows (DoD, 2025a).

Recovery requires restoring predictability through explicit version control, well defined transition windows, and clear communication of the rationale behind each change. Programs need to know **what changed**, **why it changed**, **how long they must adapt**, and **what interim accommodations exist**. This mirrors DoD's configuration control principles for technical baselines.

22.4 Failure Mode 4: Role Ambiguity

Role ambiguity arises when decision rights and responsibilities are poorly defined. This leads to slow, inconsistent governance as stakeholders make conflicting assumptions about who approves waivers, interprets criteria, or adjudicates disputes. GAO has repeatedly identified unclear roles as a root cause of acquisition delays and inconsistent oversight (GAO, 2025).

Recovery requires revisiting and explicitly documenting authority matrices, escalation protocols, and role descriptions. These must then be socialized through training and reinforced through repeated practice in governance forums. Clear lines of responsibility and accessible escalation paths restore timely and coherent decisionmaking.

22.5 Failure Mode 5: Data Fragmentation

Conformance assessments depend on access to coherent, integrated evidence. When data are fragmented across incompatible systems or siloed repositories, reviewers cannot form a reliable picture of risk. Evidence may exist, but it cannot be correlated or analyzed effectively. This mirrors challenges identified in DoD's digital engineering adoption, where inconsistent identifiers and disconnected tools undermine traceability (DoD, 2025a).

Recovery requires prioritizing schema normalization and platform integration so that key evidence types share common identifiers, structures, and access paths. This may involve consolidating repositories, establishing APIs, or introducing a unifying evidence model that supports crosscutting analytics.

22.6 Recovery Playbook

Because governance degradation is inevitable over time, a **recovery playbook** should be defined as part of the conformance design. A typical playbook begins with a rapid diagnostic review to identify which failure modes are present and at what scale. Immediate stabilization actions may include pausing certain waivers, tightening review thresholds, or revalidating criteria for contentious domains.

Mediumterm actions focus on process redesign, role clarification, and tool adjustments, with explicit rebaselining of expectations across stakeholders. Importantly, recovery should be treated as an opportunity to learn and improve not merely to reset to the previous state that allowed drift to occur. This aligns with DoD's continuous improvement ethos in digital engineering and cybersecurity (DoD, 2021b).

23. Operational Scenarios and Mission Threads

Operational scenarios and mission threads provide the bridge between abstract conformance criteria and realworld mission outcomes. DoD operational test guidance emphasizes that modularity and openness must be validated under realistic mission conditions, not just through static analysis (DoD, 2020c). Scenariobased evaluation ensures that MOSA principles translate into operational adaptability, resilience, and mission assurance.

23.1 Scenario Methodology

A structured scenario methodology begins by defining mission threads that are operationally meaningful and technically traceable. Each thread should include:

- A **baseline case** under expected conditions
- A **contested case** involving adversary actions or environmental stress
- A **degraded case** involving impaired infrastructure or components
- One or more **rapid reconfiguration events**

Conformance success is assessed based on whether modular substitutions, updates, or reconfigurations preserve mission objectives, maintain acceptable risk levels, and demonstrate the adaptability MOSA is intended to provide. This aligns with DoD's emphasis on missioncentric evaluation in operational test and evaluation (OT&E).

23.2 Scenario A: Sensor Module Substitution Under Time Pressure

This scenario evaluates the ability to integrate a new sensor processing module from an alternate supplier under tight operational timelines. Conformance success requires:

- Integration within the target time window
- Preservation of data semantics across the processing chain
- No critical regressions in downstream decision functions

This scenario tests interface clarity, onboarding package completeness, and the effectiveness of continuous verification mechanisms. It reflects realworld operational pressures where rapid capability insertion is essential.

23.3 Scenario B: Cyber Vulnerability Response

This scenario focuses on response to a newly discovered vulnerability in a shared module. Conformance success requires:

- Rapid patching or replacement
- Reproducible build and test evidence
- Preservation of missionthread integrity

Evaluators assess whether the program can identify the scope of impact, prioritize affected systems, and validate that mitigations do not break interoperability. This scenario highlights the importance of modular boundaries, dependency transparency, and Zero Trustaligned controls (DoD, 2025a).

23.4 Scenario C: Coalition Data Exchange Update

This scenario evaluates the ability to adapt to new coalition data exchange requirements while maintaining domestic mission assurance and classification compliance. Conformance success requires:

- Controlled interface updates
- Profile versioning
- Missionthread tests including coalition proxies

This scenario tests multiparty negotiation, shared standards, and evolvable interoperability key elements of joint and coalition operations.

23.5 Scenario D: Supplier Exit Event

This scenario represents a supplier exit event, where a key vendor becomes unavailable. Conformance success requires:

- Transferready packages
- Sufficient technical data and test artifacts
- Bounded onboarding time for alternate suppliers

This scenario tests legal technical coupling, rights sufficiency, and competition planning areas repeatedly highlighted in GAO's MOSA assessments (GAO, 2025).

23.6 Scenario Learning Loop

Across all scenarios, the most important design element is the **learning loop**. Scenario outcomes should feed directly into criteria refinement, test asset enhancement, and targeted training updates. Highvalue scenarios those that reveal recurring weaknesses should be reused and adapted across domains. This iterative approach ensures that MOSA conformance evolves in step with emerging threats, technologies, and mission concepts.

24. MBSE, Digital Thread, and Toolchain Integration

Modelbased systems engineering (MBSE), digital thread approaches, and federated toolchains can significantly strengthen MOSA conformance if integrated thoughtfully. DoD's digital engineering strategy emphasizes that models, data, and tools must support traceability, automation, and authoritative evidence (DoD, 2025a). This section examines how MBSE artifacts can serve as structured evidence, how to connect models to runtime behavior, and how to manage diverse toolchains.

24.1 MBSE as Evidence Source

MBSE artifacts provide structured evidence about architecture boundaries, interface allocations, requirement decomposition, and system behavior assumptions. When models capture module definitions, interface contracts, and dependencies consistently, they offer a more analyzable view of architecture than free text documents. The conformance harness should treat MBSE artifacts as **firstclass evidence sources**, not optional supplements.

This requires mapping architecture models, interface models, and behavioral models to specific conformance criteria. Automated extraction from model repositories reduces transcription errors and increases precision. This aligns with DoD's push for authoritative digital artifacts (DoD, 2025a).

24.2 Traceability from Model to Runtime

MBSE is most persuasive when assertions about the system are linked to runtime behavior. Conformance confidence increases when requirements and architecture models trace to implementation artifacts, test cases, and telemetry. Without this, models risk becoming aspirational rather than authoritative.

The harness should support traceability chains connecting model elements to runtime evidence. This transforms MBSE from design time documentation into a **living evidence backbone** for conformance.

24.3 Digital Thread Considerations

A digital thread is useful only when identifiers, version lineage, and configuration context are consistent. Without this, evidence cannot be correlated across lifecycle phases. DoD's digital

engineering strategy emphasizes that digital thread value depends on coherence, not tool uniformity (DoD, 2025a).

For conformance, the digital thread must allow queries such as: *Which criteria were validated for this configuration, under what test conditions, and based on which models?*

This is the practical threshold for a digital thread that genuinely supports governance.

24.4 Toolchain Federation Challenges

Defense programs use diverse toolchains. Attempting to standardize everyone on a single tool stack is unrealistic and counterproductive. A more sustainable approach is **toolchain federation**, where the harness prioritizes interoperable data contracts and evidence schemas over forced tool standardization.

This preserves flexibility while enabling enterpriselevel conformance analytics.

24.5 Automation Boundaries

Automation is essential for scaling conformance checks, but there are natural boundaries. Schema validation, interface consistency checks, and dependency analysis can be automated. However, semantic completeness, mission tradeoffs, and safetycritical judgments require expert review.

Governance should explicitly define where automation ends and human judgment begins. This avoids overreliance on raw metrics and ensures that automation does not replace expert assessment.

24.6 MBSE Adoption Risks

MBSE introduces risks such as inconsistent model quality, stale baselines, and weak synchronization between models and implementation. Conformance checks should include **model hygiene controls**, such as verifying version alignment, completeness of key elements, and traceability to implement artifacts.

Programs may be required to demonstrate periodic modeltoimplementation reconciliation. This ensures that MBSE strengthens conformance rather than introducing new failure modes.

25. Discussion, Limitations, and Future Research

25.1 Synthesis of Public and Experimental Evidence

The public evidence review and controlled experiment answer different questions. Public cases address whether modular and open system practices have been associated with integration, cost, refresh, or supplier outcomes in actual defense contexts. The prototype addresses whether a concrete evidence mechanism can distinguish known conformant and nonconformant modules under controlled conditions. Neither source alone is sufficient. Together they support a more defensible chain of reasoning: MOSA benefits are possible, public measurement is uneven, and an executable harness can improve the observability and diagnosis of the technical mechanisms on which those benefits depend.

The experiment produced a clear internal result. The harness detected all eighteen injected fault classes and generated the intended root cause diagnosis for each, while the baseline detected five. This outcome is consistent with modularity and conformance theory: a declared interface is not enough when semantics, dependencies, state, provenance, evidence, and boundary conditions remain untested. The result also shows that evidence depth imposes measurable overhead, but in the prototype the absolute median increase was milliseconds rather than minutes or hours. Field studies must determine whether automation reduces or redistributes human effort.

25.2 Threats to Validity

Table 25. Threats to validity and mitigation strategies

Validity threat	Effect on interpretation	Mitigation and remaining limitation
Synthetic fault model	Injected defects may not represent the frequency or interaction of field defects.	Eighteen diverse single fault classes and transparent ground truth improve mechanism coverage; multi fault and field validation remain future work.
Small control sample	Two accepted controls cannot establish a precise false positive rate.	Wilson intervals are reported and specificity is described as a control acceptance proxy.
Author built artifact and candidates	Design knowledge may influence tests and diagnosis rules.	Ground truth metadata, source release, automated results, and independent reproducibility enable external challenge; independent replication is still needed.
Runtime environment	Container timing may not represent production toolchains or hardware.	Paired execution and randomized order support internal comparison; only relative computational overhead is interpreted.
Construct scope	Software interface conformance does not equal full MOSA effectiveness.	Claims are limited to detection, diagnosis, evidence, and decision behavior; economics and mission outcomes are not inferred.
Public evidence heterogeneity	Program reported estimates and different baselines limit cross case synthesis.	Claims are classified by realization and evidence strength rather than pooled into a single effect size.
Weight model	Four dimensions and a symmetric Dirichlet distribution are analytical choices.	Ten thousand draws demonstrate the compensation problem; operational profiles require calibrated weights and gates.
Evidence digest instability	Dynamic findings may change serialized evidence even when decisions remain stable.	Decision stability and digest stability are reported separately; nondeterminism is treated as a finding.

25.3 Generalizability and Boundary Conditions

Generalizability depends on module type, interface maturity, assurance burden, deployment environment, upgrade frequency, supplier market, technical data, and certification constraints. Stateless software services are more amenable to automated substitution than flight critical hardware or tightly integrated cyber physical subsystems. The prototype's exact thresholds and checks should therefore be treated as a reference profile. The broader contribution is the evidence pattern: versioned criteria, explicit scope, known configuration, executable semantics, provenance, independent change, non compensable gates, and reproducible results.

Economic openness remains the most difficult dimension to validate in a laboratory. A technically replaceable module does not create a qualified supplier, funding, facility access, data rights, or certification capacity. Conversely, a program can have multiple nominal suppliers but retain hidden integration lock in. External pilots should link the technical evidence package to observed solicitation participation, onboarding time, change cost, schedule, defect escape, and post fielding performance.

25.4 Implications for Acquisition and Engineering

The findings support four implementation principles. First, MOSA assessments should require executable evidence at selected interfaces rather than relying only on architecture descriptions. Second, critical conditions such as invalid provenance, unsupported interface versions, semantic incompatibility, insufficient rights, or prohibited dependencies should be gates, not merely weighted deductions. Third, evidence completeness and confidence should be shown separately from performance scores. Fourth, programs should preserve machine readable evidence and raw measurements so that cost benefit claims can be revisited as the system evolves.

25.5 Future Research Agenda

The next research phase should apply the harness to multiple releasable module interface change events with independent integrators. A minimum dataset should capture requirements, configuration, interface maturity, integration labor, elapsed time, defects, test outcomes, rights posture, supplier characteristics, certification effort, and post change operational performance. Repeated events would support hierarchical analysis that separates program, domain, module, and event effects. Additional experiments should introduce multi fault candidates, hardware in the loop

constraints, evolving interface versions, cybersecurity remediation scenarios, and blinded assessor calibration.

A second line of work should evaluate governance. Multiple assessors can independently score the same evidence package to estimate agreement; alternative weight profiles can be calibrated through structured expert elicitation; and waiver histories can be tested as leading indicators of architectural degradation. A third line should connect conformance to lifecycle economics by comparing the cost of producing evidence with avoided reintegration, shortened onboarding, reduced defect escape, and preserved competition options.

26. Conclusion

This thesis began with a practical problem: MOSA is mandatory to the maximum extent practicable, yet the evidence needed to determine whether a program is genuinely modular, interoperable, replaceable, and competitively sustainable is inconsistent. The public record contains meaningful success cases, but it also contains weak baselines, estimated counterfactuals, and limited lifecycle cost benefit analysis. A policy statement or interface document therefore cannot, by itself, establish an open system outcome.

The research responded by moving from proposal to implementation. It defined an evidence model, built a minimum viable conformance harness, constructed two conformant controls and eighteen faulted candidates, and executed a paired controlled evaluation. The baseline detected five fault classes; the harness detected all eighteen. Both accepted the two controls. The harness produced correct actionable diagnoses for all fault classes, complete control evidence, stable decisions, and gate robust results across 10,000 random weight sets. The exact paired correctness test yielded $p = 0.000244$. These gains required more checks and approximately a twentyfold median runtime increase, but the primary median harness runtime remained about seven milliseconds in the synthetic environment.

The strongest conclusion is therefore bounded but substantive: an evidence driven conformance harness is technically feasible and can make important forms of hidden coupling and interface failure materially more observable than a minimal documentary and happy path process. The experiment does not prove field cost savings, operational mission effects, or universal performance across DoD systems. It does provide an executable research artifact, transparent data, and a reproducible foundation for the external pilots needed to answer those larger questions.

For defense acquisition, the implication is direct. MOSA should be governed as an evidence producing lifecycle capability. Programs should define the change they expect, acquire the rights and artifacts required to perform it, exercise representative substitution and mission threads, preserve the resulting evidence, and use non compensable gates for decisive defects. When those practices are institutionalized, the Department can move from asking whether a program claims MOSA to asking what can be changed, by whom, under what conditions, with what risk, and supported by what proof.

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Appendix A. Illustrative Criteria Catalog

This appendix provides a compact example of how a versioned criterion can connect policy intent, engineering evidence, test method, threshold, and decision consequence. A production catalog would be machine readable and would include inheritance, profiles, dependencies, waiver rules, and change history.

Table 26. Illustrative criteria catalog excerpt.

ID / dimension	Criterion	Evidence and method	Threshold / decision effect
STR IF 01 Structural	Major interface is defined in a machine readable form.	Schema or IDL, version, and semantic supplement; inspection plus automated validation.	All required fields present and no unresolved critical ambiguity. Blocks Level 1.
OPS SUB 02 Operational	An alternate module can be integrated without privileged knowledge.	Released package, integration logs, and acceptance results; independent substitution test.	Passes mission tests within profile time and defect limits. Blocks Level 2.
ECO SUP 03 Economic	A critical module has credible competitive alternatives.	Qualification records, onboarding data, and rights posture; market and readiness analysis.	Meets the profile defined effective supplier count or has an approved risk plan. Risk finding if unmet.
SEC PROV 04 Security	Software and firmware provenance is traceable.	SBOM, hashes, attestations, and supplier records; automated provenance validation.	Required coverage and signature checks pass. Blocks release when a critical gap remains.

Appendix B. Minimum Evidence Schema

The schema below is technology neutral. JSON, YAML, XML, or an OSCAL aligned representation may be used, provided the fields remain stable, versioned, and machine readable.

Table 27. Minimum evidence schema fields.

Field	Purpose
criterion_id / version	Identifies the exact assertion and interpretation used
profile_id / version	Records the domain and risk tailored selection of criteria
system_boundary	Defines the component, interface, mission thread, and excluded scope
configuration_id	Links results to software, hardware, data, and environment baselines
evaluation_method	Automated test, inspection, analysis, demonstration, or independent substitution
artifact uri / hash	Locates each evidence item and protects integrity
provenance	Records producer, tool, time, chain of custody, and signature or attestation
result / units	Stores observed value, pass/fail status, units, and tolerance
baseline / comparator	Defines the counterfactual or prior state used for outcome claims
confidence_components	Stores reliability, baseline, traceability, and verification factors
finding / severity / owner	Connects deficiencies to remediation and decision authority
waiver / expiry	Records rationale, compensating controls, approval, and review trigger

Appendix C. Worked Conformance Scoring Example

The notional example demonstrates calculation mechanics only. The module is a mission processing service assessed against four dimensions. Values have no relationship to an actual defense program.

Table 28. Worked scoring example for a notional mission processing module.

Dimension	Criterion	Observed x_i	Confidence e_i	Evidence adjusted value
Structural	Interface completeness	0.90	0.95	0.855
Structural	Coupling control	0.72	0.85	0.612
Operational	Substitution test	0.80	0.90	0.720
Operational	Rollback test	1.00	0.95	0.950
Economic	Alternate supplier onboarding	0.50	0.70	0.350
Security	Provenance and boundary controls	0.88	0.95	0.836

Applying the approved criterion and dimension weights produces a composite score of 64.4 in the illustrative worked example in Appendix C. An unresolved rights defect carries an eight point penalty, reducing the decision score to 56.4. If the profile defines rights sufficiency as a non compensable gate, the module cannot be certified regardless of the arithmetic score. This example demonstrates why scores must remain traceable to findings, confidence, and decision rules.

Appendix D. Prototype Artifact and Reproducibility Package

The companion archive, MOSA_Conformance_Harness_Prototype_2026, is the executable research artifact used for the controlled evaluation. It contains no operational DoD data. The package is organized so that the experiment, analysis, figures, and automated tests can be reproduced from the recorded version rather than inferred from screenshots or narrative description.

Table 29. Companion artifact contents

Path or artifact	Purpose
schemas/	Machine readable input, output, and manifest contracts.
candidate_modules/	Two conformant controls and eighteen single fault candidate packages.
src/mosa_harness/	Baseline evaluator, harness evaluator, experiment runner, analysis, and figure generation.
test_vectors/	Eight fixed contract and semantic vectors.
results/primary_results.csv	One primary observation for each candidate method pair.
results/repeated_runs.csv	Twenty repeated runtime and stability observations per candidate method pair.
results/evidence_packages/	Structured harness evidence for every candidate.
results/statistical_summary.json	Binomial estimates, paired tests, reliability, runtime, and weight sensitivity.
tests/	Executable checks for controls, fault detection, and evidence behavior.
EXPERIMENT_PROTOCOL.md and DATA_DICTIONARY.md	Pre specified conditions and field definitions.
SHA256SUMS.txt	Integrity hashes for the packaged files.

The recorded reproduction sequence is: create a Python environment, install the pinned requirements, run the experiment with 20 repeats and seed 20260623, run the analysis, generate the figures, and execute the automated tests. The archive includes a convenience shell script. A reproducer should compare decisions and principal statistics rather than expect identical wall clock durations on different hardware.

Appendix E. Detailed Fault and Result Inventory

Table 30. Candidate level primary decisions and harness diagnosis

Candidate	Ground truth class	Status	Baseline	Harness	Harness diagnosis	Score
Configuration Coupling	Configuration Coupling	Faulted	Pass	Fail	Configuration Coupling	8.28
Control Vendor A	Control	Conformant	Pass	Pass	None	00.00
Control Vendor B	Control	Conformant	Pass	Pass	None	00.00
Edge Case Exception	Edge Case Exception	Faulted	Pass	Fail	Edge Case Exception	8.62
Event Id Not Preserved	Event Id Not Preserved	Faulted	Pass	Fail	Event Id Not Preserved	1.38
Forbidden Network Import	Forbidden Network Import	Faulted	Pass	Fail	Forbidden Network Import	6.25
Hidden File Dependency	Hidden File Dependency	Faulted	Pass	Fail	Hidden File Dependency	4.53
Latency Violation	Latency Violation	Faulted	Pass	Fail	Latency Violation	9.31
Missing Manifest	Missing Manifest	Faulted	Fail	Fail	Missing Manifest	9.17
Missing Output Field	Missing Output Field	Faulted	Fail	Fail	Missing Output Field	1.38
Missing Sbom	Missing Sbom	Faulted	Pass	Fail	Missing Sbom	6.25
Nondeterministic Output	Nondeterministic Output	Faulted	Pass	Fail	Nondeterministic Output	0.69
Provenance Hash Mismatch	Provenance Hash Mismatch	Faulted	Pass	Fail	Provenance Hash Mismatch	8.33
Semantic Threshold Shift	Semantic Threshold Shift	Faulted	Pass	Fail	Semantic Threshold Shift	1.04
State Leakage	State Leakage	Faulted	Pass	Fail	State Leakage	1.36
Undeclared Dependency	Undeclared Dependency	Faulted	Pass	Fail	Undeclared Dependency	6.25
Units Error	Units Error	Faulted	Pass	Fail	Units Error	2.76
Wrong Interface Version	Wrong Interface Version	Faulted	Fail	Fail	Wrong Interface Version	8.33
Wrong Output Type	Wrong Output Type	Faulted	Fail	Fail	Wrong Output Type	1.38
Wrong Schema Version	Wrong Schema Version	Faulted	Fail	Fail	Wrong Schema Version	1.38

Note. The two controls are independently implemented conformant modules. Faulted candidates each contain one intended root cause class.

Appendix F. Statistical Output and Reproducibility Record

Table 31. Recorded statistical outputs

Statistic	Value
Baseline sensitivity	27.8% (5/18); Wilson 95% CI 12.5% 50.9%
Harness sensitivity	100.0% (18/18); Wilson 95% CI 82.4% 100.0%
Exact McNemar	Table [[7, 0], [13, 0]]; p = 0.000244
Primary median runtime	Baseline 0.348 ms; harness 7.042 ms
Paired runtime test	Wilcoxon p = 9.537e 07; Cliff's delta = 0.905
Decision stability	Baseline 100.0%; harness 100.0%
Digest stability	Harness controls 100.0%; all candidates 90.0%
Weight sensitivity	Score only maximum fault pass 100.0%; gate adjusted maximum fault pass 0.0%
Automated tests	3 passed in the recorded environment.

The primary CSV, repeated run CSV, JSON statistical summary, source code, and generated figures are included in the companion archive. The thesis reports rounded values for readability; the machine readable files retain additional precision. Timing observations will vary across processors and operating environments, but accept/reject decisions and the non timing statistics should reproduce when the same version, seed, and dependencies are used.

Appendix G. Targeted Evidence Review Protocol and Search Log

The evidence synthesis was executed as a targeted review designed for traceability rather than as an exhaustive systematic review. The search date for the final update was June 23, 2026. Sources were prioritized in this order: statute and official DoD policy; independent GAO oversight; official program and service documentation; NIST conformance and security publications; peer reviewed modularity, software architecture, design science, and empirical testing literature; and publicly accessible thesis repositories.

Table 32. Evidence review search log

Repository or source	Representative search terms	Use
U.S. Code and DoD OUSD(R&E)	MOSA, modular open systems, conformance, interface, data rights, cost benefit	Requirements and implementation expectations.
GAO	modular open systems, open architecture, weapon systems, acquisition, cost, schedule	Independent case evidence and measurement gaps.
NIST CSRC and publications	conformance testing, OSCAL, SBOM, supply chain, continuous monitoring	Evidence schemas, testing, provenance, and assurance.
AFIT Scholar and NPS Calhoun	MOSA thesis, open architecture acquisition, SBIR open systems	Comparable defense acquisition thesis methods.
Aaltodoc and TU Delft repositories	contract testing framework, automated testing, requirement verification	Comparable implemented artifact evaluations.
ACM, IEEE, and journal sources	modularity, design science, reflexion models, mutation testing, effect size	Academic theory and empirical method.

Table 33. Evidence screening and coding rules

Rule	Application
Include named and traceable cases	The source identifies a program, demonstration, dataset, implemented artifact, or formal policy.
Separate claim types	Realized, demonstrated, estimated, expected, and policy claims are not merged.
Record the baseline	The comparator is identified as observed, planned, modeled, or absent.
Assess source independence	Program reported outcomes are distinguished from independent analysis.
Preserve negative and mixed results	Findings contrary to expected MOSA benefits are retained.
Avoid unsupported aggregation	Heterogeneous cases are synthesized narratively rather than pooled into a single effect size.
Document the boundary	Classified, proprietary, and inaccessible program data remain outside the review.

The review can be extended by preserving the same fields: source, date, program or artifact, intervention, outcome domain, observed value, baseline, claim class, evidence strength factors, and limitations. A future systematic review should add duplicate screening, documented search result counts, independent coding, and inter rater agreement.